M.A.M SCHOOL OF ENGINEERING

(An Autonomous Institution)
(Accredited by NAAC || Approved by AICTE || Affiliated to Anna University)
Trichy – Chennai Trunk Road, Siruganur, Tiruchirappalli – 621 105

UG CURRICULUM

(I to VIII SEMESTERS)

B.E. AERONAUTICAL ENGINEERING

Choice Based Credit System (CBCS)

&

Outcome Based Education (OBE)

(For the students admitted from the academic year 2024– 25)

REGULATIONS 2024

M.A.M SCHOOL OF ENGINEERING

(AUTONOMOUS)

REGULATIONS 2024

CHOICE BASED CREDIT SYSTEM

B.E. AERONAUTICAL ENGINEERING

I PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

1. To grasp all aspects of aeronautical engineering along with a good grounding in communication and interpersonal skills which will make them an asset to any organization
2. To secure jobs in private or public organizations and will be able to apply their skills in designing and creating innovative solutions
3. To express ethical values which will make them evolve as good human beings apart from being talented Aeronautical Engineers.

II PROGRAM OUTCOMES (POs)

PO1	**Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	**Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	**Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	**Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7	**Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	**Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

III.PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1	Ability to choose appropriate materials for designing, fabricating aircraft parts and structures to meet the societal and industrial needs.
PSO2	Ability to model and analyse the structures of aircrafts and other aeronautical elements using standard software packages.
PSO3	Ability to design and develop Unmanned Aerial Vehicles (UAVs) for several applications, with ethical practices.

CURRICULUM

M.A.M SCHOOL OF ENGINEERING

DEPARTMENT OF AERONAUTICAL ENGINEERING

REGULATIONS 2024

CHOICE BASED CREDIT SYSTEM

(Students admitted from the Academic Year 2024 – 25 onwards)

I TO VIII SEMESTERS CURRICULUM

Induction Program (Mandatory)	3 Weeks Duration
Induction program for students to be offered right at the start of the first year	• Physical activity • Creative Arts • Universal Human Values • Literary • Proficiency Modules • Lectures by Eminent People • Visits to local Areas • Familiarization to Dept./Branch & Innovations

B.E. AERONAUTICAL ENGINEERING

SEMESTER I

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1	24HS101	Communicative English	3	0	0	3	40	60	100	HS
2	24BS101	Matrices & Calculus	3	1	0	4	40	60	100	BS
3	24ES101	Problem solving and Python Programming	3	0	0	3	40	60	100	ES
4	24HS102	Heritage of Tamil	1	0	0	1	40	60	100	HS
THEORY COURSE WITH LABORATORY COMPONENT										
5	24BS103	Engineering Physics	3	0	2	4	50	50	100	BS
LABORATORY COURSES										
6	24HS103	Communicative English Laboratory	0	0	2	1	60	40	100	HS
7	24ES102	Problem solving and Python Programming Laboratory	0	0	4	2	60	40	100	ES
8	24ES103	Engineering Graphics	0	0	4	2	60	40	100	ES
		TOTAL	**13**	**1**	**12**	**20**				

SEMESTER II										
S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1	24BS201	Transforms and Partial Differential Equations	3	1	0	4	40	60	100	BS
2	24HS202	Professional English	2	0	0	2	40	60	100	HS
3	24ES207	Engineering Mechanics	3	0	0	3	40	60	100	ES
4	24HS201	தமிழரும்தொழில்நுட்பமும் / Tamils and Technology	1	0	0	1	40	60	100	HS
5	24ES201	Design Thinking	2	0	0	2	40	60	100	ES
THEORY COURSE WITH LABORATORY COMPONENT										
6	24BS203	Chemistry for Engineers	3	0	2	4	50	50	100	BS
7	24ES214	Basic Electrical & Electronics Engineering	3	0	2	4	50	50	100	ES
LABORATORY COURSES										
8	24ES211	Foundation skills Lab	0	0	2	1	60	40	100	ES
9	24ES212	Basic Engineering skills lab	0	0	2	1	60	40	100	ES
10	24TP201	Aptitude Skills & Communication Skills 1	0	0	2	1	100	-	100	EEC
		TOTAL	17	1	10	23				

SEMESTER III										
S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1	24BS301	Statistics & Numerical Methods	3	1	0	4	40	60	100	BS
2	24AE301	Elements of Aeronautical Engineering	3	0	0	3	40	60	100	PC
3	24AE302	Aircraft Systems & Instrumentation	3	0	0	3	40	60	100	PC
4	24AE303	Manufacturing Technology	3	0	0	3	40	60	100	PC
THEORY COURSE WITH LABORATORY COMPONENT										
5	24AE304	Fluid Mechanics and Strength of Materials	3	0	2	4	50	50	100	PC
6	24AE305	Aero Thermodynamics	3	0	2	4	50	50	100	PC
LABORATORY COURSES										
7	24AE306	Manufacturing Processes Lab	0	0	3	2	60	40	100	PC
8	24TPS02	Aptitude Skills and Communication skills II	0	0	2	1	60	40	100	EEC
		TOTAL	13	1	9	24				

SEMESTER IV

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1	24AE401	Aerodynamics	3	0	0	3	40	60	100	PC
2	24AE402	Aircraft Engine Maintenance & Repair	3	0	0	3	40	60	100	PC
3	24AE403	Mechanics of Machines	3	1	0	4	40	60	100	PC
4	24MC401	Environmental Sciences	3	0	0	0	40	60	100	MC
5	24MC402	Disaster Management and Preparedness	1	0	0	2	40	60	100	MC
THEORY COURSE WITH LABORATORY COMPONENT										
6	24AE404	Aircraft Structures	3	0	2	4	50	50	100	PC
7	24AE405	Propulsion	3	0	2	4	50	50	100	PC
LABORATORY COURSES										
8	24AE406	Aerodynamics Lab	0	0	4	2	50	50	100	PC
9	24AE407	Aero Engines & Aero Frame Lab	0	0	4	2	60	40	100	PC
10	24TP401	Aptitude Skills III & Technical Skills I	0	0	2	1	60	40	100	EEC
		TOTAL	**18**	**1**	**14**	**25**				

SEMESTER V

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1	24AE501	Flight Dynamics	3	1	0	4	40	60	100	PC
2	24AE502	Rocket Propulsion	3	0	0	3	40	60	100	PC
3	-	Professional Elective - I	3	0	0	3	40	60	100	PE
4	-	Professional Elective - II	3	0	0	3	40	60	100	PE
5	-	Open Elective - I	3	0	0	3	40	60	100	OE
THEORY COURSE WITH LABORATORY COMPONENT										
6	24AE503	Finite Element Analysis	3	0	2	4	50	50	100	PC
LABORATORY COURSES										
7	24AE504	Aircraft Systems Lab	0	0	4	2	60	40	100	PC
8	24AE505	CAD Modelling Lab	0	0	2	1	60	40	100	PC
9	24TP501	Aptitude Skills IV & Technical Skills II	0	0	2	1	60	40	100	EEC
		TOTAL	**18**	**1**	**10**	**24**				

SEMESTER VI

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1	24AE601	Aircraft Design	3	0	0	3	40	60	100	PC
2	24AE602	Airport Management	3	0	0	3	40	60	100	PC
3	-	Professional Elective - III	3	0	0	3	40	60	100	PE
4	-	Professional Elective - IV	3	0	0	3	40	60	100	PE
5	-	Open Elective - II	3	0	0	3	40	60	100	OE
THEORY COURSES WITH LABORATORY COMPONENT										
6	24AE603	Basics and Applications of AI & ML in Aeronautical Engg	3	0	2	4	50	50	100	PC
LABORATORY COURSES										
7	24TP601	Aptitude Skills V & Technical Skills III	0	0	2	1	60	40	100	EEC
		TOTAL	18	0	4	20				

SEMESTER VII

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1	24HS701	Universal Human Values	3	0	0	3	40	60	100	HS
2	24AE701	Wind Tunnel Techniques	3	0	0	3	40	60	100	PC
3	-	Professional Elective - V	3	0	0	3	40	60	100	PE
4	-	Open Elective - III	3	0	0	3	40	60	100	OE
LABORATORY COURSES										
5	24TP701	Internship	0	0	0	2	100	-	100	EEC
6	24AE702	Aircraft & UAV Design Project	0	0	6	3	60	40	100	EEC
		TOTAL	12	0	0	17				

Note: Four weeks Summer Internship carries Two credit and it will be done during VI semester summer vacation and same will be evaluated in VII semester.

SEMESTER VIII

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
LABORATORY COURSES										
1	24AE801	Project Work	0	0	20	10	60	40	100	EEC
		TOTAL	0	0	20	10				

VERTICAL I – PROFESSIONAL ELECTIVE I

AERODYNAMICS & FLIGHT MECHANICS

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1	24AEX01	Highspeed Aerodynamics	3	0	0	3	40	60	100	PE
2	24AEX02	Aerodynamic Testing and Instrumentation	3	0	0	3	40	60	100	PE
3	24AEX03	Computational Fluid Dynamics	3	0	0	3	40	60	100	PE
4	24AEX04	Helicopter Theory	3	0	0	3	40	60	100	PE
5	24AEX05	Rockets and Missiles	3	0	0	3	40	60	100	PE
6	24AEX06	Aircraft Stability and Control	3	0	0	3	40	60	100	PE

VERTICAL II – PROFESSIONAL ELECTIVE II

PROPULSION

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1	24AEX07	Advanced Propulsion Systems	3	0	0	3	40	60	100	PE
2	24AEX08	Design of Gas Turbine Engine Components	3	0	0	3	40	60	100	PE
3	24AEX09	Turbo Machines	3	0	0	3	40	60	100	PE
4	24AEX10	Cryogenic Engineering	3	0	0	3	40	60	100	PE
5	24AEX11	Combustion Techniques	3	0	0	3	40	60	100	PE
6	24AEX12	Aircraft Fuels and Propellants	3	0	0	3	40	60	100	PE

VERTICAL III – PROFESSIONAL ELECTIVE III

AIRCRAFT STRUCTURES & MATERIALS

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1	24AEX13	Fatigue and Fracture Mechanics	3	0	0	3	40	60	100	PE
2	24AEX14	Composite Materials and Structures	3	0	0	3	40	60	100	PE
3	24AEX15	Additive Manufacturing	3	0	0	3	40	60	100	PE
4	24AEX16	Aerospace Materials	3	0	0	3	40	60	100	PE
5	24AEX17	Experimental Stress Analysis	3	0	0	3	40	60	100	PE
6	24AEX18	Smart Materials and Structures	3	0	0	3	40	60	100	PE

VERTICAL IV – PROFESSIONAL ELECTIVE IV

DIVERSIFIED GROUP

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1	24AEX19	Aircraft General Engineering & Maintenance Practices	3	0	0	3	40	60	100	PE
2	24AEX20	Design of UAV systems	3	0	0	3	40	60	100	PE
3	24AEX21	Non-Destructive Testing and Evaluation	3	0	0	3	40	60	100	PE
4	24AEX22	Aircraft Hydraulics Systems	3	0	0	3	40	60	100	PE
5	24AEX23	Air Traffic Control	3	0	0	3	40	60	100	PE
6	24AEX24	Aircraft Rules and Regulations	3	0	0	3	40	60	100	PE
7	24AEX25	MEMS	3	0	0	3	40	60	100	PE

VERTICAL V – PROFESSIONAL ELECTIVE V

MANAGEMENT ELECTIVES

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1	24MGT01	Total Quality Management	3	0	0	3	40	60	100	PE
2	24MGT02	Principles of Management	3	0	0	3	40	60	100	PE
3	24MGT03	Industrial Management	3	0	0	3	40	60	100	PE
4	24MGT04	Safety Management	3	0	0	3	40	60	100	PE
5	24MGT05	Operational Management	3	0	0	3	40	60	100	PE

OPEN ELECTIVES

(To be offered by Faculty of Aeronautical Engineering)

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1		Drone Technologies	3	0	0	3	40	60	100	PE
2		Aviation Management	3	0	0	3	40	60	100	PE
3		Basics of Aeroplanes	3	0	0	3	40	60	100	PE
4		Satellite System and Application	3	0	0	3	40	60	100	PE
5		Entrepreneurship Development	3	0	0	3	40	60	100	PE
6		Project Report Writing	3	0	0	3	40	60	100	PE
7		Urban Agriculture	3	0	0	3	40	60	100	PE
8		Drinking water Supply & Treatment	3	0	0	3	40	60	100	PE

S.No.	Category	Credits Per Semester								Total Credit	Credits in%
		I	II	III	IV	V	VI	VII	VIII		
1	HS	5	3					3		11	06.75 %
2	BS	8	8	4						20	12.27 %
3	ES	7	11							18	11.04 %
5	PC			19	22	14	10	3		68	41.72 %
6	PE					6	6	3		15	09.20 %
7	OE					3	3	3		09	05.52 %
8	EEC		1	1	1	1	1	5	10	20	12.27 %
9	MC				2					2	01.23 %
Total		20	23	24	25	24	20	17	10	163	100.0%

HS	-	Humanities and Social Science
BS	-	Basic Science
ES	-	Engineering Science
PC	-	Professional Core
PE	-	Professional Elective
OE	-	Open Elective
EEC	-	Employability Enhancement Course
MC	-	Mandatory Course
CA	-	Continuous Assessment
ES	-	End Semester Examination

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Trichy – Chennai Trunk Road, Siruganur, Tiruchirappalli – 621 105

B.E. AERONAUTICAL ENGINEERING

SEMESTER I

S.No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1.	24HS101	Communicative English	3	0	0	3	40	60	100	HS
2.	24BS101	Matrices & Calculus	3	1	0	4	40	60	100	BS
3.	24ES101	Problem solving and Python Programming	3	0	0	3	40	60	100	ES
4.	24HS102	Heritage of Tamil	1	0	0	1	40	60	100	HS
THEORY COURSE WITH LABORATORY COMPONENT										
5.	24BS103	Engineering Physics	3	0	2	4	50	50	100	BS
LABORATORY COURSES										
6.	24HS103	Communicative English Laboratory	0	0	2	1	60	40	100	HS
7.	24ES102	Problem solving and Python Programming Laboratory	0	0	4	2	60	40	100	ES
8.	24ES103	Engineering Graphics	0	0	4	2	60	40	100	ES
		TOTAL	13	1	12	20				

REGULATIONS 2024

Recommended by I[st] BOS held on 05.9.24 & Approved by I[st] Academic Council held on 25.11.24

R 2024	ENGLISH					SEMESTER: I
24HS101	COMMUNICATIVE ENGLISH	L	T	P	C	HS
		3	0	0	3	

COMMON TO ALL DEPARTMENTS

COURSE OBJECTIVES:

The objectives of learning this course are to:

- Enable learners use words appropriately in their communication.
- Enhance learners' grammatical accuracy in communication.
- Develop learners' ability to read and listen to texts in English.
- Strengthen the communication skills of the learners.
- Help learners write appropriately in professional contexts

COURSE OUTCOMES:

At the end of this course, students are able to

CO1: Understand the basic grammatical structures and apply them in right context

CO2: Identify and report cause and effects in events, industrial processes through technical texts.

CO3: Apply appropriate words in a professional context.

CO4: Interpret information presented in tables, charts and other graphic forms.

CO5: Draft effective resumes in the context of job search.

UNIT: I	BASICS OF LANGUAGE	9

Reading - Reading brochures (technical context), telephone messages advertisements, user manuals. **Writing** - Sequential Writing – connecting ideas using transitional words (Jumbled Sentences), **Gramma**r – basics; parts of speech, Simple Tenses – Form, Function and Meaning; **Vocabulary** - Synonyms; One word substitution

UNIT: II	INTRODUCTION TO FUNDAMENTALS OF COMMUNICATION	9

Reading - Reading biographies, travelogues, newspaper reports, **Writing** -Cause and EffectEssays, **Grammar**: Continuous Tenses, Subject-Verb Agreement, Idioms; **Vocabulary:** Antonyms, Language puzzles.

UNIT: III	NARRATION AND SUMMATION	9

Reading – Reading advertisements, Case Studies, **Writing**- Check-list, Instructions. **Grammar**: Perfect Tenses, Imperatives; Adjectives, **Vocabulary:** Language Games/ Group Discussion.

UNIT: IV	REPORTING OF EVENTS AND RESEARCH	9

Reading –Newspaper articles; **Writing** – Recommendations, Transcoding **Grammar** – Reported Speech, Pronouns - Possessive & Relative pronouns, **Vocabulary:** Oral Presentation.

UNIT: V	THE ABILITY TO PUT IDEAS OR INFORMATION COGENTLY	9

Reading – Company profiles, Statement of Purpose, (SOP), an excerpt of interview with professionals; **Writing** – Job / Internship application – Cover letter & Resume; **Grammar** – Numerical adjectives, Relative Clauses. Degrees of comparison, Phrasal Verbs; **Vocabulary:** Informal Vocabulary and Formal Substitutes.

Total Periods :45

Pedagogical Tools:	Black board, chalk, group discussion, role play, youtube videos, NPTEL videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Raymond, Murphy	English Grammar in Use (5^{th} Edition)	Cambridge Press: New York	2019
2	Dr. KN. Shoba, and Dr. Lourdes Joevani	English for Science & Technology	Cambridge University Press	2021

REFERENCE BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of Publication**
1	Meenakshi Raman & Sangeeta Sharma	Technical Communication Principles And Practices	Oxford Univ. Press	2016
2	Lakshmi Narayanan	A Course Book on Technical English	Scitech Publications (India) Pvt. Ltd.	2017
3	Kulbhusan Kumar	Effective Communication Skill	R S Salaria, Khanna Publishing House.	2018

WEB LEARNING RESOURCES:	
1	https://store.acolad.com/products/english-for-engineering
2	https://www.cambridge.es/en/catalogue/business-english/other-titles/cambridge-english-for/engineering
3	https://shipcon.eu.com/english-for-engineers/
4	https://www.udemy.com/course/english-for-engineers/
5	https://store.acolad.com/products/english-for-engineering

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	1	1	-	-	-	-	3	-	-	-
CO2	-	3	-	-	-	-	3	3	-	3	-	3	-	-	-
CO3	-	-	-	-	2	-	2	-	-	3	-	3	-	-	-
CO4	-	-	-	-	-	3	-	1	2	3	-	3	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	3	3	-	-	-
AVG	-	3	-	-	2	2	2	2	2	3	3	3	-	-	-

R 2024	MATHEMATICS					SEMESTER: I
24BS101	MATRICES AND CALCULUS	L	T	P	C	BS
		3	1	0	4	

COMMON TO ALL DEPARTMENTS

COURSE OBJECTIVES:

The objectives of learning this course are to:

- Develop the use of matrix algebra techniques that is needed by engineers for practical applications.
- Familiarize the student with functions of several variables. this is needed in many branches of engineering.
- Make the students understand various techniques of integration.
- Acquaint the student with mathematical tools needed in evaluating multiple integrals and their applications.
- Make the student acquire sound knowledge of techniques in solving ordinary differentialequations that model engineering problems.

COURSE OUTCOMES:

At the end of this course, students are able to

CO1: Apply the knowledge of matrices with the concepts of eigenvalues to study their problems in coreareas
CO2: Apply the basic techniques and theorems function of several variables in other areas ofmathematics
CO3: Apply different methods of integration in solving practical problems.
CO4: Apply multiple integral ideas in solving areas, volumes and other practical problems.
CO5: Solve basic application problems described by second and higher order linear differential equationswith constant coefficients.

UNIT: I	MATRICES	9+3

Eigen values and Eigenvectors of a real matrix - Properties of Eigen values and Eigenvectors (without proof) - Statement and applications of Cayley- Hamilton theorem (without proof) - Diagonalization of matrices- Reduction of a quadratic form to canonical form by orthogonal transformation-Nature of quadratic forms.

UNIT: II	FUNCTIONS OF SEVERAL VARIABLES	9+3

Partial derivatives - Total derivative - Jacobian and properties - Taylor's series expansion for function of two variables - Extreme values of functions of two variables - Lagrange multipliers method.

UNIT: III	INTEGRAL CALCULUS	9+3

Definite and indefinite integrals - Substitution rule - Techniques of Integration: Integration by parts, Trigonometric integrals, Trigonometric substitutions, Integration of rational functions by Partial fraction, Integration of irrational functions

UNIT: IV	MULTIPLE INTEGRALS	9+3

Double integrals - Change of order of integration - Double integrals in polar coordinates - Triple integrals - Applications in area and volume (except spherical , cylindrical coordinates)

UNIT: V	ORDINARY DIFFERENTIAL EQUATIONS	9+3

Second and higher order linear differential equations with constant coefficients - Variable coefficients - Euler Cauchy equation - method of variation parameters.

Total Periods :60

Pedagogical Tools:	Chalk & Board, PPT, NPTEL video, you tube video

Recommended by I st BOS held on 05.9.24 & Approved by I st Academic Council held on 25.11.24

TEXT BOOKS:				
Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Kreyszig.E	Advanced Engineering Mathematics	John Wiley and sons, New Delhi	2016
2	Grewal B.S	Higher Engineering Mathematics	Khanna Publishers, New Delhi	2018
3	James Stewart	Calculus: Early Transcendentals	Cengage Learning, New Delhi	2015
REFERENCE BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of Publication**
1	Bali.N, M.Goyal And Watkins.C	Advanced Engineering Mathematics	Lakshmi Publications, New Delhi	2015
2	Ramana B.V	Higher Engineering Mathematics	McGraw Hill Education, New Delhi	2016
3	Narayanan.S, Manicavasagam Pillai.T.K	Calculus	S.Vishwanathan Publishers, Chennai	2009

WEB LEARNING RESOURCES:	
1	https://nptel.ac.in/courses/111108157
2	https://nptel.ac.in/courses/111104125
3	https://nptel.ac.in/courses/111105121
4	https://nptel.ac.in/courses/111104085
5	https://nptel.ac.in/courses/111104521
6	https://www.brainkart.com/subject/Matrices-and-Calculus_454/

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	1	1	-	-	-	-	-	-	-	3	-	-	-
CO2	3	3	1	1	-	-	-	-	-	-	-	3	-	-	-
CO3	3	3	1	1	-	-	-	-	-	-	-	3	-	-	-
CO4	3	3	1	1	-	-	-	-	-	-	-	3	-	-	-
CO5	3	3	3	3	-	-	-	-	-	-	-	2	-	-	-
AVG	3	3	1.4	1.4	-	-	-	-	-	-	-	2.8	-	-	-

<table>
<tr><td>R 2024</td><td colspan="4">COMPUTER SCIENCE AND ENGINEERING</td><td>SEMESTER: I</td></tr>
<tr><td rowspan="2">24ES101</td><td rowspan="2">PROBLEM SOLVING AND PYTHON PROGRAMMING</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">ES</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td colspan="7">Common to AERO, BME, ECE, EEE, MECH AND MCT Departments</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are:

- To understand the basics of algorithmic problem solving.
- To learn to solve problems using Python conditionals and loops.
- To define Python functions and use function calls to solve problems.
- To use Python data structures - lists, tuples, dictionaries to represent complex data.
- To do input/output with files in Python

COURSE OUTCOMES:

At the end of this course, students able to
CO1: Develop algorithmic solutions to simple computational problems and execute simple Python programs.
CO2: Write simple Python programs using conditionals and loops for solving problems.
CO3: Decompose a Python program into functions.
CO4: Represent compound data using Python lists, tuples, dictionaries etc.
CO5: Read and write data from/to files in Python programs.

UNIT: I	COMPUTATIONAL THINKING AND PROBLEM SOLVING	9

Fundamentals of Computing – Identification of Computational Problems -Algorithms, building blocks of algorithms (statements, state, control flow, functions), notation (pseudo code, flow chart, programming language), algorithmic problem solving, simple strategies for developing algorithms (iteration, recursion). Illustrative problems: Flowchart to find minimum in a list, Flowchart to insert a card in a list of sorted cards, Pseudo code to find an integer number in a range, Pseudo code to find the position of the largest element in an list of n numbers, Towers of Hanoi.

UNIT: II	DATA TYPES, EXPRESSIONS, STATEMENTS	9

Python interpreter and interactive mode, debugging; values and types: int, float, boolean, string, and list; variables, expressions, statements, packing and unpacking arguments, precedence of operators, comments; Illustrative programs: swap the values of two variables, circulate the values of n variables, distance between two points, reverse the string.

UNIT: III	CONTROL FLOW, FUNCTIONS, STRINGS	9

Conditionals: Boolean values and operators, conditional (if), alternative (if-else),chained conditional (if-elif-else);Iteration: state, while, for, break, continue, pass; Fruitful functions: return values, parameters, local and global scope, function composition, recursion; Strings: string slices, immutability, string functions and methods, string module; Lists as arrays. Illustrative programs: square root, gcd, exponentiation, sum an array of numbers, factorial, fibonacci series, palindrome, linear search, binary search.

UNIT: IV	LISTS, TUPLES, DICTIONARIES	9

Lists: list operations, list slices, list methods, list loop, mutability, aliasing, cloning lists, list parameters; Tuples: tuple assignment, tuple as return value; Dictionaries: operations and methods; advanced list processing - list comprehension; Illustrative programs: Bubble sorting, Insertion, selection, merge sort, histogram, Add Two Matrices, Transpose a Matrix, Students marks statement, Retail bill preparation.

UNIT: V	FILES, MODULES, PACKAGES	9

Files and exceptions: text files, reading and writing files, format operator; command line arguments, errors and exceptions, handling exceptions, modules (numpy, pandas, scipy, matplotlib, statmodels), packages; Illustrative programs: word count, copy file, check voting eligibility, count the number of each vowel in a string, random number generation, time series analysis, Marks range validation (0-100).

Total Periods: 45

Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, You Tube Videos, NPTEL videos.

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Allen B. Downey	Think Python: How to Think like a Computer Scientist	O'Reilly Publishers	2016
2	Karl Beecher	Computational Thinking: A Beginner's Guide to Problem Solving and Programming	BCS Learning & Development Limited	2017

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Paul Deitel and Harvey Deitel	Python for Programmers	Pearson Education	2021
2	G Venkatesh and Madhavan Mukund	Computational Thinking: A Primer for Programmers and Data Scientists	Notion Press	2021
3	John V Guttag	Introduction to Computation and Programming Using Python: With Applications to Computational Modeling and Understanding Data	MIT Press	2021

WEB LEARNING RESOURCES:

1. https://www.python.org/
2. https://www.geeksforgeeks.org/python-programming-language-tutorial/
3. https://www.w3schools.com/python/

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	-	-	-	-	-	2	2	3	3	-
CO2	3	3	3	3	2	-	-	-	-	-	2	2	3	-	-
CO3	2	2	-	2	2	-	-	-	-	-	1	-	3	-	-
CO4	1	2	-	-	1	-	-	-	-	-	1	-	2	-	-
CO5	2	2	-	2	2	-	-	-	-	-	1	-	3	-	-
AVG	2	2	2	2	2	-	-	-	-	-	1	1.5	3	1	-

<table>
<tr><td>R 2024</td><td colspan="5">TAMIL</td><td>SEMESTER: I</td></tr>
<tr><td rowspan="2">24HS102</td><td rowspan="2">தமிழர் மரபு / Heritage of Tamil</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">HS</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td colspan="7">COMMON TO ALL DEPARTMENTS</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are to

- Learn the Extensive literature of classical Tamil
- Review the fine arts heritage of Tamil culture
- Realize the contribution of Tamil in Indian freedom struggle

COURSE OUTCOMES:

At the end of this course, students are able to

CO1: Understand the weaving and ceramic technology of ancient Tamil people nature.

CO2: Understand the construction technology, building materials in sangam period and case studies.

CO3: Infer the metal process, coin and beads manufacturing with relevant archaeological evidence.

CO4: Realize the agriculture methods, irrigation technology and pearl diving.

CO5: Apply the knowledge of scientific Tamil and Tamil computing.

UNIT: I	**LANGUAGE AND LITERATURE**	**3**

Dravidian Languages – Tamil as a Classical Language - Classical Literature in Tamil – Distributive Justice in Sangam Literature - Management Principles in Thirukural - Tamil Epics and Impact of Buddhism & Jainism in Tamil Land - Bakthi Literature Azhwars and Nayanmars - Forms of minor Poetry - Development of Modern literature in Tamil - Contribution of Bharathiyar and Bharathidhasan

UNIT: II	**HERITAGE - ROCK ART PAINTINGS TO MODERN ART – SCULPTURE**	**3**

Hero stone to modern sculpture - Bronze icons - Tribes and their handicrafts - Art of temple car making - - Massive Terracotta sculptures, Village deities, Thiruvalluvar Statue at Kanyakumari, Making of musical instruments - Mridangam, Parai, Veenai, Yazh and Nadhaswaram - Role of Temples in Social and Economic Life of Tamils.

UNIT: III	**FOLK AND MARTIAL ARTS**	**3**

Therukoothu, Karakattam, VilluPattu, KaniyanKoothu, Oyillattam, Leather Puppetry, Silambattam, Valari, Tiger dance - Sports and Games of Tamils.

UNIT: IV	**THINAI CONCEPT OF TAMILS**	**3**

Flora and Fauna of Tamils &Agam and Puram Concept from Tholkappiyam and Sangam Literature - Aram Concept of Tamils - Education and Literacy during Sangam Age - Ancient Cities and Ports of Sangam Age - Export and Import during Sangam Age - Overseas Conquest of Cholas.

UNIT: V	**CONTRIBUTION OF TAMILS TO INDIAN NATIONAL MOVEMENT AND INDIAN CULTURE**	**3**

Contribution of Tamils to Indian Freedom Struggle - The Cultural Influence of Tamils over the other parts of India – Self-Respect Movement - Role of Siddha Medicine in Indigenous Systems of Medicine – Inscriptions & Manuscripts – Print History of Tamil Books.

Total Periods: 15

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:				
Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Dr.K.K.Pillay	Tamilnadu history people and culture	Tamilnadu Textbook and Education works Corporation	2019
2	EL Sundaram	Computer Tamil	Vikatanprasuram	2016
3	Dr.S.Singaravelu	Social Life of the Tamils - The Classical Period	International Institute of Tamil Studies.	2001
4	Dr.S.V.Subatamanian, Dr.K.D. Thirunavukkarasu	Historical Heritage of the Tamils	International Institute of Tamil Studies	2010
5	Dr.M.Valarmathi	The Contributions of the Tamils to Indian Culture	International Institute of Tamil Studies	2001
6	-	Keeladi - 'Sangam City Civilization on the banks of river Vaigai'	Department of Archaeology & TamilNadu Text Book and Educational Services Corporation, Tamil Nadu	2019

REFERENCE BOOKS:				
1	Dr.K.K.Pillay	Studies in the History of India with Special Reference to Tamil Nadu	The Author	1979
2	-	Porunai Civilization	Department of Archaeology & TamilNadu Text Book and Educational Services Corporation, Tamil Nadu	2019
3	R.Balakrishnan	Journey of Civilization Indus to Vaigai	RMRL	2019
4	Dr.K.K.Pillay	Social Life of Tamils	A joint publication of TNTB & ESC and RMRL	1975

WEB LEARNING RESOURCES:

1. https://youtu.be/8J3UJXu4JZ0?si=ekqrc_x3J79C_MwI
2. https://www.youtube.com/live/WbnNQM2LNQA?si=S5YS3vXjIotIuDxp
3. https://www.youtube.com/live/10Z7NdBPAYU?si=Xbvjmr9wzfQBCHH6

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
CO2	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
CO3	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
CO4	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
CO5	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-

R 2024	DEPARTMENT OF SCIENCE AND HUMANITIES					SEMESTER: 1
24HS102	தமிழர் மரபு	L	T	P	C	SH
		3	0	0	3	

COURSE OBJECTIVES:

The main objectives of this course are to:

- செம்மொழியான தமிழில் விரிவான இலக்கியத்தை கற்க கற்க வேண்டும்.
- தமிழ் கலாச்சாரத்தின் நுன்களை பாரம்பரியத்தை மீட்டெடுக்க வேண்டும்.
- இந்திய சுதந்திரப் போராட்டத்தில் தமிழரின் பங்களிப்பை உணர்த்த வேண்டும்.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: தமிழின் விரிவான இலக்கியங்களையும் அதன் செம்மொழி தன்மையையும் புரிந்து கொள்வார்கள்

CO2: 2பண்டைய மக்களின் சிற்பம் ஓவியம் மற்றும் இசைக்கருவிகள் பாரம்பரியத்தை புரிந்து கொள்வார்கள்
CO3: 3தமிழ் மக்களின் நாட்டுப்புற மற்றும் தற்காப்பு கலைகளுக்கு புத்துயிர் அளிப்பார்கள்
CO4: 4திணை கருத்துக்கள் வர்த்தகம் மற்றும் சோழ வம்சத்தின் வெற்றி ஆகியவற்றை புரிந்து கொள்வார்கள்
CO5: 5இந்திய சுதந்திரப் போராட்டம் சுயமரியாதை இயக்கத்திலும் சித்த மருத்துவத்திலும் தமிழரின் பங்களிப்பை புரிந்து கொள்வார்கள்

UNIT: I	மொழி மற்றும் இலக்கியம் திராவிட மொழிகள்	3

தமிழ் ஒரு செம்மொழி -தமிழ் செவ்விளக்கியங்கள் -சங்க இலக்கியத்தில் பகிர்தல் அறம் - திருக்குறளில் மேலாண்மை கருத்துக்கள் -தமிழ் காப்பியங்கள் -தமிழகத்தில் சமண பௌத்த சமயங்களின் தாக்கம் -பக்தி இலக்கியம் ஆழ்வார்கள் மற்றும் நாயன்மார்கள் -சிற்றிலக்கியங்கள் - தமிழில் நவீன இலக்கியத்தின் வளர்ச்சி- தமிழ் இலக்கிய வளர்ச்சியில் பாரதியார் மற்றும் பாரதிதாசன் ஆகியோரின் பங்களிப்பு.

Pedagogical Tools	கரும்பலகை ,வினாடி- வினா, பிபிடி ,யூடியூப் வீடியோஸ்,, கலந்தாலோசித்தல்

UNIT: II	மரபு -பாறை ஓவியங்கள் முதல் நவீன ஓவியங்கள் வரை சிற்பக்கலை	3

நடுகள் முதல் நவீன சிற்பங்கள் வரை- ஐம்பொன் சிலைகள் -பழங்குடியினர் மற்றும் அவர்கள் தயாரிக்கும் கைவினைப் பொருள்கள்- பொம்மைகள்- தேர் செய்யும் கலை- சுடுமண் சிற்பங்கள் - நாட்டுப்புற தெய்வங்கள்- குமரி முனையில் திருவள்ளுவர் சிலை- இசைக்கருவிகள்- மிருதங்கம்- பறை- வீணை -யாழ் –நாதஸ்வரம்- தமிழர்களின் சமூக பொருளாதார வாழ்வில் கோவில்களின் பங்கு.

Pedagogical Tools	கரும்பலகை ,வினாடி- வினா, பிபிடி ,யூடியூப் வீடியோஸ்,, கலந்தாலோசித்தல்

UNIT: III	நாட்டுப்புறக் கலைகள் மற்றும் வீர விளையாட்டுக்கள்	3

தெருக்கூத்து- கரகாட்டம்- வில்லுப்பாட்டு -கணியான் கூத்து -ஒயிலாட்டம் -தோல்பாவை கூத்து சிலம்பாட்டம் –வளரி- புலியாட்டம் -தமிழர்களின் விளையாட்டுக்கள்.

Pedagogical Tools	கரும்பலகை ,வினாடி- வினா, பிபிடி ,யூடியூப் வீடியோஸ்,, கலந்தாலோசித்தல்	
UNIT: IV	தமிழர்களின் திணை கோட்பாடுகள்	**3**
தொல்காப்பியம் மற்றும் சங்க இலக்கியத்தில் அகம் மற்றும் புற கோட்பாடுகள்- தமிழர்கள் போற்றிய அறக்கோட்பாடு -சங்ககாலத்தில் தமிழகத்தில் எழுத்தறிவும் கல்வியும்- சங்க கால நகரங்களும் துறைமுகங்களும் -சங்க காலத்தில் ஏற்றுமதி மற்றும் இறக்குமதி -கடல் கடந்த நாடுகளில் சோழர்களின் வெற்றி.		
Pedagogical Tools	கரும்பலகை ,வினாடி- வினா, பிபிடி ,யூடியூப் வீடியோஸ்,, கலந்தாலோசித்தல்	
UNIT: V	இந்திய தேசிய இயக்கம் மற்றும் இந்திய பண்பாட்டிற்கு தமிழர்களின் பங்களிப்பு	**3**
இந்திய விடுதலைப் போரில் தமிழர்களின் பங்கு- இந்தியாவில் பிற பகுதிகளில் தமிழ் பண்பாட்டின் தாக்கம்- சுயமரியாதை இயக்கம்- இந்திய மருத்துவத்தில் சித்த மருத்துவத்தின் பங்கு- கல்வெட்டுக்கள்- கையெழுத்து படிகள்- தமிழ் புத்தகங்களின் அச்சு வரலாறு.		
Pedagogical Tools	கரும்பலகை ,வினாடி- வினா, பிபிடி ,யூடியூப் வீடியோஸ்,, கலந்தாலோசித்தல்	
		Total Periods : 15

R 2024	Physics					SEMESTER: I
24BS103	ENGINEERING PHYSICS	L	T	P	C	BS
		3	0	2	4	

COMMON TO: BME ,EEE, AERONAUTICAL, MECHANICAL and MECHATRONICS ENGINEERING

COURSE OBJECTIVES:

The objectives of learning this course are to:

- Achieve an understanding of rotational dynamics of multi-particles
- Acquire the knowledge of transfer of heat in conductors and insulators
- Introduce the basics of oscillations, optics and lasers
- Equip the students to understand the importance of quantum physics
- Introduce and classify crystal structures of materials

COURSE OUTCOMES:

At the end of this course, students can able to
CO1: Understand and analyze the rotational dynamics of multi-particles
CO2: Apply the concepts of heat transfer in various applications.
CO3: Demonstrate a strong foundational knowledge in oscillations, optics and lasers
CO4: Recognize the basics of quantum physics.
CO5: Differentiate crystal structures of materials

UNIT: I	MECHANICS	9

Multi-particle dynamics: Center of mass (C.M) – CM of continuous bodies – motion of the CM – kinetic energy of system of particles. Rotation of rigid bodies: Rotational kinematics – rotational kinetic energy and moment of inertia - theorems of M .I –moment of inertia of continuous bodies – M.I of a diatomic molecule - torque – rotational dynamics of rigid bodies – conservation of angular momentum – rotational energy state of a rigid diatomic molecule - gyroscope - torsional pendulum – double pendulum –Introduction to nonlinear oscillations.

UNIT: II	THERMAL PHYSICS	9

Transfer of heat energy – thermal expansion of solids and liquids – expansion joints - bimetallic strips - thermal conduction, convection and radiation –rectilinear heat flow – thermal conductivity - Forbe's and Lee's disc method: theory and experiment-conduction through compound media (series and parallel)–thermal insulation – applications: heat exchangers, refrigerators, ovens and solar water heaters.

UNIT: III	OSCILLATIONS, OPTICS AND LASERS	9

Simple harmonic motion - resonance –analogy between electrical and mechanical oscillating systems - waves on a string - standing waves - traveling waves - Energy transfer of a wave - sound waves - Doppler effect. Reflection and refraction of light waves - total internal reflection - interference –Michelson interferometer – Theory of air wedge and experiment. Theory of laser - characteristics - Spontaneous and stimulated emission - Einstein's coefficients - population inversion - Nd-YAG laser, CO_2 laser, semiconductor laser –Basic applications of lasers in industry.

UNIT: IV	BASIC QUANTUM MECHANICS	9

Photons and light waves - Electrons and matter waves – Compton effect - The Schrodinger equation (Time dependent and time independent forms) - meaning of wave function - Normalization –Free particle - particle in a infinite potential well: 1D,2D and 3D Boxes- Normalization, probabilities and the correspondence principle

UNIT: V	CRYSTAL STRUCTURE	9

Introduction – Classification of solids –Space lattice –Basis-Lattice parameter – Unit cell – Crystal system – Miller indices –d-spacing in cubic lattice - Calculation of number of atoms per unit cell – Atomic radius- Coordination number – Packing factor for SC, BCC, FCC and HCP structures – crystal imperfection – Burger vector.

Total Periods: 45

Pedagogical Tools	Chalk & board, PPT, NPTEL videos and You Tube videos

Description of the Experiments: **(Any six experiments to be conducted)**	**Total Periods: 30**
1. Non-uniform bending - Determination of Young's modulus	
2. Uniform bending – Determination of Young's modulus	
3. Torsional pendulum - Determination of rigidity modulus of wire and moment of inertia of regular and irregular objects.	
4. Laser- Determination of the wave length of the laser using grating	
5. Optical fibre -Determination of numerical aperture (NA) and acceptance angle (AA)	
6. Air wedge - Determination of thickness of a thin sheet/wire	
7. Ultrasonic interferometer – determination of the velocity of sound and compressibility of liquids	
8. Acoustic grating- Determination of velocity of ultrasonic waves in liquids.	
9. Simple harmonic oscillations of cantilever.	

LIST OF EQUIPMENT FOR A BATCH OF 30 STUDENTS

S. No	NAME OF THE EQUIPMENT	Qty Required
1	Torsional pendulum stop-clock suspension metallic wire two different thickness two identical cylindrical mass screw gauge wooden scale	3
2	Nonuniform bending 1-meter wooden scale two knife edges travelling microscope weight hanger with slotted weights screw gauge vernier caliper pin	3
3	Uniform bending 1-meter wooden scale two knife edges travelling microscope two weight hangers with slotted weights screw gauge vernier caliper pin	3
4	Laser wavelength he ne laser red grating screen	3
5	Numerical aperture and acceptance angle diode laser fiber optic cable movable arrangement with screen	3
6	Ultrasonic interferometer apparatus	3

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	D.Kleppner and R.Kolenkow	An Introduction to Mechanics	McGraw Hill Education (Indian Edition)	2017
2	Gaur,R.K.andGupta,S.L	Engineering Physics	DhanpatRai Publishers	2018
3	D.Halliday, R.Resnick and J.Walker	Principles of Physics	Wiley (Indian Edition)	2015
4	Arthur Beiser, ShobhitMahajan, S.RaiChoudhury	Concepts of Modern Physics	McGraw-Hill (Indian Edition)	2017
5	M.Arumugam	Engineering Physics	Anuradha publications	2010
6	Gaur,R.K.andGupta,S.L	Engineering Physics	DhanpatRai Publishers	2018

REFERENCE BOOKS:				
1	R.Wolfson	Essential University Physics. Volume 1 & 2	Pearson Education (Indian Edition)	2020
2	K.Thyagarajan and A.Ghatak	Lasers: Fundamentals and Applications	Laxmi Publications, (Indian Edition)	2019
3	R.K.Rajput	Thermal Engineering	Laxmi Publications,	2011
4	S.O.Pillai,	Solid State Physics	New Age International, (Multicolour Edition)	2018

WEB LEARNING RESOURCES:

1. https://youtu.be/fDJeVR0o__w?list=PLyQSN7X0ro203puVhQsmCj9qhIFQ-As8e
(Rotating Objects, Moment of Inertia, Rotational KE)
2. https://archive.nptel.ac.in/courses/104/104/104104085/ (Lasers)
3. https://www.youtube.com/playlist?list=PL1gyM10tgL1hK9666oGndGIWDQdpQzkY9
(NPTEL: Heat transfer lectures by Dr.Gangesh A. Viswanathan, IITB)
4 https://archive.nptel.ac.in/courses/115/101/115101107/ (Quantum mechanics)
5 https://youtu.be/5EiZjZjG-IY
(NPTEL lectures: Crystal Structure - 2 (Unit Cell, Lattice, Crystal)
6. https://www.youtube.com/watch?v=mx2P1_M-7UA&list=PLFE3074A4CB751B2B&index=9
(Rotations, Part I: Dynamics of Rigid Bodies)
7. https://www.youtube.com/watch?v=UzrZxpup3rc&list=PLFE3074A4CB751B2B&index=10
(Rotations, Part II: Parallel Axis Theorem)
8. https://youtu.be/7Bj3N1E7vZk?list=PLZOZfX_TaWAHZOgn8CRjpqRElp5Dd-GaY
(Introduction to heat transfer, conduction, convection, and radiation)
9. https://youtu.be/dRpyfm66GxM
(Particle in an Infinite Potential Well (QUANTUM MECHANICS)

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	1	1	1	-	-	-	-	-	-	-	-	-
CO2	3	-	1	1	-	-	-	-	-	-	-	1	-	-	-
CO3	3	3	2	1	2	1	-	-	-	-	-	-	-	-	-
CO4	3	3	1	1	2	1	-	-	-	-	-	-	-	-	-
CO5	3	1	-	-	-	-	-	-	-		-	-	-	-	-
AVG	3	2.5	1.5	1	1.6	1						1			

R2024	ENGLISH					SEMESTER: I
24HS103	COMMUNICATIVE ENGLISH LABORATORY	L	T	P	C	BS
		0	0	2	1	
COMMON TO: ALL DEPARTMENTS						

COURSE OBJECTIVES:

The objectives of learning this course are to:

- Improve the communicative competence of learners
- Help learners use language effectively in academic /work contexts
- Develop various listening strategies to comprehend various types of audio materials like
- Build on students' English language skills by engaging them in listening, speaking
- Use language efficiently in expressing their opinions via various media.

COURSE OUTCOMES:

At the end of this course, students are able to

CO1: Identify varied group discussion skills and apply them to take part in effective
CO2: Listen to and understand different points of view in a discussion
CO3: Speak fluently and accurately in formal and informal communicative contexts
CO4: Describe products and processes and explain their uses and purposes clearly and accurately
CO5: Express their opinions effectively in both formal and informal discussions

LIST OF EXPERIMENTS

1. Write about a self-introduction for your future job opportunities
2. Write a telephonic conversation between a father and a son on "career"
3. Write a product description for a fire extinguisher
4. Give any one product user manual
5. Prepare a TED talk about artificial intelligence
6. Describe a famous person's inspirational you heard before in your life
7. Write about panel discussion
8. Write your view and opinion the solve the water scarcity

Total Periods: 30

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	1	1	-	-	-	-	3	-	-	-
CO2	-	3	-	-	-	-	3	3	-	3	-	3	-	-	-
CO3	-	-	-	-	2	-	2	-	-	3	-	3	-	-	-
CO4	-	-	-	-	-	3	-	1	2	3	-	3	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	3	3	-	-	-
AVG	-	-	-	-	-	1.4	-	1.4	1.4	-	3	3	-	-	-

R 2024	DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING					SEMESTER: I
24ES102	PROBLEM SOLVING AND PYTHON PROGRAMMING LABORATORY	L	T	P	C	ES
		0	0	4	2	

Common to AERO, BME, ECE, EEE , MECH AND MCT Departments

COURSE OBJECTIVES:

The objectives of learning this course are:

- To understand the problem-solving approaches.
- To learn the basic programming constructs in Python.
- To practice various computing strategies for Python-based solutions to real world problems.
- To use Python data structures - lists, tuples, dictionaries.
- To do input/output with files in Python.

COURSE OUTCOMES:

At the end of this course, students able to

CO1: Develop algorithmic solutions to simple computational problems

CO2: Implement programs in Python using conditionals and loops for solving problems.

CO3: Deploy functions to decompose a Python program.

CO4: Process compound data using Python data structures.

CO5: Utilize Python packages in developing software applications.

PRACTICAL EXERCISES:

1. Identification and solving of simple real life or scientific or technical problems, and developing flow charts for the same. (Electricity Billing, Retail shop billing, Sin series, weight of a motorbike, Weight of a steel bar, compute Electrical Current in Three Phase AC Circuit, etc.)
2. Python programming using simple statements and expressions (exchange the values of two variables, circulate the values of n variables, distance between two points).
3. Scientific problems using Conditionals and Iterative loops. (Number series, Number Patterns, pyramid pattern) 4. Implementing real-time/technical applications using Lists, Tuples. (Items present in a library/Components of a car/ Materials required for construction of a building – operations of list & tuples)
4. Implementing real-time/technical applications using Sets, Dictionaries. (Language, components of an automobile, Elements of a civil structure, etc.- operations of Sets & Dictionaries)
5. Implementing real-time/technical applications using Sets, Dictionaries. (Language, components of an automobile, Elements of a civil structure, etc.- operations of Sets & Dictionaries)
6. Implementing programs using Functions. (Factorial, largest number in a list, area of shape)
7. Implementing programs using Strings. (reverse, palindrome, character count, replacing characters)
8. Implementing programs using written modules and Python Standard Libraries (pandas, numpy. Matplotlib, scipy)
9. Implementing real-time/technical applications using File handling. (copy from one file to another, word count, longest word)
10. Implementing real-time/technical applications using Exception handling. (divide by zero error, voter's age validity, student mark range validation)
11. Exploring Pygame tool.
12. Mini Project - Developing a game activity using Pygame like bouncing ball, car race, Cricket alerts etc.

Total Periods: 60	
LIST OF EQUIPMENTS	
INTEL based desktop PC with min. 8 GB RAM and 500GB HDD, 17"or higher TFT Monitor, Keyboard and mouse.	30
Windows10 or higher operating system / Linux Ubuntu 20 or higher.	30
Python3.9 or above	30

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	2	2	2	1	2	1	1	1	2	-	3	2	2	2	-
CO2	2	3	2	1	2	1	1	1	2	-	3	2	2	2	-
CO3	3	2	2	1	3	1	1	1	2	-	3	3	2	2	-
CO4	2	3	3	1	2	1	2	1	2	-	3	2	2	3	-
CO5	2	3	3	1	2	1	-	-	2	1	2	2	2	2	-
AVG	2	3	2	1	2	1	1	1	2	1	3	2	2	2	-

R 2024	MECHANICAL ENGINEERING					SEMESTER: I
24ES103	ENGINEERING GRAPHICS	L	T	P	C	PC
		0	0	4	2	

COMMON TO: AERONAUTICAL, MECHANICAL and MECHATRONICS ENGINEERING

COURSE OBJECTIVES:

The main objectives of this course are to:

- To learn conventions and use of drawing tools in making engineering drawings
- To draw orthographic projection of points and lines
- To understand the projection of planes and simple solids
- To teach the section of solids and obtain the development of surfaces of given solids
- To deliver how to draw isometric and perspective projections of the given solids

COURSE OUTCOMES:

Upon completion of the course, the student will be able to

CO1: Recognize the conventions and construct basic engineering curves.

CO2: Draw the projection of points and lines.

CO3: Sketch the projection of planes and simple solids.

CO4: Produce the projection section of solids and development of surfaces of given solids

CO5: Develop the isometric projection and Perspective projections of the given objects

PRACTICAL EXERCISES:

1. Fundamental of drawing: Importance of graphics in engineering applications–Use of drafting instruments–BIS conventions and specifications – Size, layout and folding of drawing sheets – Lettering and dimensioning. (Not for examination)
2. Fundamental of drawing: Importance of graphics in engineering applications–Use of drafting instruments–BIS conventions and specifications – Size, layout and folding of drawing sheets – Lettering and dimensioning. (Not for examination)
3. Projection of straight lines (only First angle projection) inclined to both the principal planes - Determination of true lengths and true inclinations by rotating line method.
4. Projection of polygonal plane surface inclined to both the principal planes by rotating object method (Pentagonal and Hexagonal plane surface)
5. Projection of Circular plane inclined to both the principal planes by rotating object method.
6. Projection of simple prisms (Hexagon and pentagon) when the axis is inclined to one of the principal planes.
   Note: One problem has to be drawn using CAD software.
7. Projection of simple prisms (Hexagon and pentagon) when the axis is inclined to one of the principal planes.
   Note: One problem has to be drawn using CAD software.
8. Projection of simple pyramids (Hexagon and pentagon), cylinder and cone when the axis is inclined to one of the principal planes.
   Note: One problem has to be drawn using CAD software.
9. Projection of cylinder and cone when the axis is inclined to one of the principal planes.
   Note: One problem has to be drawn using CAD software.
10. Projection of sectioned solids in simple vertical position when the cutting plane is inclined to the one of the principal planes and perpendicular to the other – obtaining true shape of section (Prism or Pyramid)
    Note: One problem has to be drawn using CAD software.

11. Development of lateral surfaces of simple and sectioned solids (Prism or Pyramid) Note: One problem has to be drawn using CAD software. 12. Draw the isometric view of frustum of solids like Prism or Pyramid of pentagonal or hexagonal base. Note: One problem has to be drawn using CAD software. 13. Perspective projection of simple solids-Prisms, pyramids and cylinders by visual ray method. Note: One problem has to be drawn using CAD software.
Total Periods: 60

LIST OF EQUIPMENTS

Sl. No	Description of Equipment	Required numbers (for batch of 30 students)
1	Computers	30 Nos
2	Laser Printer	1 Nos
3	CAD software (Free CAD / BRL CAD/ Equivalent software)	30 Nos

CO PO PSO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	3	3	3	3	2	-	-	-	-	3	-	2	1	3	2
CO2	3	3	3	3	2	-	-	-	-	3	-	2	1	3	2
CO3	3	3	3	3	2	-	-	-	-	3	-	2	1	3	2
CO4	3	3	3	2	2	-	-	-	-	3	-	2	1	3	2
CO5	3	3	3	2	2	-	-	-	-	3	-	2	1	3	2
AVG	3	3	3	2	2	-	-	-	-	3	-	2	1	3	2

M.A.M SCHOOL OF ENGINEERING

(An Autonomous Institution)

(Accredited by NAAC || Approved by AICTE || Affiliated to Anna University)

Trichy – Chennai Trunk Road, Siruganur, Tiruchirappalli – 621 105

B.E. AERONAUTICAL ENGINEERING

SEMESTER II

THEORY COURSES										
S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1.	24BS201	Transforms and Partial Differential Equations	3	1	0	4	40	60	100	BS
2.	24HS202	Professional English	2	0	0	2	40	60	100	HS
3.	24ES207	Engineering Mechanics	3	0	0	3	40	60	100	ES
4.	24HS201	தமிழரும்தொழில்நுட்பமும் / Tamils and Technology	1	0	0	1	40	60	100	HS
5	24ES201	Design Thinking	2	0	0	2	40	60	100	ES
THEORY COURSES WITH LABORATORY COMPONENT										
6.	24BS203	Chemistry for Engineers	3	0	2	4	50	50	100	BS
7.	24ES214	Basic Electrical & Electronics Engineering	3	0	2	4	50	50	100	ES
LABORATORY COURSES										
8.	24ES211	Foundation skills Lab	0	0	2	1	60	40	100	ES
9.	24ES212	Basic Engineering skills lab	0	0	2	1	60	40	100	ES
10.	24TP201	Aptitude Skills & Communication Skills 1	0	0	2	1	100	-	100	EEC
		TOTAL	17	1	10	23				

REGULATIONS 2024

Recommended by 1st BOS held on 17.08.2024 and approved by 1st Academic council held on 25.11.2024

R 2024	MATHEMATICS					SEMESTER: II
24BS201	TRANSFORMS AND PARTIAL DIFFERENTIAL EQUATIONS	L	T	P	C	BS
		3	1	0	4	

COMMON TO: AERO, BME, ECE, EEE, MECH & MCT

COURSE LEARNING OBJECTIVES:

The objectives of learning this course are:

- To Introduce the basic concepts of PDE for solving standard partial differential equations.
- To Make the student appreciate the purpose of using transforms to create a new domain in which it is easier to handle the problem that is being investigated.
- To Introduce fourier series analysis which is central to many applications in engineering apart from its use in solving boundary value problems.
- To Acquaint the student with fourier transform techniques used in wide variety of situations.
- To introduce the effective mathematical tools for the solutions of partial differential equations that model several physical processes and to develop Z-Transform techniques for discrete time systems.

COURSE OUTCOMES:

At the end of this course, students are able to :

CO1: understand how to solve the given standard partial differential equations.
CO2: apply Laplace transform techniques in solving linear differential equations.
CO3: apply Fourier series techniques in engineering applications.
CO4: use the Fourier transforms techniques in solving engineering problems.
CO5: use the Z-Transforms techniques in solving difference equations.

UNIT: I	**PARTIAL DIFFERENTIAL EQUATIONS**	**9+3**

Formation of partial differential equations –Solutions of standard types of first order partial differential equations - Lagrange's linear equation - Homogeneous Linear partial differential equations of second and higher order with constant coefficients

UNIT: II	**LAPLACE TRANSFORMS**	**9+3**

Existence theorem - Transform of standard functions – Transform of Unit step function and Dirac delta function – Basic properties - Shifting theorems - Transforms of derivatives and integrals – Transform of periodic functions - Inverse Laplace Transforms- Convolution theorem (without proof) – Solving differential equations using Laplace Transform techniques.

UNIT: III	**FOURIER SERIES**	**9+3**

Dirichlet's conditions – General Fourier series – Odd and even functions – Half range sine series and cosine series – Root mean square value – Parseval's identity – Harmonic analysis.

UNIT: IV	**FOURIER TRANSFORMS**	**9+3**

Fourier integral theorem – Fourier transform pair - Fourier sine and cosine transforms – Properties – Transform of elementary functions - Convolution theorem (without proof) – Parsevals's identity.

UNIT: V	**Z – TRANSFORMS**	**9+3**

Z-transforms - Elementary properties – Inverse Z-transform using partial fraction and convolution theorem - Formation of difference equations – Solution of difference equations using Z - transforms.

Total Periods :60

Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.

TEXT BOOKS:

Sl.No	Authors	Title of the Book	Publisher	Year of publication
1	Kreyszig.E	Advanced Engineering Mathematics	John Wiley and sons, New Delhi	2016
2	Grewal B.S	Higher Engineering Mathematics	Khanna Publishers, New Delhi	2018

REFERENCE BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of publication**
1	Bali.N, M.Goyal	A text book of Engineering Mathematics	Lakshmi Publications, Reprint,New Delhi	2015
2	Jain R.K. and Iyengar S.R.K.	Advanced Engineering Mathematics	Narosa Publications, New Delhi , 3rd Edition	2017
3	Ramana B.V.	Higher Engineering Mathematics	Tata McGraw Hill Co. Ltd., 11th Reprint, New Delhi	2010
5	Peter V. O'Neil	Advanced Engineering Mathematics	Cengage Learning India Pvt., Ltd,7th Edition, New Delhi	2012

WEB LEARNING RESOURCES:

1. https://www.brainkart.com/subject/Transforms-and-Partial-Differential-Equations_93/
2. https://nptel.ac.in/courses/111105093
3. https://nptel.ac.in/courses/111102129
4. https://nptel.ac.in/courses/111105123
5. https://nptel.ac.in/courses/111106046
6. https://youtu.be/Sb6qrdMPRPE?si=2kqgDNOyQYkh1UJC
7. https://youtu.be/I4pFAAR5km8?si=wcPssWizT66RCYiP
8. https://youtu.be/NNTJ5VinRPU?si=dOq5vs2VbpJx0cOo
9. https://youtu.be/PG_-ax_HmS0?si=bCbVoOtY68o0uZC0
10. https://youtu.be/kum70H2NcqU?si=WelThJwV8X_ysdDa

CO – PO – PSO MAPPING

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	2	3	-	-	-	-	-	-	-	-	3	-	-	-
CO2	3	2	3	-	-	-	-	-	-	-	-	3	-	-	-
CO3	3	2	3	-	-	-	-	-	-	-	-	3	-	-	-
CO4	3	2	3	-	-	-	-	-	-	-	-	3	-	-	-
CO5	3	2	3	-	-	-	-	-	-	-	-	3	-	-	-
Avg	3	2	3									3			

R 2024	ENGLISH					SEMESTER: II
24HS202	Professional English	L	T	P	C	HS
		2	0	0	2	

COMMON TO ALL DEPARTMENTS

COURSE OBJECTIVES:

The objectives of learning this course are to:

- Enable learners use words appropriately in their communication.
- Enhance learners grammatical accuracy in communication.
- Develop learners ability to read and listen to texts in English.
- Strengthen the communication skills of the learners.
- Help learners write appropriately in professional contexts.

COURSE OUTCOMES:

At the end of this course, students are able to:

CO1: Apply their comprehension skills and interpret different contents effortlessly

CO2: Participate effectively in diverse speaking situations

CO3: Apply technical information and knowledge in practical documents.

CO4: Demonstrate appropriate language use in extended discussions.

CO5: Present, discuss and coordinate with their peers in workplace using their language skills.

UNIT: I	GRAMMATICAL LANGUAGE	6

Reading - Intentional Reading - Short Narratives and Passages. **Writing** - Writing emails / letters introducing oneself., **Gramma**r – Sentence Patterns, Why/ Yes or No/ and Tags; **Vocabulary** - Word formation – Prefix Suffix

UNIT: II	LANGUAGE DEVELOPMENT	6

Reading - Excerpts from literature, and travel & technical blogs. **Writing** - Note Making, Note Taking – Paragraph Writing, **Grammar**: Prepositions, Articles, Model verbs; **Vocabulary:** Verbal Analogy / Cloze Exercise.

UNIT: III	GRAMMAR TRANSFORMATION	6

Reading – Timed Reading, Filling KWL, **Writing**- Writing responses to complaints. **Grammar**: Active Passive Voice transformations, Punctuations. **Vocabulary:** Different forms of the same words.

UNIT: IV	REPORTING WRITING	6

Reading – Extensive reading (Jigsaw Reading, Short Stories, Novels); **Writing** – Problem solution essay / Argumentative Essay **Grammar** – Error correction, Infinitive and Gerunds **Vocabulary:** Compound Words.

UNIT: V	FORMAL WRITING AND VERBAL ABILITY	6

Reading– Reading editorials; and Opinion Blogs – Reading editorials; and Opinion Blogs; **Writing** – Paragraph writing Short Report on an event (field trip etc.); **Grammar** – Concord, If conditionals; Vocabulary**:** Dialogue writing.

Total Periods :30

Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, YouTube Videos, NPTEL videos .

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	M. Ashraf Rizvi	Effective Technical Communication	Orient Blackswan Private Ltd.	2020
2	Dr. KN. Shoba, and Dr. Lourdes Joevani	English for Science & Technology	Cambridge University Press	2021

REFERENCE BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of publication**
1	Meenakshi Raman &Sangeeta Sharma	Technical Communication – Principles And Practices	Oxford Univ. Press	2016
2	Lakshmi Narayanan	A Course Book on Technical English	Scitech Publications (India) Pvt. Ltd.	2017
3	Kulbhusan Kumar	Effective Communication Skill	R S Salaria, Khanna Publishing House.	2018

WEB LEARNING RESOURCES:	
1	https://store.acolad.com/products/english-for-engineering
2	https://www.cambridge.es/en/catalogue/business-english/other-titles/cambridge-english-for/engineering
3	https://shipcon.eu.com/english-for-engineers/
4	https://www.udemy.com/course/english-for-engineers/
5	https://store.acolad.com/products/english-for-engineering

CO – PO – PSO MAPPING

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	3	-	-	-	3	-	3	-	-	-
CO2	-	-	-	-	-	-	-	-	3	3	-	3	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	3	3	-	-	-
CO4	-	-	-	-	-	-	-	-	-	3	3	3	-	-	-
CO5	-	-	-	-	-	-	-	-	3	3	3	3	-	-	-
AVG	-	-	-	-	-	-	-	-	3	3	3	3	-	-	-

R 2024	MECHANICAL ENGINEERING					SEMESTER: II
24ES207	ENGINEERING MECHANICS	L	T	P	C	ES
		3	0	0	3	

COMMON TO: AERONAUTICAL, MECHANICAL and MECHATRONICS

COURSE OBJECTIVES:

The objectives of learning this course are:

- To Learn the use scalar and vector analytical techniques for analyzing forces in statically determinate structures
- To introduce the equilibrium of rigid bodies, vector methods and free body diagram
- To study and understand the distributed forces, surface, loading on beam and intensity.
- To learn the principles of friction, forces and to determine the apply the concepts of frictional forces at the contact surfaces of various engineering systems.
- To develop basic dynamics concepts – force, momentum, work and energy;

COURSE OUTCOMES:

At the end of this course, students can able to

CO1 Students will understand the concepts of engineering mechanics

CO2 Students will understand the vectorial representation of forces and moments

CO3 Students will gain knowledge regarding center of gravity and moment of inertia and apply them for practical problems.

CO4 Student will gain knowledge on friction on equilibrium and its application.

CO5 Student will gain knowledge in solving problems involving work and energy

UNIT: I	STATICS OF PARTICLES	9

Fundamental Concepts and Principles, Systems of Units, Method of Problem Solutions, Statics of Particles -Forces in a Plane, Resultant of Forces, Resolution of a Force into Components, Rectangular Components of a Force, Unit Vectors. Equilibrium of a Particle- Newton’s First Law of Motion, Space and Free-Body Diagrams, Forces in Space, Equilibrium of a Particle in Space.

UNIT: II	STATICS OF RIGID BODIES	9

Principle of Transmissibility, Equivalent Forces, Moment of a Force about a Point, Varignon’s Theorem, Rectangular Components of the Moment of a Force, Moment of a Force about an Axis, Couple - Moment of a Couple, Equivalent Couples, Addition of Couples, Resolution of a Given Force into a Force -Couple system, Further Reduction of a System of Forces, Equilibrium in Two and Three Dimensions - Reactions at Supports and Connections.

UNIT: III	PROPERTIES OF SURFACES AND SOLIDS	9

Centroids of lines and areas – symmetrical and unsymmetrical shapes, Determination of Centroids by Integration, Theorems of Pappus s-Guldinus, Centre of Gravity of a Three-Dimensional Body, Centroid of a Volume, Composite Bodies, Determination of Centroids of Volumes by Integration. Moments of Inertia of Areas and Mass - Determination of the Moment of Inertia of an Area by Integration, Polar Moment of Inertia, Radius of Gyration of an Area, Parallel-Axis Theorem, Moments of Inertia of Composite Areas, Moments of Inertia of a Mass - Moments of Inertia of Thin Plates.

UNIT: IV	FRICTION	9

The Laws of Dry Friction, Coefficients of Friction, Angles of Friction, Wedge friction, Wheel Friction, Rolling Resistance, Ladder friction.

UNIT: V	DYNAMICS OF PARTICLES	9

Kinematics - Rectilinear Motion and Curvilinear Motion of Particles. Kinetics- Newton’s Second Law of Motion -Equations of Motions, Dynamic Equilibrium, Energy and Momentum Methods - Work of a Force, Kinetic Energy of a Particle, Principle of Work and Energy, Principle of Impulse and Momentum, Impact of bodies.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL video, you tube video			
TEXT BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of publication**
1	Beer Ferdinand P, Russel Johnston Jr., David F Mazurek, Philip J Cornwell, Sanjeev Sanghi,	Vector Mechanics for Engineers: Statics and Dynamics	McGraw Higher Education	12thEdition, 2019
2	Vela Murali	Engineering Mechanics-Statics and Dynamics	Oxford University Press	2018
3	Rajasekaran S & Sankarasubramanian G	Fundamentals of Engineering Mechanics	3rd Edition, Vikas Publishing, Chennai,	2017.

REFERENCE BOOKS:				
1	Hibbeller, R.C.	Engineering Mechanics: Statics, and Engineering Mechanics: Dynamics	13 th edition, Prentice Hall,	2013
2	Irving H. Shames, Krishna Mohana Rao G	Engineering Mechanics – Statics and Dynamics	4th Edition, Pearson Education Asia Pvt. Ltd.,	2005
3	Timoshenko S, Young D H, Rao J V and Sukumar Pati	Engineering Mechanics	5th Edition, McGraw Hill Higher Education,	2013

WEB LEARNING RESOURCES:

1. NPTEL - https://www.youtube.com/watch?v=A-
2. 3W1EbQ13k&list=PLyqSpQzTE6M_MEUdn1izTMB2yZgP1NLfs
3. https://www.youtube.com/watch?v=I0gY6k6Oc1I
4. https://www.youtube.com/watch?v=GUvoVvXwoOQ&list=PLUl4u3cNGP62esZEwffjMAsEMW_YArxYC

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	1	2	-	-	-	-	-	-	2	-	-	-
CO2	3	2	2	1	2	-	-	-	-	-	-	2	-	-	-
CO3	3	2	2	1	2	-	-	-	-	-	-	2	-	-	-
CO4	3	2	2	1	2	-	-	-	-	-	-	2	-	-	-
CO5	3	2	2	1	2	-	-	-	-	-	-	2	-	-	-
Avg	3	2	2	1	2	-	-	-	-	-	-	2	-	-	-

<table>
<tr><td>R 2024</td><td colspan="5">TAMIL</td><td>SEMESTER: II</td></tr>
<tr><td rowspan="2">24HS201</td><td rowspan="2">Tamils and Technology</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">HS</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td colspan="7">COMMON TO ALL BRANCHES</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are to:

- Learn weaving, ceramic and construction technology of Tamil.
- Understand the agriculture, irrigation and manufacturing technology of tamil.
- Realize the development of scientific Tamil and computing.

COURSE OUTCOMES:

At the end of this course, students can able to :

CO1: Understand the weaving and ceramic technology of ancient Tamil people nature.

CO2: Understand the construction technology, building materials in sangam period and case studies.

CO3: Infer the metal process, coin and beads manufacturing with relevant archaeological evidence.

CO4: Realize the agriculture methods, irrigation technology and pearl diving.

CO5: Apply the knowledge of scientific Tamil and Tamil computing.

UNIT: I	**WEAVING AND CERAMIC TECHNOLOGY**	**3**

Weaving Industry during Sangam Age – Ceramic technology – Black and Red Ware Potteries (BRW) – Graffiti on Potteries.

UNIT: II	**DESIGN AND CONSTRUCTION TECHNOLOGY**	**3**

Designing and Structural construction House & Designs in household materials during Sangam Age - Building materials and Hero stones of Sangam age – Details of Stage Constructions in Silapathikaram - Sculptures and Temples of Mamallapuram - Great Temples of Cholas and other worship places - Temples of Nayaka Period - Type study Madurai Meenakshi Temple)- ThirumalaiNayakarMahal - Chettinad Houses, Indo - Saracenic architecture at Madras during British Period

UNIT: III	**MANUFACTURING TECHNOLOGY**	**3**

Art of Ship Building - Metallurgical studies - Iron industry - Iron smelting,steel -Copper and gold- Coins as source of history - Minting of Coins – Beads making-industries Stone beads - Glass beads - Terracotta beads -Shell beads/ bone beats - Archeological evidences - Gemstone types described in SilapathikaramTherukoothu, Karakattam, VilluPattu, KaniyanKoothu, Oyilattam, Leather Puppetry, Silambattam, Valari, Tiger dance - Sports and Games of Tamils.

UNIT: IV	**AGRICULTURE AND IRRIGATION TECHNOLOGY**	**3**

Flora and Fauna of Tamils &Agam and Puram Concept from Tholkappiyam and Sangam Literature - Aram Concept of Tamils - Education and Literacy during Sangam Age - Ancient Cities and Ports of Sangam Age - Export and Import during Sangam Age - Overseas Conquest of Cholas

UNIT: V	**SCIENTIFIC TAMIL & TAMIL COMPUTING**	**3**

Development of Scientific Tamil - Tamil computing – Digitalization of Tamil Books – Development of Tamil Software – Tamil Virtual Academy – Tamil Digital Library – Online Tamil Dictionaries – Sekai Project.

Total Periods :15

Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, You Tube Videos, NPTEL videos.

TEXT CUM REFERENCE BOOKS:

Sl.No	Authors	Title of the Book	Publisher	Year of publication
1	Dr.K.K.Pillay	Tamilnadu history people and culture	Tamilnadu Textbook and Education works Corporation	2019
2	EL Sundaram	Computer Tamil	Vikatanprasuram	2016

3	Dr.S.Singaravelu	Social Life of the Tamils - The Classical Period	International Institute of Tamil Studies.	2001
4	Dr.S.V.Subatamanian, Dr.K.D. Thirunavukkarasu	Historical Heritage of the Tamils	International Institute of Tamil Studies	2010
5	Dr.M.Valarmathi	The Contributions of the Tamils to Indian Culture	International Institute of Tamil Studies.	2001
6	Dr. R. Sivanantham	Keeladi - 'Sangam City Civilization on the banks of river Vaigai'	Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu	2019
7	Dr.K.K.Pillay	Studies in the History of India with Special Reference to Tamil Nadu	The Author	1979
8		Porunai Civilization	Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu	2019
9	R.Balakrishnan	Journey of Civilization Indus to Vaigai	RMRL	2019
10	Dr.K.K.Pillay	Social Life of Tamils	A joint publication of TNTB & ESC and RMRL	1975

WEB LEARNING RESOURCES:

1	https://youtu.be/jteRvnNiD6w?si=HmAS7a_gng6hYcL_
2	https://youtu.be/WZwdo20QgP8?si=2oTevNPCiGzTPi0-
3	https://youtu.be/05e3v0xGA9k?si=SHa2vsQG39RpDPtZ
4	https://youtu.be/bxYdHw4rvec?si=Eryg0PF72BPhbRBH
5	https://youtu.be/MRfbeJvJZ0k?si=YpAYFFEpLdV8FlrX
6	https://youtu.be/BS_BSDZp6HA?si=D_QdZn1Zr6X3C95p

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
CO2	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
CO3	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
CO4	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
CO5	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
Avg	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-

R 2024	DEPARTMENT OF SCIENCE AND HUMANITIES					SEMESTER: 2
24HS201	தமிழரும் தொழில்நுட்பமும்	L	T	P	C	SH
		3	0	0	3	

COURSE OBJECTIVES:

The main objectives of this course are to:

- தமிழர்களின் மற்றும் செராமிக் மற்றும் கட்டுமான தொழில்நுட்பங்களை அறிதல்
- தமிழர்களின் வேளாண்மை பாசன மற்றும் உற்பத்தி தொழில்நுட்பத்தை புரிந்து கொள்ளுதல்.
- அறிவியல் தமிழ் மற்றும் கணினி தமிழில் வளர்ச்சியைஉணர்ந்து கொள்ளுதல்

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: சங்க கால தமிழ் மக்களின் நெசவு மற்றும் செராமிக் தொழில்நுட்பத்தை குறித்தும் அதன் இயல்பை குறித்தும் புரிந்து கொள்வார்கள்

CO2: கட்டுமான தொழில்நுட்பம் கட்டிட பொருட்கள் மற்றும் கட்டுமான பணிகளை புரிந்து கொள்வார்கள்

CO3: உலோகத்தை உருக்குதல், நாணய உருவாக்கம், மாணிக்கம் , பளிங்கு, பொன் உற்பத்தி போன்ற பொருட்களின் ஆதாரங்களை விவரிக்கலாம்.

CO4: வேளாண்மை முறைகள் பாசன தொழில்நுட்பம் மற்றும் முத்து குளித்தலாகியவற்றைப் பற்றிய அறிவைப் பெறுவார்கள்.

CO5: அறிவியல் தமிழ் மற்றும் கணினி தமிழில் வளர்ச்சியை பயன்படுத்த கூடியகூடியவராக மாறுவார்கள்.

UNIT: I	நெசவு மற்றும் பானைத் தொழில் நுட்பம்	3
சங்ககாலத்தில் நெசவுத்தொழில்- பானைத் தொழில் நுட்பம் -கருப்பு சிவப்பு பாண்டங்கள் - பண்டங்களில் கீறல் குறியீடுகள்		
Pedagogical Tools	கரும்பலகை ,வினாடி- வினா, பிபிடி ,யூடியூப் வீடியோஸ்,, கலந்தாலோசித்தல்.	
UNIT: II	வடிவமைப்பு மற்றும் கட்டிடத் தொழில் நுட்பம்	3
சங்ககாலத்தில் வடிவமைப்பு மற்றும் கட்டுமானங்கள்- சங்ககாலத்தில் வீட்டு பொருள்களில் வடிவமைப்பு -சங்ககாலத்தில் கட்டுமான பொருள்களும் நடுகல்லும்- சிலப்பதிகாரத்தில் மேடை அமைப்பு பற்றிய விவரங்கள்- மாமல்லபுர சிற்பங்களும் கோவில்களும்- சோழர் காலத்து பெருங்கோயில்கள் மற்றும் பிற வழிபாட்டுத்தலங்கள்- நாயக்கர் கால கோயில்கள் மாதிரி கட்டமைப்புகள் பற்றி அறிதல்- மதுரை மீனாட்சி அம்மன் ஆலயம் மற்றும் திருமலை நாயக்கர் மஹால் -செட்டிநாட்டு வீடுகள் -பிரிட்டிஷ் காலத்தில் சென்னையில் இந்தோ சரோசெனிக் கட்டிடக்கலை		
Pedagogical Tools	கரும்பலகை ,வினாடி- வினா, பிபிடி ,யூடியூப் வீடியோஸ்,, கலந்தாலோசித்தல்	
UNIT: III	உற்பத்தி தொழில்நுட்பம்	3
கப்பல் கட்டும் கலை –உலோகவியல்- இரும்பு தொழிற்சாலை -இரும்பை உருக்குதல், எஃகு - வரலாற்றுச் சான்றுகளாக செம்பு மற்றும் தங்க நாணயங்கள்- நாணயங்கள் அச்சடித்தல்- மணி உருவாக்கும் தொழிற்சாலைகள் -கல்மணிகள் .கண்ணாடி மணிகள் -சுடுமண் மணிகள் - சங்கு மணிகள் -எலும்பு துண்டுகள் -தொல்லியல் சான்றுகள் சிலப்பதிகாரத்தில் மணிகளின் வகைகள்		
Pedagogical Tools	கரும்பலகை ,வினாடி- வினா, பிபிடி ,யூடியூப் வீடியோஸ்,, கலந்தாலோசித்தல் Black.	
UNIT: IV	வேளாண்மை மற்றும் நீர்ப்பாசன தொழில்நுட்பம்	3

அணை, ஏரி, குளங்கள் ,மதகு, சோழர் கால குமிழித்தூம்பின் முக்கியத்துவம்- கால்நடை பராமரிப்பு- கால்நடைகளுக்காக வடிவமைக்கப்பட்ட கிணறுகள்- வேளாண்மை மற்றும் வேளாண்மை சார்ந்த செயல்பாடுகள்- கடல்சார் அறிவு மீன்வளம்- முத்து மற்றும் முத்து குளித்தல் -பெருங்கடல் குறித்த பண்டைய அறிவுசார் சமூகம்.		
Pedagogical Tools	கரும்பலகை ,வினாடி- வினா, பிபிடி ,யூடியூப் வீடியோஸ்,, கலந்தாலோசித்தல்	
UNIT: V	அறிவியல் தமிழ் மற்றும் கணித்தமிழ்	**3**
அறிவியல் தமிழின் வளர்ச்சி -கணித்தமிழ் வளர்ச்சி -தமிழ் நூல்களை மின்பதிப்பு செய்தல் -தமிழ் மென்பொருள்கள் உருவாக்கம் -தமிழ் இணையக் கல்விக் கழகம் -தமிழ் மின் நூலகம் இணையத்தில் தமிழ் அகராதிகள் சொற்குவை திட்டம்.		
Pedagogical Tools	கரும்பலகை ,வினாடி- வினா, பிபிடி ,யூடியூப் வீடியோஸ்,, கலந்தாலோசித்தல்	
		Total Periods : 15

<table>
<tr><td>R 2024</td><td colspan="5">COMPUTER SCIENCE AND ENGINEERING</td><td>SEMESTER II</td></tr>
<tr><td rowspan="2">24ES201</td><td rowspan="2">DESIGN THINKING</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">ES</td></tr>
<tr><td>2</td><td>0</td><td>0</td><td>2</td></tr>
<tr><td colspan="7">Common to All Departments</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are to:

- Provide the new ways of creative thinking.
- Learn the innovation cycle of Design Thinking process for developing innovative products.
- Learn which useful for a student in preparing for an engineering career and to apply them for the prospective.
- Learn various types of design thinking models.
- Learn how to apply the Design Thinking Principles, through real world case studies.

COURSE OUTCOMES:

At the end of this course, students able to

CO1 Understand the Concept of Design Thinking through its principles.
CO2 Learn the tools and techniques of Design Thinking and to apply them in real life cases.
CO3 Understand the different stages of Structured Models used in Design thinking.
CO4 Apply the perspectives of design thinking in the entrepreneurial activities.
CO5 Learn from the real world case studies about how to apply the concept of design thinking in product development.

UNIT: I	**OVERVIEW OF DESIGN THINKING**	**6**

Introduction to Design Thinking – Conceptual Understanding, Evolution of Design Thinking, Attributes, Principles (Human Rule, Ambiguity Rule, Re-Design Rule and Tangibility Rule) – Cycle of Design Thinking – Resources (3Ps) – Applications.

UNIT: II	**TOOLS AND TECHNIQUES FOR DESIGN THINKING**	**6**

Personas, Visualization, Stakeholder Mapping, Journey Mapping, Mind Mapping, Star Bursting, Divergent Thinking, Convergent Thinking, Ethnography, Brainstorming, Story Telling, Role Playing, User Interviews. (All these techniques shall be taught only to level of understanding the core concept)

UNIT: III	**DESIGN THINKING MODELS**	**6**

Double Diamond Model – Phases of Discover, Define, Develop and Deliver – Feedback Mechanism.
Stanford 5 Phase Model – Empathize, Define, Ideate, Prototype and Test.

UNIT: IV	**DESIGN THINKING FOR ENTREPRENEURS**	**6**

Idea of Growth Design, Comparison of Growth Design and Product Design, Growth Process Model : What is? - What if? - What Wows? - What Works, Principle of Optimism. Ethics in Design Thinking : 5 Approaches – Utilitarian, Rights, Fairness, Common Good and Virtue - Ethical Issues – Ethical Design Test.

UNIT: V	**CASE STUDIES**	**6**

1. Why Patients were not visiting a healthcare center for a free health checkup - Karnataka Health Promotion Trust
2. Why Sales Officers were not accessing help even though it was available and were still abandoning sales when a difficult objection was raised in a sales call -Shriram Life Insurance Corporation.
3. My City Savior APP - Government of Odisha.
4. Designing of a Banking APP – Kotak Mahindra Bank

Total Periods: 30

Pedagogical Tools	Chalk & Board, PPT, Brainstorming, TEDx like public Speech

TEXT BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of publication**
1	E Bala Guruswamy, Bindu Vijayakumar	Design Thinking – A Business Perspective	McGraw Hill Education (India) Private Limited.	2024
REFERENCE BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of publication**
1	David Lee	Design Thinking In the Class Room	Ulysses Press	2018
WEB LEARNING RESOURCES:				
1.https://youtube/6-5J6YTrYf4?si=WE9MLo-2tbccTWNG				
2.https://youtube/4nTh3AP6knM?si=rdEHE4yGxSJ4zDji				
3.https://youtube/j6Ro7TPzRoo?si=wa75cakOWyR0dSZC				
4.https://youtube/DmLVfQfxtPU?si=q6NyR6yCmir3Y2ia				
5.https://youtube/OE2ooXUEAwc?si=A3yYLYTOKvuYx_Cn				

CO – PO - PSO MAPPING

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	2	-	-	2	-	-	-	-	-	-	2	-	-	-
CO2	3	2	-	-	2	-	-	-	-	-	-	2	-	-	-
CO3	3	2	-	-	2	-	-	-	-	-	-	2	-	-	-
CO4	3	2	-	-	2	-	-	-	-	-	-	2	-	-	-
CO5	3	2	-	-	2	-	-	-	-	-	-	2	-	-	-
AVG	3	2	-	-	2	-	-	-	-	-	-	2	-	-	-

R 2024	Chemistry					SEMESTER: II
24BS203	CHEMISTRY FOR ENGINEERS	L	T	P	C	BS
		3	0	2	4	

COMMON TO: BME, EEE, AERONAUTICAL, MECHANICAL and MECHATRONICS ENGINEERING

COURSE OBJECTIVES:

The objectives of learning this course are to:

- Inculcate sound understanding of water quality parameters and water treatment techniques.
- Introduce the basic concepts and applications of phase rule and alloys.
- Facilitate the understanding of different types of fuels, their preparation, properties and combustion characteristics.
- Familiarize the students with the different energy sources, operating principles, working processes and applications of energy conversion and storage devices.
- Impart knowledge on the basic principles and preparatory methods of nanomaterials.

COURSE OUTCOMES:

At the end of this course, students are able to :

CO1: Understand the quality of water from quality parameter data, analyze and propose the suitable treatment methodologies to treat water.

CO2: Recognize different forms of energy resources and apply them for suitable applications in energy sectors.

CO3: Apply the knowledge of phase rule and alloys for material selection requirements.

CO4: Analyze and recommend suitable fuels for engineering processes and applications.

CO5: Apply basic concepts of nanoscience and nanotechnology in designing the synthesis of nanomaterials

UNIT: I	WATER TECHNOLOGY	9

Water: Sources, impurities and water quality parameters, Hardness of water – types – expression of hardness – units, Boiler troubles: Scale and sludge, Priming &foaming. Need for water treatment, Treatment of boiler feed water: Internal treatment (phosphate, colloidal, sodium aluminate and calgon conditioning) and External treatment (Ion exchange or demineralization and zeolite process), Municipal water treatment: primary treatment and disinfection (UV, Ozonation, break-point chlorination). Desalination of brackish water: Reverse Osmosis.

UNIT: II	ENERGY SOURCES AND STORAGE DEVICES	9

Nuclear energy: light water nuclear power plant, breeder reactor. Solar energy conversion: Principle, working and applications of solar cells; Recent developments in solar cell materials. Wind energy; Basic Electrochemical Terminologies, Batteries: Types of batteries, Primary battery (dry cell), Secondary battery (lead acid battery and lithium-ion-battery); Electric vehicles— working principles; Fuel cells: H_2-O_2 fuel cell, Bio Fuel,microbial fuel cell; Super capacitors: Storage principle, types and examples.

UNIT: III	PHASE RULE AND ALLOYS	9

Phase rule: Introduction, definition of terms with examples. One component system - water system, sulphur system; Reduced phase rule; Construction of a simple eutectic phase diagram – Two component system: lead-silver system-Pattinson's process,$FeCl_2$-H_2O system.

Alloys: Introduction- Definition- properties of alloys- significance of alloying, Alloys-Nichrome and stainless steel (18/8) – heat treatment of steel. Introduction to composites – definition-types-uses.

UNIT: IV	FUELS AND COMBUSTION	9

Fuels: Introduction: Classification of fuels; Coal and coke: Analysis of coal (proximate and ultimate),Carbonization, Manufacture of metallurgical coke (Otto Hoffmann method). Petroleum and Diesel: Manufacture of synthetic petrol (Bergius process), Property - Knocking, Power alcohol and biodiesel (trans-esterification).Combustion of fuels: Introduction: Calorific value - higher and lower calorific values, Flue gas analysis-ORSAT Method.CO_2 emission and carbon footprint.

UNIT: V	NANO TECHNOLOGY	9

Basics: Distinction between molecules, nanomaterials and bulk materials; Size-dependent properties (optical, electrical, mechanical and magnetic); Types of nanomaterials: Definition, properties and uses of – nanoparticle, nanocluster, nanorod, nanowire and nanotube. Preparation of nanomaterials: sol-gel, laser ablation, chemical vapour deposition, Analytical techniques- SEM, TEM, Applications of nanomaterials

Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos, Nptel videos.
	Total Periods :45

Description of the Experiments: **(Any six experiments to be conducted)**	**Total Periods: 30**
1. Preparation of Na_2CO_3 as a primary standard and determination of types and amount of alkalinity in water sample	
2. Determination of total, temporary & permanent hardness of water by EDTA method.	
3. Determination of chloride content of water sample by Argentometric method.	
4. Estimation of sodium /potassium present in water using a flame photometer.	
5. Estimation of copper content of the given solution by Iodometry	
6. Determination of strength of given hydrochloric acid using pH meter.	
7. Determination of strength of acids in a mixture of acids using conductivity meter.	
8. Estimation of iron content of the given solution using potentiometer	
9. Estimation of Nickel in steel	

LIST OF EQUIPMENTS FOR A BATCH OF 30 STUDENTS

S.No.	Description of Equipment	Quantity
1.	pH meter	10 nos
2.	Conductivity meter	10 nos
3.	Potentiometer	10 nos
4.	Flame Photometer	1 no
5.	Electronic Balance (Four digit)	1 no
6.	Hot plate with temperature controller	5 nos
7	Hot Air Oven	1no
8	Magnetic stirrer	1 no

TEXT BOOKS:

Sl.No	Authors	Title of the Book	Publisher	Year of publication
1	P.C.Jain and Monica Jain	Engineering Chemistry	16thEdition,DhanpatRaiPublishingCompany (P)Ltd, New Delhi	2018
2	S.S. Dara	A Text book of Engineering Chemistry	S.Chand Publishing,12thEdition	2018
3	Vairam S, Kalyani P and Suba Ramesh	Engineering Chemistry	2nd Edition, Wiley India Pvt. Ltd., New Delhi	2014
4	J Mendham RC Denn MJK Thomas David J Barnes	Vogel's Text book of Quantitative Chemical Analysis	Pearson Education	2018

REFERENCE BOOKS:

Sl.No	Authors	Title of the Book	Publisher	Year of publication
1	B.S.Murty,P. Shankar, Baldev Raj,B. B.Rath and James Murday	Text book of nano science and nanotechnology	Universities Press-IIM Series in Metallurgy and Materials Science	2018
2	Shikha Agarwal	Engineering Chemistry-Fundamentals and Applications	Cambridge University Press, Delhi, Second Edition	2019

3	O.G. Palanna	Engineering Chemistry	McGraw Hill Education (India) Private Limited, 2nd Edition	2017
4	Prasanta Rath	Engineering Chemistry	Cengage Learning India, Pvt., Ltd., Delhi. 1st Edition	2015

WEB LEARNING RESOURCES:

1	https://nptel.ac.in/courses/105106119 (Unit 1)
2	https://nptel.ac.in/courses/103103206 (Unit 2)
3	https://www.brainkart.com>article phase rule (Unit 3)
4	https://nptel.ac.in/courses/113/104/113104008/ (Unit 4)
5	https://nptel.ac.in/courses/104103019 (Unit 5)
6	https://www.brainkart.com/subject/engineering-chemistry_264/ (All Units)
7	https://www.youtube.com/watch?v=4RDA_B_dRQ0(Reverse Osmosis)
8	https://www.youtube.com/watch?v=XUzpG1-rJLA Bergius Process)

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	1	-	1	1	-	-	-	-	2	-	-	-
CO2	3	1	2	1	-			-	-	-	-	2	-	-	-
CO3	3	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	1	1	-	-	1	2	-	-	-	-	-	-	-	-
CO5	2	1	1		-	-	-	-	-	-	-	-	-	-	-
AVG	3	1	1	1	-	1	1	-	-	-	-	2	-	-	-

<table>
<tr><td>R 2024</td><td colspan="5">ELECTRICAL AND ELECTRONICS ENGINEERING</td><td>SEMESTER : II</td></tr>
<tr><td rowspan="2">24ES 214</td><td rowspan="2">BASIC ELECTRICAL AND ELECTRONICS ENGINEERING</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">ES</td></tr>
<tr><td>3</td><td>0</td><td>2</td><td>4</td></tr>
</table>

COMMON TO MECHANICAL, MECHATRONICS AND AERONAUTICAL ENGINEERING

COURSE OBJECTIVES:

The objectives of learning this course are to

- Equip students with a foundational understanding of electric circuits and analysis.
- Impart knowledge in the basics of working principles and application of electrical machines.
- Introduce analog devices and digital electronics and their characteristics.
- Foster Proficiency in handling the measuring instruments and its working.
- Clarify the power generation, distribution and safety.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Compute the electric circuit parameters for simple problems.
CO2: Understand the working principle and applications of electrical machines.
CO3: Analyze the characteristics of analog electronic and digital electronics .
CO4: Know the basic concepts of functional elements and working of measuring instruments
CO5: Realize the concepts of power generation, distribution and safety.

UNIT: I	**ELECTRICAL CIRCUITS**	**9**

DC Circuits: Circuit Components: Conductor, Resistor, Inductor, Capacitor – Ohm's Law - Kirchhoff's Laws ,series and parallel combinations of resistance–voltage and current division .Independent and Dependent Sources – Simple problems- Nodal Analysis, Mesh analysis with Independent sources only (Steady state) Steady state analysis for sinusoidal excitation:(simple Problems only) Introduction –sinusoidal function-sinusoidal steady state analysis – power in sinusoidal steady state – nodal and mesh analysis (Simple problems only)

UNIT: II	**ELECTRICAL MACHINES**	**9**

Construction and Working principle of DC motors - EMF equation, Types and Applications. Torque Equation, Types and Applications. Construction, Working principle and Applications of Transformer, Synchronous motor and Three Phase Induction Motor. (Qualitative treatment only)

UNIT: III	**ANALOG & DIGITAL ELECTRONICS**	**9**

Semiconductor Materials: Silicon &Germanium – PN Junction Diodes, Zener Diode –Characteristics Applications – Bipolar Junction Transistor-Biasing, SCR,– Types, I-V Characteristics and Applications, Rectifier (half wave& Full wave centre tapped, Bridge type) (Qualitative treatment only) Number systems, Hexa Decimal system and Octal number system ,Logic Gates AND Universal Gates

UNIT: IV	**MEASUREMENTS AND INSTRUMENTATION**	**9**

Electrical and electronics instruments, classification of instruments, Types of indicating instruments, Electro dynamometer instruments, Energy meter, instruments Transformer CT and PT.

Unit V	**GENERATION TRANSMISSION DISTRIBUTION OF POWER SYSTEM**	**9**

Power system structure -Generation , Transmission and distribution , Various voltage levels, Earthing – Methods of earthing, protective devices- switch fuse unit- Miniature circuit breaker moulded case circuit breaker- earth leakage circuit breaker, safety precautions and First Aid (Qualitative treatment only)

TOTAL PERIODS :45

Pedagogical Tools	Black board, chalk, Group Discussion, Youtube Videos, Nptel videos.

LIST OF EXPERIMENTS:	**TOTAL PERIODS :30**

1.	Verification of ohms and Kirchhoff's Laws.
2	Load test on DC Shunt Motor.

3	Load test on Single phase Transformer.
4	Load Test on three phase Induction Motor.
5	Speed control of DC shunt motor.
6	Load test on Single phase induction motor.

LIST OF EQUIPMENTS:

1	Multimeter	3 Nos
2	Ammeter, Voltmeters, Wattmeter	5 Nos
3	Digital Tachometer	3 Nos
4	Regulated Power Supply	2 Nos
5	Breadboard	4 Nos
6	Resistors, Capacitors, Inductors	3 Nos
7	DC Shunt Motor	2 Nos
8	Single Phase Transformer	2 Nos
9	Single Phase Induction Motor	1 Nos
10	Three-Phase Induction Motor	1 Nos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Kothari DP and I.J Nagrath	Basic Electrical and Electronics Engineering	Second edition McGraw Hill Education,	2020
2	S.K. Bhattacharya	Basic Electrical and Electronics Engineering	Pearson Education, Second Edition,	2017.
3	C.L.Wadhwa	"Generation, Distribution and Utilisation of Electrical Energy",	New Age International pvt.ltd.	2015

REFERENCE BOOKS:

1	Thomas L. Floyd,	Digital Fundamentals11th Edition	Pearson Education	2017
2	H.S. Kalsi	Electronic Instrumentation	Tata McGraw-Hill, New Delhi	2010
3	Albert Malvino, David Bates,	Electronic Principles, 7th edition	McGraw Hill Education	2017.

WEB LEARNING RESOURCES :

1. https://ocw.mit.edu/courses/6-002-circuits-and-electronics-spring-2007/
2. https://www.khanacademy.org/science/electrical-engineering
2. https://www.coursera.org/browse/physical-science-and-engineering/electrical-engineering
3. https://ethw.org/Category:Engineering_fundamentals?gad_source=1&gclid=Cj0KCQjwtZK1BhDuARIsAAy2VzthS61SRxHUpVR7d7ruGv1_Gz_SlwmTjGusTibTehRbl2z1ZPRCilsaAoM2EALw_wcB

4. https://ethw.org/Category:Engineering_fundamentals?gad_source=1&gclid=Cj0KCQjwtZK1BhDuARIsAAy2Vzv97YRwkx1whxouLZli7v5gQF6lwqAdMDvpdfn_8pbiWvycM68h80MaArogEALw_wcB
5. https://www.wolframalpha.com/examples/science-and-technology/engineering/electrical-engineering/electric-circuits/

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	2	1	-	-	-	-	1	-	-	-	2	-	-	1
CO2	3	2	1	-	-	-	-	1	-	-	-	2	-	-	1
CO3	3	2	1	-	-	-	-	1	-	-	-	2	-	-	1
CO4	3	2	1	-	-	-	-	1	-	-	-	2	-	-	1
CO5	3	2	1	-	-	-	-	1	-	-	-	2	-	-	1
Avg	3	1.8	1	-	-	-	-	1	-	-	-	2	-	-	1

<table>
<tr><td>R 2024</td><td colspan="5">MECHANICAL ENGINEERING</td><td>SEMESTER: II</td></tr>
<tr><td rowspan="2">24ES211</td><td rowspan="2">FOUNDATION SKILLS</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">PC</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>1</td></tr>
<tr><td colspan="7">COMMON TO: AERONAUTICAL ENGINEERING, MECHANICAL ENGINEERING and MECHATRONICS ENGINEERING</td></tr>
</table>

COURSE OBJECTIVES:

The main objectives of this course are to:

- Practice few basic engineering operations in welding, and sheet metal works.
- Make the specified skills in fitting operations.
- Perform few basic operations to produce wooden joints
- Make pipe connections for household applications.

COURSE OUTCOMES:

Upon completion of this course, the students will be able to:
CO1-Draw pipe line plan; lay and connect various pipe fittings used in common household plumbing work
CO2Saw; plan; make joints in wood materials used in common household wood work.
CO3-Weld various joints in steel plates using arc welding work;
CO4-Make a tray out of metal sheet using sheet metal work.
CO5-Prepare metal joints using fitting tools

PRACTICAL EXERCISES:

1. Plumbing Works: Basic pipe connections – Mixed pipe material connection – Pipe connections with different joining components for pumping water from sump to overhead tank and pipe connections from overhead tank to bath shower and wash basin.
2. Carpentry using modern tools only: T – Joint , Mortise and Tenon Joint Dove Tail Joint.
3. Carpentry using modern tools only: Mortise and Tenon Joint Dove Tail Joint.
4. Welding: Preparation of butt joints, lap joints and T- joints by Arc welding
5. Welding: Preparation of butt joints, lap joints and T- joints by Gas welding
6. Sheet Metal Work: Model making – Trays
7. Sheet Metal Work: Model making – Funnel
8. Fitting: Preparation of Square fitting and V – fitting models.
9. Machining – Plain turning and Facing
10. Machining – Step turning

Total Periods: 30

List of Equipments:

Sl. No	Description of Equipment	Required numbers (for batch of 30 students)
1.	Assorted components for plumbing consisting of metallic pipes, plastic pipes, flexible pipes, couplings, unions, elbows, plugs and other fittings.	5 Sets
2.	Carpentry vice (fitted to work bench)	5 Nos
3.	Standard wood working tools	5 Nos
4.	Models of industrial trusses, door joints, furniture joints	1 Each
5.	Power Tools:	
	(a) Rotary Hammer	1 Nos
	(b) Circular Saw	1 Nos
	(c) Planer	1 Nos

	(d) Hand Drilling Machine	1 Nos
	(e) Jigsaw	1 Nos
6.	Arc welding transformer with cables and holders	3 Nos
7.	Welding accessories like welding shield, chipping hammer, wire brush, etc.	3 Sets
8.	Oxygen and acetylene gas cylinders, blow pipe and other welding outfit	1 Nos
9.	Standard fitting tools	3 Sets
10.	Centre lathe	1 Nos

CO PO PSO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1
CO2	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1
CO3	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1
CO4	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1
CO5	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1
AVG	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1

<table>
<tr><td>R 2024</td><td colspan="4">ELECTRICA AND ELECTRONICS ENGINEERING</td><td></td><td>SEMESTER: II</td></tr>
<tr><td rowspan="2">24ES212</td><td rowspan="2">BASIC ENGINEERING SKILLS</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">ES</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>1</td></tr>
</table>

COMMON TO : AERONAUTICAL, MECHANICAL and MECHATRONICS ENGINEERING Departments

COURSE OBJECTIVES:

The main objectives of this course are to:

- Study the various basic domestic wiring circuits and measure the electrical parameters.
- Impart the Knowledge about the stair case wiring, wiring layout and its connections
- Impart the knowledge of various basic electronic components .
- Know about Solder and test simple electronic circuits; Assemble and test simple electronic components on PCB.
- Study about the operation of various Boolean operations in electronics.

LCOURSE OUTCOMES:

At the end of this course, students are able to:

CO1: Wire various electrical joints in common household electrical wire work.
CO2: Understand the stair case wiring, wiring layout and its connections
CO3: Measure the electrical quantities using ammeter, voltmeter, wattmeter and energy meter
CO4: Study the construction, working principle and wiring of single phase energy meter.
CO5: Solder and test simple electronic circuits; Assemble and test simple electronic components on PCB.

LIST OF EXPERIMENTS:

I ELECTRICAL ENGINEERING PRACTICE

1. Residential house wiring using switches, fuse, indicator, lamp and energy meter.
2. Fitting and Installation of household appliances- LED TV,Fan
3. Stair case wiring.
4. Measurement of electrical quantities – voltage, current, power & power factor in RLC circuit
5. Measurement of energy using single phase energy meter.

II ELECTRONIC ENGINEERING PRACTICE

1. Study of Electronic components and equipments – Resistor, colour coding, Measurement of AC signal parameter (peak-peak, rms period, frequency) using CRO.
2. Verification of logic gates AND, OR, EX-OR and NOT.
3. Generation of Clock Signal.
4. Soldering simple electronic circuits and checking continuity.
5. Assembling and testing electronic components on a small PCB.

Total Periods :30

LIST OF EQUIPMENTS:

Sl. No	Description of Equipment	Required numbers (for batch of 30 students)
1.	Assorted electrical components for house wiring	10 sets
2.	Ammeter, voltmeter, wattmeter. Energy meter.	Each 3 Nos
3.	LED TV, Fan	Each 2 Nos
4.	Clock timer.	2 nos
5.	Soldering guns and PCBs	10 Nos.
6.	Assorted electronic components for making circuits	20 Nos.
7.	Inductor, Capacitor, (various Values)	Each 5 Nos.
8.	Multimeter	5 Nos.
9.	Functional Generator, CRO	Each 3nos
10.	Logic gates ICs IC 7408, IC7432, IC 7404 and IC 7486.	Each 5 Nos

CO PO PSO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1
CO2	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1
CO3	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1
CO4	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1
CO5	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1
AVG	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1

M.A.M SCHOOL OF ENGINEERING

(An Autonomous Institution)

(Accredited by NAAC || Approved by AICTE || Affiliated to Anna University)

Trichy – Chennai Trunk Road, Siruganur, Tiruchirappalli – 621 105

B.E. AERONAUTICAL ENGINEERING

SEMESTER III

THEORY COURSES										
S.No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1.	24BS301	Statistics & Numerical Methods	3	1	0	4	40	60	100	BS
2.	24AE301	Elements of Aeronautical Engineering	3	0	0	3	40	60	100	PC
3.	24AE302	Aircraft Systems & Instrumentation	3	0	0	3	40	60	100	PC
4.	24AE303	Manufacturing Technology	3	0	0	3	40	60	100	PC
THEORY COURSES WITH LABORATORY COMPONENT										
5.	24AE304	Fluid Mechanics and Strength of Materials	3	0	2	4	50	50	100	PC
6.	24AE305	Aero Thermodynamics	3	0	2	4	50	50	100	PC
LABORATORY COURSES										
7.	24AE306	Manufacturing Processes Lab	0	0	3	2	60	40	100	PC
8.	24TPS02	Aptitude Skills and Communication skills II	0	0	2	1	60	40	100	EEC
		TOTAL	13	1	9	24				

REGULATIONS 2024

Recommended by 2nd BOS held on 10.05.2025 and Approved in 2nd Academic Council held on 12.06.2025

R 2024	MATHEMATICS					SEMESTER: III
24BS301	STATISTICS AND NUMERICAL METHODS	L	T	P	C	BS
		3	1	0	4	
COMMON TO ALL DEPARTMENTS						

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To provide the necessary basic concepts of a few statistical and numerical methods and give procedures for solving numerically different kinds of problems occurring in engineering and technology.
2. To acquaint the knowledge of testing of hypothesis for small and large samples which plays an important role in real life problems.
3. To introduce the basic concepts of solving algebraic and transcendental equations.
4. To introduce the numerical techniques of interpolation in various intervals and numerical techniques of differentiation and integration which plays an important role in engineering and technology disciplines.
5. To acquaint the knowledge of various techniques and methods of solving ordinary differential equations.

COURSE OUTCOMES:

At the end of this course, students can be able to

CO1: Apply the concept of testing of hypothesis for small and large samples in real life problems.

CO2: Apply the basic concepts of classifications of design of experiments in the field of agriculture.

CO3: Apply knowledge about solving algebraic and transcendental by using numerical techniques.

CO4: Appreciate the numerical techniques of interpolation in various intervals and apply the numerical techniques of differentiation and integration for engineering problems.

CO5: Solve the ordinary differential equations with initial conditions by using certain techniques with engineering applications

UNIT: I	TESTING OF HYPOTHESIS	9+3

Sampling distributions-Tests for single mean and difference of means (small samples)–Tests for single variance and equality of variances– Chi-square test for goodness of fit Independence of attributes

UNIT: II	DESIGN OF EXPERIMENTS	9+3

One way and two-way classifications-Completely randomized design–Randomized block design- Latin square design.

UNIT: III	SOLUTION OF EQUATIONS AND EIGEN VALUE PROBLEMS	9+3

Solution of algebraic and transcendental equations-Fixed point iteration method–Newton Raphson Method - Solution of linear system of equations-Gauss elimination method–Gauss Jordan method – Iterative Methods of Gauss Jacobi and Gauss Seidel - Eigenvalues of a matrix by Power method

UNIT: IV	INTERPOLATION, NUMERICAL DIFFERENTIATION AND NUMERICAL INTEGRATION	9+3

Lagrange's and Newton's divided difference interpolations– Newton's forward and backward difference interpolation – Approximation of derivates using interpolation polynomials – Numerical single and double integrations using Trapezoidal and Simpson's 1/3 rules.

UNIT: V	NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS	9+3

Single step methods: Taylor's series method – Euler's method - Modified Euler's method – Fourth order Runge-Kutta method for solving first order differential equations - Multi step methods: Milne's and Adams-Bash forth predictor corrector methods for solving first order differential equations.

	Total Periods: 60
Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Grewal,B.S.,and Grewal,J.S.,	Numerical Methods in Engineering and Science	Khanna Publishers	2015
2	Johnson,R.A., Miller,I andFreundJ	Millerand Freund's Probability and Statistics for Engineers	Pearson Education	2015

REFERENCE BOOKS:

1	Burden,R. LandFaires,J.D	Numerical Analysis	Cengage Learning	2016
2	Gupta S.C. and Kapoor V.K	Fundamentals of Mathematical Statistics	Sultan Chand & Sons	2020
3	Gerald.C.F. and Wheatley.P.O	Applied Numerical Analysis	Pearson Education	2007

WEB LEARNING RESOURCES:

1. https://nptel.ac.in/courses/111106415
2. https://nptel.ac.in/courses/111101165
3. https://youtu.be/PwuZ3nir7d4?si=33sS2AuQX5p6tJma
4. https://www.youtube.com/watch?v=9OFfJlZLh0w&list=PLuwKjRfi2s1WuOA6zbPuNQXp_psFsdkxM
5. https://www.youtube.com/watch?v=Sv7ZDb6r3gA&list=PLuwKjRfi2s1UgA-RamRM5SVxkplhzB1do

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	3	1	1	1	-	-	-	-	-	-	3	-	-	-
CO2	3	3	1	1	1	-	-	-	-	-	-	3	-	-	-
CO3	3	3	1	1	1	-	-	-	-	-	-	3	-	-	-
CO4	3	3	1	1	1	-	-	-	-	-	-	3	-	-	-
CO5	3	3	1	1	1	-	-	-	-	-	-	3	-	-	-
Avg	3	3	1	1	1	-	-	-	-	-	-	3	-	-	-

<table>
<tr><td>R 2024</td><td colspan="5">AERONAUTICAL ENGINEERING</td><td>SEMESTER: III</td></tr>
<tr><td rowspan="2">24AE301</td><td rowspan="2">ELEMENTS OF AERONAUTICAL ENGINEERING</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">PC</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td colspan="7">COURSE OBJECTIVES:</td></tr>
<tr><td colspan="7">The objectives of learning this course are:
1. To acquire the knowledge on the Historical evaluation of Airplanes
2. To know the concepts of basic properties and principles behind the flight
3. To learn the various types of power plants used in aircrafts
4. To learn the basics of different structures & construction
5. To learn the different component systems and functions</td></tr>
<tr><td colspan="7">COURSE OUTCOMES:
At the end of this course, students can be able to
CO1: Illustrate the history of aircraft & developments over the years
CO2: Explain the basic concepts of flight & Physical properties of Atmosphere
CO3: Identify the types of components & distinguish the types of Engines
CO4: Describe the types of fuselage and constructions.
CO5: Understand the working of aircraft systems</td></tr>
<tr><td>UNIT: I</td><td colspan="5">EVOLUTION OF FLIGHT</td><td>9</td></tr>
<tr><td colspan="7">History and Evolution of Aviation - Basics of Aerospace Engineering - Classification of Aircraft and Spacecraft - Lighter Than Air & Heavier Than Air - Balloon Flight - Early Airplanes by Wright Brothers – Historic Progress in aerodynamics, materials, structures and propulsion over the years</td></tr>
<tr><td>UNIT: II</td><td colspan="5">BASICS OF AERODYNAMICS</td><td>9</td></tr>
<tr><td colspan="7">Atmosphere and Its Properties - Layers of the Atmosphere - Newton’s Law of Motions applied to Aeronautics - Temperature, pressure and altitude relationships - Forces acting on an aircraft – Airfoil Nomenclature – Mach Number and its regimes</td></tr>
<tr><td>UNIT: III</td><td colspan="5">AIRCRAFT PROPULSION SYSTEMS</td><td>9</td></tr>
<tr><td colspan="7">Different types of flight vehicles - Components of an airplane and their functions - Types of Propulsion Systems: Piston Engines, Turboprops, Turbojets, Turbofans, and Rockets - Thrust Equation and Specific Fuel Consumption - Introduction to Rocket Propulsion and Space Propulsion Systems</td></tr>
<tr><td>UNIT: IV</td><td colspan="5">BASICS OF AIRCRAFT STRUCTURES</td><td>9</td></tr>
<tr><td colspan="7">General types of construction - Monocoque, Semi-Monocoque and Geodesic Constructions - Typical Wing and Fuselage structure - Materials Used in Aerospace: Aluminum Alloys, Composites, Titanium, etc. - Stresses and Strains - Hooke’s law - Stress-Strain Diagrams- Elastic Constants - Factor of Safety.</td></tr>
<tr><td>UNIT: V</td><td colspan="5">OVERVIEW OF AIRCRAFT SYSTEMS</td><td>9</td></tr>
<tr><td colspan="7">Conventional Control - Powered Control - Basic instruments for Flying -Typical systems for control actuation - Aircraft Maneuvering: Turns, Takeoff, and Landing Performance - Aircraft Electrical & Hydraulic Systems - Future Trends: UAVs, Electric Aircraft, and Hypersonic Flight</td></tr>
<tr><td colspan="7" align="right">Total Periods: 45</td></tr>
<tr><td>Pedagogical Tools</td><td colspan="6">Chalk & Board, PPT, NPTEL Videos, YouTube Videos</td></tr>
</table>

TEXT BOOKS:				
Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Anderson. J.D	Introduction to Flight	McGraw-Hill, 8th edition	2015
2	Rathakrishnan. E	Introduction to Aerospace Engineering: Basic Principles of Flight	John Wiley, NJ	2021
3	S. M. Yahya	Fundamentals of Compressible Flow: With Aircraft and Rocket Propulsion	New Age International	2010

REFERENCE BOOKS:				
1	Stephen.A. Brandt	Introduction to aeronautics: A design perspective	AIAA Education Series, 2nd Edition	2004
2	Kermode, A.C.	Flight without Formulae	Pearson Education 11th Edition	2011

WEB LEARNING RESOURCES:

1. https://www.youtube.com/watch?v=zKzCd1mbrb4&list=PLOzRYVm0a65ey1nPhnbfrz59Hvu-NVob7
2. https://ocw.mit.edu/courses/16-00-introduction-to-aerospace-engineering-and-design-spring-2003/
3. https://online-learning.tudelft.nl/courses/introduction-aeronautical-engineering/
4. https://www.youtube.com/watch?v=Oe0gWCZ1TcE&list=PLH75gcHDYGOedMSicSp9zqB14cfHnnymH
5. https://www.youtube.com/watch?v=oWiOEMaMy3c&list=PLfYIUGIhe03hCiEKvxEvbpnOVHioR_OLE

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	1	-	-	-	-	-	-	-	-	-	-	-	2	1	-
CO2	1	2	2	2	2	-	-	-	-	-	1	-	2	1	-
CO3	1	2	2	2	2	-	-	-	-	-	1	-	2	1	-
CO4	1	2	2	2	2	-	-	-	-	-	1	-	2	1	-
CO5	1	2	2	2	2	-	-	-	-	-	1	-	2	1	-
Avg	1	2	2	2	2	-	-	-	-	-	1	-	2	1	-

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: III
24AE302	AIRCRAFT SYSTEMS & INSTRUMENTATION	L	T	P	C	PC
		3	0	0	3	

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To impart knowledge of the hydraulic and pneumatic systems components
2. To familiarize the working of aircraft engines and other control systems
3. To acquire understanding of essential systems for safe aircraft operation.
4. To study the various instruments and sensors used in aircrafts

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Demonstrate the capability to design aircraft systems utilizing pneumatic and hydraulic components.

CO2: Develop knowledge on various flight control system and its recent advancements.

CO3: Explain the fundamental understanding of the operation of engine auxiliary systems.

CO4: Describe the principles behind the operation of air conditioning and safety systems

CO5: Identify various types of instruments and sensors used in aircraft

UNIT: I	CONVENTIONAL AIRCRAFT SYSTEMS	9

Hydraulic systems – Components – Hydraulic Fluids – Hydraulic systems controllers – Modes of operation – Pneumatic systems – Working principles – Braking System – Landing Gear Systems – Classification – Shock absorbers – Retractive mechanism

UNIT: II	AIRCRAFT CONTROL SYSTEMS	9

Conventional Systems – Power assisted and Fully Powered Flight Controls – Power actuated systems – Engine control systems – Push pull rod system – Modern control systems – Digital fly by wire systems – Auto Pilot system - Recent Advances in Flight Control Technologies

UNIT: III	AIRCRAFT ENGINE AND AUXILIARY SYSTEMS	9

Piston and Jet Engines – Fuel systems – Components - Multi-Engine Fuel Systems - Lubricating Systems – Starting and Ignition systems - Superchargers and Turbochargers - Thrust Reversers and Afterburners

UNIT: IV	AIRCONDITIONING AND SAFETY SYSTEMS	9

Basic Air Cycle systems – Vapour Cycle Systems, Boot-strap air cycle system – Basics of Aircraft Environmental Control Systems (ECS) – Oxygen Storage, Distribution, and Mask Systems – Fire extinguishing system and smoke detection system, Anti-Icing and De-Icing Methods - Emergency Exit Systems

UNIT: V	AIRCRAFT INSTRUMENTS & SENSORS	9

Flight Instruments and Navigation Instruments – Accelerometers, Air speed Indicators – Mach Meters – Altimeters - Gyroscopic Instruments – Study of various types of engine instruments – Tachometers – Temperature and Pressure gauges - Types of Sensors Used in Aircraft Systems

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	E. H. J. Pallett	Aircraft Instruments & Principles	Pitman & Co, 2nd Edition	1993

2	S. Nagabhushana L. K. Sudha	Aircraft Instrumentation and Systems	TechSar Pvt. Ltd	2013
3	Ian Moir & Allan Seabridge	Aircraft Systems: Mechanical, Electrical, and Avionics Subsystems Integration	John Wiley & Sons	2011

REFERENCE BOOKS:				
1	McKinley, J.L. and Bent R.D.	Aircraft Maintenance & Repair	. McGraw Hill	1993
2	Lloyd Dingle & Mike Tooley	Aircraft Engineering Principles	Routledge	2005

WEB LEARNING RESOURCES:

1. https://aerocastle.wordpress.com/wp-content/uploads/2012/04/aircraft_systems.pdf
2. https://onlinecourses.nptel.ac.in/noc25_ae07/preview
3. https://www.youtube.com/playlist?list=PLcW-kjuuRl2KdkWwg7ZJO9eO3rPNAqLa-
4. https://www.faa.gov/sites/faa.gov/files/09_phak_ch7.pdf
5. https://www.youtube.com/watch?v=eAIGNjcqRX4

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	2	2	2	2	2	1	1	-	-	-	-	3	1	1
CO2	3	3	2	2	2	1	1	1	-	-	-	-	3	1	1
CO3	3	2	2	2	3	2	1	1	-	-	-	-	3	1	1
CO4	3	3	2	2	2	2	1	1	-	-	-	-	3	1	1
CO5	3	3	2	2	2	1	1	1	-	-	-	-	3	1	1
Avg	3	3	2	2	2	2	1	1	-	-	-	-	3	1	1

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<tr><td>R 2024</td><td colspan="5">AERONAUTICAL ENGINEERING</td><td>SEMESTER: III</td></tr>
<tr><td rowspan="2">24AE303</td><td rowspan="2">MANUFACTURING TECHNOLOGY</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">PC</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To illustrate various metal casting processes.
2. To learn and apply the working principles of various metal joining processes.
3. To analyze the methods of bulk deformation of metals.
4. To learn the working principles of sheet metal forming process.
5. To study and practice the working principles of plastics molding.

COURSE OUTCOMES:

At the end of this course, students can be able to

CO1: Explain the principle of different metal casting processes.
CO2: Describe the various metal joining processes.
CO3: Illustrate the different bulk deformation processes.
CO4: Apply the various sheet metal forming process.
CO5: Apply suitable moulding technique for manufacturing of plastics components.

UNIT: I	**METAL CASTING PROCESS**	**9**

Sand Casting – Type of patterns - Pattern Materials – Pattern allowances – Molding sand Properties and testing – Cores – Cupola furnaces - Shell, Investment casting process – Ceramic mould – Pressure Die Casting - Centrifugal Casting - Defects in Sand casting process - Remedies

UNIT: II	**METAL JOINING PROCESSES**	**9**

Oxy Fuel Welding – Arc Welding – Gas Metal Arc Welding – Plasma Arc Welding – Resistance Welding Processes - Electron Beam Welding – Laser Beam Welding – Friction Welding – Friction Stir Welding – Weld Defects – Inspection & Remedies

UNIT: III	**BULK DEFORMATION PROCESSES**	**9**

Hot Working and Cold Working of Metals – Forging Processes – Open, Impression And Closed Die Forging – Typical Forging Operations – Rolling of Metals – Types of Rolling – Principle of Rod and Wire Drawing – Tube Drawing – Principles of Extrusion – Types – Hot And Cold Extrusion. Introduction to Shaping Operations.

UNIT: IV	**SHEET METAL PROCESSES**	**9**

Sheet metal characteristics – Typical shearing, bending and drawing operations – Stretch forming operations – Hydro forming – Rubber pad forming – Metal spinning –Explosive forming – Incremental forming - Applications of sheet metal in Aerospace industries

UNIT: V	**MANUFACTURE OF PLASTIC COMPONENTS**	**9**

Types and characteristics of plastics – Molding of thermoplastics & Thermosetting polymers –– Injection molding – Plunger and screw machines – Compression moulding, Transfer Moulding –– Rotational molding – Extrusion – Thermoforming – Manufacture of Composite Materials

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of Publication**
1	Kalpakjian. S,	Manufacturing Engineering and Technology	Pearson Education India	2013
2	Hajra Choudhury	Elements of Workshop Technology I & II	Media Promoters and Publishers	2023
3	Sharma, P.C	A Text book of Production Technology	S.Chand and Co	2022

REFERENCE BOOKS:				
1	P.N .Rao	Manufacturing Technology Volume 1	Mc Grawhill Education	2018
2	Roy. A. Linberg	Process and Materials of Manufacture	Prentice Hall India	2000

WEB LEARNING RESOURCES:

1. https://onlinecourses.nptel.ac.in/noc22_me28/preview
2. https://www.coursera.org/learn/manufacturing-industry-101
3. https://www.youtube.com/watch?v=YftQGPTMUYs&list=PL-JvKqQx2Atd59aCxARxsGtx62TtTy3iN
4. https://www.edx.org/learn/manufacturing/massachusetts-institute-of-technology-manufacturing-process-control-i
5. https://www.youtube.com/watch?v=Um_g8sQ_p3Y

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	-	2	-	-	2	3	1	1	-	-	1	3	1	2
CO2	3	-	2	-	-	2	2	1	1	-	-	1	3	1	2
CO3	3	-	2	-	-	2	2	1	1	-	-	1	3	1	2
CO4	3	-	2	-	-	2	2	1	1	-	-	1	3	1	2
CO5	3	-	2	-	-	2	2	1	1	-	-	1	3	1	2
Avg	3	-	2	-	-	2	2	1	1	-	-	1	3	1	2

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: III
24AE304	FLUID MECHANICS AND STRENGTH OF MATERIALS	L	T	P	C	PC
		3	0	2	4	

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To introduce the students about properties of the fluids, behaviour of fluids under static conditions.
2. To impart basic knowledge of the dynamics of fluids and working of pumps
3. To enable understanding of the behaviour and response of materials
4. To familiarize with the different methods used for beam deflection analysis.

COURSE OUTCOMES:

At the end of this course, students can be able to

CO1: Understand the conservation laws applicable to fluids and its application through fluid dynamics

CO2: Explain the working principles of centrifugal, reciprocating and rotary pumps

CO3: Describe mechanical behaviour of materials.

CO4: Design members under axial loading and torsional loading

CO5: Understand concepts of deflection of beam and buckling of column

UNIT: I	FLUID PROPERTIES AND FLOW CHARACTERISTICS	9

Properties of fluids – Fluid statics - Pressure Measurements - Buoyancy and floatation - Flow characteristics - Eulerian and Lagrangian approach - Concept of control volume and system - Reynold’s transportation theorem - Continuity equation, energy equation and momentum equation - Applications.

UNIT: II	FLOW THROUGH PIPES AND PUMPS	9

Reynold’s Experiment - Laminar flow through circular conduits - Darcy Weisbach equation - Friction Factor - Moody Diagram - Major and minor losses - Pipes in series and parallel - Classification of pumps - Centrifugal pumps - Heads and efficiencies - Work done by the impeller - Performance curves - Reciprocating Pumps - Rotary Pumps.

UNIT: III	STRESS, STRAIN AND MATERIAL BEHAVIOR	9

Simple stress and Strain, Mechanical Properties of Materials, Statically Determinate Problems and Elastic Constants, Tension, Compression, and Shear, Elasticity, Plasticity and Creep. Allowable stresses - Thermal Stresses in Aircraft Structures

UNIT: IV	SHEAR FORCE, BENDING & TORSION	9

Axially loaded members - Statically indeterminate structures - Torsion of circular bar, Transmission of power by circular shafts, Pure bending and Nonuniform bending, Design of beams for bending stresses, Shear stresses in beams of rectangular cross section.

UNIT: V	BEAMS AND COLUMNS	9

Plane stress, Principal stresses, Mohr’s circle and Hooke’s law for plane stresses. Spherical and Cylindrical pressure vessels. Deflection of beams, Column buckling.

Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

Practical Exercises: **Periods: 30**

Fluid Mechanics Section:

1. Verification of Bernoulli's theorem
2. Flow through Orifice/Venturi meter
3. Friction factor for flow through pipes
4. Characteristics of Centrifugal pump
5. Characteristics of Reciprocating pump

Strength of Materials Section:

1. Tension test on a mild steel rod
2. Torsion test on mild steel rod
3. Impact test on metal specimen
4. Rockwell Hardness test on metals
5. Deflection test on beams

Total Periods: 75

LIST OF EQUIPMENTS:

1.	Bernoulli's Experiment Apparatus	:	1 Nos
2.	Orifice/Venturi meter Apparatus	:	1 Nos
3.	Friction Factor Measurement Setup	:	1 Nos
4.	Centrifugal Pump Setup	:	1 Nos
5.	Reciprocating Pump Setup	:	1 Nos
6.	Universal Tensile Testing Machine	:	1 Nos
7.	Torsion Testing Machine	:	1 Nos
8.	Impact Testing Machine	:	1 Nos
9.	Rockwell Hardness Testing Machine	:	1 Nos
10.	Beams with weight hangers	:	1 Nos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Fox W.R. and McDonald A.T	Introduction to Fluid Mechanics	John-Wiley and Sons	2020
2	Ferdinand Beer, E. Russell Johnston	Mechanics of Materials	McGraw Hill Education	2020
3	R. K. Bansal	A Text Book of Fluid Mechanics and Hydraulic Machines	Laxmi Publications	2019

REFERENCE BOOKS:

1	Russell C Hibbeler	Mechanics of Materials	Pearson	2013
2	R.K.Rajput	A Text Book of Strength of Materials	S.Chand Publishers	2021

WEB LEARNING RESOURCES:

1. https://archive.nptel.ac.in/courses/112/105/112105269/
2. https://nptel.ac.in/courses/112107146
3. https://www.youtube.com/watch?v=8iZe_UiBtTc&list=PLZ5iF05Ly-kgGWarGh0iIdUIu4cz7Hrdw
4. https://www.youtube.com/watch?v=aQf6Q8t1FQE&list=PLEYqyyrm-hQ3wtF34smyJSAOqUJqnf1ch
5. https://www.youtube.com/watch?v=La4UEa7hA7Q&list=PLJoALJA_KMOARYNi50T6b488kPUBbOIsX

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	3	2	2	-	-	-	1	2	-	-	-	3	2	3
CO2	3	3	2	2	-	-	-	1	2	-	-	-	3	2	3
CO3	3	3	2	2	-	-	-	1	2	-	-	-	3	3	3
CO4	3	3	2	2	-	-	-	1	2	-	-	-	3	2	3
CO5	3	3	2	2	-	-	-	1	2	-	-	-	3	2	2
Avg	3	3	2	2	-	-	-	1	2	-	-	-	3	2	3

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: III
24AE305	AERO THERMODYNAMICS	L	T	P	C	PC
		3	0	2	4	

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand the processes for transformation of energy and between work & heat.
2. To understand the Laws of thermodynamics
3. To develop basic concept of air cycle, gas turbine engines and heat transfer.
4. To identify the different components of Jet Engines

COURSE OUTCOMES:

At the end of this course, students can be able to

CO1: Apply the laws of thermodynamics in real time problems.
CO2: Understand the Reversibility and Concept of Entropy
CO3: Demonstrate the efficiency of different air standard cycles and understand the properties of steam
CO4: Illustrate working of Airconditioning & Refrigeration Systems
CO5: Solve heat transfer problems in complex systems

UNIT: I	FUNDAMENTAL CONCEPTS AND FIRST LAW	9

Concept of continuum, macroscopic approach, thermodynamic systems – closed, open and isolated. Property, state, path and process, quasi-static process, work, internal energy, enthalpy, SFEE, First law of thermodynamics, relation between P,V,T for various processes, Zeroth law of thermodynamics.

UNIT: II	SECOND LAW AND ENTROPY	9

Second law of thermodynamics – Kelvin Planck and Clausius statements of second law. Reversibility and Irreversibility, Thermal Reservoir, Carnot theorem. Carnot cycle, COP, Thermodynamic temperature scale - Clausius inequality, Entropy changes for various processes

UNIT: III	AIR STANDARD CYCLES & VAPOUR STANDARD CYCLES	9

Otto, Ericsson and Brayton cycles - Air standard efficiency – Mean effective pressure - Thermodynamic properties of steam - Standard Rankine cycle - Heat rate, Specific steam consumption,

UNIT: IV	REFRIGERATION AND AIR-CONDITIONING	9

Principle of Refrigeration ‐ Vapour compression & Vapour absorption types - Coefficient of performance, Properties of refrigerants, Psychrometry ‐ Relative Humidity ‐ WBT/DBT - Principle of Air conditioning ‐ Types of AC Units - Case study: Aircraft cabin air-conditioning.

UNIT: V	HEAT TRANSFER	9

Heat Transfer: Conduction, convection and radiation heat transfer - heat conduction through composite walls, Different types of heat exchangers - heat transfer coefficient for parallel, counter and cross flow type heat exchanger

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

Periods: 45

Practical Exercises: **Periods: 30**

1. Performance test on a 4-stroke engine
2. Valve timing of a 4 – stroke engine
3. Determination of effectiveness of heat exchanger – Parallel & Counter Flow
4. Determination of flash point & fire point of a fuel
5. Determination of thermal conductivity of solid.

6. Determination of thermal resistance of a composite wall.
7. COP test on a refrigeration test rig
8. COP test on a air-conditioning test rig

Total Periods: 75

LIST OF EQUIPMENTS:

1. 4-Stroke Diesel Engine : 1 Nos
2. Cut section model of 4 stroke diesel engine : 1 Nos
3. Parallel and counter flow heat exchanger test rig : 1 Nos
4. Flash Point / Fire Point Apparatus : 1 Nos
5. Lagged Pipe Apparatus : 1 Nos
6. Composite Wall Apparatus : 1 Nos
7. Vapour Compression Refrigeration Test Rig : 1 Nos
8. Vapour Compression Air-Conditioning Test Rig : 1 Nos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Yunus A. Cengel & Michael A. Boles	Thermodynamics: An Engineering Approach	McGraw-Hill	2019
2	Nag.P.K	Engineering Thermodynamics	Prentice-Hall India	2017
3	Ganesan V	Internal Combustion Engine	McGraw-Hill, 4th edition	2017

REFERENCE BOOKS:

1	Rathakrishnan E.	Fundamentals of Engineering Thermodynamics	Prentice-Hall India	2005
2	Holman.J.P	Thermodynamics	McGraw-Hill	2007

WEB LEARNING RESOURCES:

1. https://archive.nptel.ac.in/courses/112/106/112106310/
2. https://archive.nptel.ac.in/courses/101/104/101104063/
3. https://www.youtube.com/watch?v=kLqduWF6GXE&list=PLF6C6594F42ECEE0D
4. https://www.youtube.com/watch?v=gBs2vAt_L1I&list=PLjtQ3BMex7hsDok5zFFRydcMnQGBqk-CI
5. https://www.youtube.com/watch?v=B-rFIdOi-No&list=PLkUEX3IbW7lfdC2ieft_9FH5zAAvUfZAn

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
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CO3	3	2	2	1	-	-	1	-	-	1	1	2	3	2	-
CO4	3	3	3	1	-	-	1	-	-	1	1	2	3	1	1
CO5	3	2	2	1	-	-	1	-	-	1	1	2	3	1	-
Avg	3	2	2	1	-	-	1	-	-	1	1	2	3	1	1

<table>
<tr><td rowspan="1">R 2024</td><td colspan="5">AERONAUTICAL ENGINEERING</td><td>SEMESTER: III</td></tr>
<tr><td rowspan="2">24AE306</td><td rowspan="2">MANUFACTURING PROCESSES LAB</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">PC</td></tr>
<tr><td>0</td><td>0</td><td>4</td><td>2</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To select appropriate tools, equipment's and machines to complete a given job.
2. To perform various welding process using GMAW
3. To fabricate gears using gear making machines.
4. To perform various machining process such as turning, shaping, drilling, milling and analyzing the defects in the cast and machined components.

COURSE OUTCOMES:

At the end of this course, students can be able to
CO1: construct simple structures using welding machine
CO2: make workpiece for given shape and size using machining processes
CO3: analyse defects in manufactured products
CO4: produce gears using gear making machines
CO5: calculate force using dynamometer in various machines

PRACTICAL EXERCISES:

1. Step Turning and Taper Turning on circular parts using lathe machine.
2. Shaping – Square and Hexagonal Heads on circular parts using shaper machine.
3. Drilling and Reaming using vertical drilling machine.
4. Contour Milling using milling machine.
5. Grinding components using cylindrical and centerless grinding machine.
6. Grinding components using surface grinding machine.
7. Generating gears using gear hobbing machine.
8. Cutting force calculation using dynamometer in lathe machine
9. 3D printing of a Spur Gear
10. Turning on circular parts using CNC lathe machine.

Total Periods: 60

LIST OF EQUIPMENTS:

1. Centre Lathes	:	7 Nos
2. Shaper	:	1 No
3. Radial Drilling Machine	:	1 No
4. Horizontal Milling Machine	:	1 No
5. Vertical Milling Machine	:	1 No
6. Surface Grinding Machine	:	1 No
7. Cylindrical Grinding Machine	:	1 No
8. Gear Hobbing Machine	:	1 No
9. Lathe Tool Dynamometer	:	1 No
10. 3D Printer	:	1 No
11. CNC Lathe	:	1 No

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	-	-	-	-	-	1	-	2	-	-	1	1	2	2
CO2	3	-	-	-	-	-	1	-	2	-	-	1	1	2	2
CO3	3	-	-	-	-	-	1	-	2	-	-	1	1	2	2
CO4	3	-	-	-	-	-	1	-	2	-	-	1	1	2	2
CO5	3	-	-	-	-	-	1	-	2	-	-	1	1	2	2
Avg	3	-	-	-	-	-	1	-	2	-	-	1	1	2	2

M.A.M SCHOOL OF ENGINEERING

(An Autonomous Institution)

(Accredited by NAAC || Approved by AICTE || Affiliated to Anna University)

Trichy – Chennai Trunk Road, Siruganur, Tiruchirappalli – 621 105

B.E. AERONAUTICAL ENGINEERING

SEMESTER IV

THEORY COURSES										
S.No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1.	24AE401	Aerodynamics	3	0	0	3	40	60	100	PC
2.	24AE402	Aircraft Engine Maintenance & Repair	3	0	0	3	40	60	100	PC
3.	24MC401	Environmental Sciences	3	0	0	0	40	60	100	MC
4.	24AE403	Mechanics of Machines	3	1	0	4	40	60	100	PC
THEORY COURSES WITH LABORATORY COMPONENT										
5.	24AE404	Aircraft Structures	3	0	2	4	50	50	100	PC
6.	24AE405	Propulsion	3	0	2	4	50	50	100	PC
LABORATORY COURSES										
7.	24AE406	Aerodynamics Lab	0	0	4	2	50	50	100	PC
8.	24AE407	Aero Engines & Aero Frame Lab	0	0	4	2	60	40	100	PC
9.	24TP401	Aptitude Skills III & Technical Skills I	0	0	2	1	60	40	100	EEC
		TOTAL	18	1	14	23				

REGULATIONS 2024

Recommended by 2nd BOS held on 10.05.2025 and Approved by 2nd Academic Council held on 12.06.2025

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: IV
24AE401	AERODYNAMICS	L	T	P	C	PC
		3	0	0	3	

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To introduce the concepts of mass, momentum and energy conservation relating to aerodynamics.
2. To learn concepts of Airfoil theory and subsonic wing theory
3. To learn the theory behind the formation of shocks and expansion fans in Supersonic flows.
4. To get knowledge on high-speed flow over air foils, wings and airplane configuration.
5. To learn the concepts of Transonic flow

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Apply the concept of 2D, inviscid incompressible flows in low-speed aerodynamics.
CO2: Solve lift generation problems using aerofoil theories
CO3: Calculate the compressible flow through a duct of varying cross section.
CO4: Estimate the properties across normal and oblique shock waves.
CO5: Predict the properties of transonic flows.

UNIT: I	INTRODUCTION TO LOW-SPEED FLOW	9

Euler equation, incompressible Bernoulli's equation. circulation and vorticity, green's lemma and Stoke's theorem, barotropic flow, kelvin's theorem, streamline, stream function, irrotational flow, elementary flows and their combinations - Ideal Flow over a circular cylinder, D'Alembert's paradox, magnus effect, Kutta Joukowski's theorem,

UNIT: II	AIRFOIL THEORY AND SUBSONIC WING THEORY	9

Cauchy-Riemann relations, complex potential, methodology of conformal transformation, Kutta- Joukowski transformation and its applications, thin airfoil theory and its applications - Vortex filament, Biot and Savart law, bound vortex and trailing vortex, horse shoe vortex, lifting line theory and its limitations.

UNIT: III	ONE DIMENSIONAL COMPRESSIBLE FLOW	9

Energy, Momentum, continuity and state equations, velocity of sound, adiabatic steady state flow equations, Flow through convergent- divergent passage, Performance under various back pressures.

UNIT: IV	NORMAL AND OBLIQUE SHOCKS	9

Prandtl equation and Rankine – Hugonoit relation, Normal shock equations, Pitot static tube, corrections for subsonic and supersonic flows, Oblique shocks and corresponding equations, Hodograph and pressure turning angle, shock polar, flow past wedges and concave corners, strong, weak and detached shocks

UNIT: V	TRANSONIC FLOW OVER WING	9

Lower and upper critical Mach numbers, Lift and drag, divergence, shock induced separation, Characteristics of swept wings, Effects of thickness, camber and aspect ratio of wings, Transonic area rule.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Anderson, J.D	Fundamentals of Aerodynamics	McGraw Hill Book	2010
2	E Rathakrishnan	Theoretical Aerodynamics	John Wiley	2013

3	Clancey, L J	Aerodynamics	Pitman	2007

REFERENCE BOOKS:				
1	Houghton, E.L., and Caruthers, N.B	Fundamentals of Aerodynamics	McGraw Hill Book	2010
2	Anderson, J.D	Modern Compressible Flows	McGraw Hill Book	2017

WEB LEARNING RESOURCES:

1. https://onlinecourses.nptel.ac.in/noc22_ae09/preview
2. https://archive.nptel.ac.in/courses/101/105/101105023/
3. https://www.youtube.com/watch?v=edLnZgF9mUg
4. https://www.youtube.com/watch?v=cZCSyElgA3U&list=PLrVwZIzUqvbOelcKH3ZGGsPOMpZEYBzeb
5. https://www.youtube.com/watch?v=TiLPrV2MpG0&list=PLhwqhcexBHpUfIlq-6XxDw3xEEN3-KN3V

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	2	2	2	2	2	1	1	-	-	-	-	3	1	1
CO2	3	3	2	2	2	1	1	1	-	-	-	-	3	1	1
CO3	3	2	2	2	3	2	1	1	-	-	-	-	3	1	1
CO4	3	3	2	2	2	2	1	1	-	-	-	-	3	1	1
CO5	3	3	2	2	2	1	1	1	-	-	-	-	3	1	1
Avg	3	3	2	2	2	2	1	1	-	-	-	-	3	1	1

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: IV
24AE402	AIRCRAFT ENGINE MAINTENANCE & REPAIR	L	T	P	C	PC
		3	0	0	3	

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To make the students to familiarize with the Aircraft engine maintenance procedure and practice.
2. To acquire knowledge of basics of Aeronautics and engine components.
3. To learn the concepts of Piston engines
4. To make students aware of aircraft propellers, aircraft jet engines and their repair
5. To update students on recent inspection, testing and overhauling procedures

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Understand the construction, types, and operating principles of piston engines sub-systems.

CO2: Perform routine maintenance tasks, inspections, and troubleshooting procedures on propeller engines

CO3: Demonstrate the ability to disassemble, inspect, repair, and reassemble jet engine components

CO4: Apply non-destructive testing procedures to identify the defects

CO5: Apply overhauling & repair procedure to aircraft engines

UNIT: I	PISTON ENGINE MAINTENANCE PRACTICES	9

Carburation and Fuel injection systems - Ignition system components - Induction, Exhaust and cooling system - Inspection and maintenance and troubleshooting - Daily and routine checks - Compression testing of cylinders - Special inspection schedules - Engine fuel, control and exhaust systems - Engine mount and super charger

UNIT: II	MAINTENANCE OF PROPELLER ENGINESS	9

Propeller theory - operation, construction assembly and installation - Pitch change mechanism - Propeller axially system- Damage and repair criteria - General Inspection procedures - Checks on constant speed propellers - Pitch setting, Propeller Balancing, Blade cuffs, Governor/Propeller operating conditions

UNIT: III	JET ENGINES MAINTENANCE	9

Inlets – compressors, turbines - exhaust section – classification and types of lubrication and fuels – Details of control, starting, running and operating procedures – Damage and repair criteria - internal inspection of engines - compressor washing - field balancing of compressor fans - Component maintenance procedures - Use of instruments for online maintenance - Special inspection procedures - Foreign Object Damage - Blade damage.

UNIT: IV	TESTING AND INSPECTION	9

Symptoms of failure - Fault diagnostics - Case studies of different engine systems – Rectification during testing equipments for overhaul: Tools and equipments requirements for various checks during overhauling - Tools for inspection - Tools for safety and for visual inspection - Methods and instruments for non-destructive testing techniques. Engine testing: Engine testing procedures and schedule preparation

UNIT: V	ENGINE OVERHAUL AND REPAIR	9

Engine Overhaul procedures - Inspections and cleaning of components – Repairs schedules for overhaul - Balancing of Gas turbine components. Procedures for trouble shooting - Condition monitoring of the engine on ground and at altitude - Dynamic balancing and calibration - Engine health monitoring and corrective methods.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of Publication**
1	Kroes & Wild	Aircraft Power Plants	McGraw Hill	2017
2	Pratt & Whitney	The Aircraft Gas Turbine Engine and its Operation	English Book Store	2003
3	William Crouse Donald Anglin	Automotive Mechanics	McGraw Hill	2017

REFERENCE BOOKS:				
1	Haguma Timothee	Fundamentals of Aircraft Maintenance Management	Notion Press	2016
2	Michael Kroes, Ronald Sterkenburg	Aircraft Maintenance and Repair	McGraw Hill	2017

WEB LEARNING RESOURCES:

1. https://archive.nptel.ac.in/courses/101/104/101104075/
2. https://archive.nptel.ac.in/courses/101/104/101104071/
3. https://www.youtube.com/watch?v=gtotJb728pI
4. https://www.youtube.com/watch?v=5xqoFdQKSfs&list=PLcw703S2LGnhWVdWJpg5EmUafCvI_nrVx
5. https://www.youtube.com/watch?v=-Wxgr4xCMUE

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	2	3	3	3	1	-	-	-	-	-	1	-	3	1	-
CO2	2	3	3	3	1	-	-	-	-	-	1	-	3	1	-
CO3	2	3	3	3	2	-	-	-	-	-	1	-	3	1	-
CO4	2	3	3	3	2	-	-	-	-	-	1	-	3	1	-
CO5	2	3	3	3	1	-	-	-	-	-	1	-	3	1	-
Avg	2	3	3	3	2	-	-	-	-	-	1	-	3	1	-

R 2024	CHEMISTRY					SEMESTER: IV
24MC401	ENVIRONMENTAL SCIENCES	L	T	P	C	MC
		3	0	0	0	

COMMON TO ALL DEPARTMENTS

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To introduce the basic concepts of environment and ecosystems
2. To introduce the basic concepts biodiversity and emphasize on the biodiversity of India and its conservation
3. To impart knowledge on the causes, effects and control or prevention measures of environmental pollution and natural disasters.
4. To facilitate the understanding of global and Indian scenario of renewable and non-renewable resources, causes of their degradation and measures to preserve them.
5. To familiarize the concept of sustainable development goals and appreciate the interdependence of economic and social aspects of sustainability, recognize and analyze climate changes, concept of carbon credit and the challenges of environmental management.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: To recognize and understand the functions of environment and ecosystem.

CO2: To develop an understanding of the roles and functions of the biodiversity and to emphasize the need for their conservation.

CO3: To identify the causes, effects of environmental pollution and natural disasters and contribute to the preventive measures in the society.

CO4: To identify and apply the understanding of renewable and non-renewable resources and contribute to the sustainable measures to preserve them for future generations

CO5: To recognize the different goals of sustainable development and apply them for suitable technological advancement and societal development.

UNIT: I	ENVIRONMENT AND ECOSYSYTEM	6

Definition, scope and importance of environment – need for public awareness - concept of an ecosystem – structure and function of an ecosystem – energy flow in the ecosystem – ecological succession – food chains, food webs and ecological pyramids – Introduction, types, characteristic features, structure and function of the (a) Terrestrial ecosystem (b) Aquatic ecosystem

UNIT: II	BIODIVERSITY	6

Definition, importance of bio-diversity-types of biodiversity: genetic, species and ecosystem diversity–values of biodiversity, India as a mega-diversity nation– hotspots of biodiversity–threats to biodiversity – endangered and endemic species of India – conservation of biodiversity: In-situ and ex-situ.

UNIT: III	ENVIRONMENTAL POLLUTION	6

Causes, Effects and Preventive measures of Water, Soil, Air, Marine and Noise Pollution. Solid, Biological, Hazardous and E-Waste management. Case studies on Occupational Health and Safety Management system (OHASMS). Environmental protection, Environment (Protection) Amendment Rules, 2024

UNIT: IV	RENEWABLE SOURCES OF ENERGY	6

Energy conservation, New Energy Sources: Need of new sources. Different types new energy sources. Wind Energy, Ocean energy resources, Tidal energy conversion, Concept, origin and power plants of geothermal energy.

UNIT: V	SUSTAINABILITY AND MANAGEMENT	6
Sustainability- concept, need for sustainability– from unsustainability to sustainability-millennium development goals - Sustainable Development Goals, Climate change - environmental issues and possible solutions-case studies on climate change.		
		Total Periods: 30
Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos	

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Anubha Kaushik and C. P. Kaushik	Perspectives in Environmental Studies	6th Edition, New Age International Publishers	2018
2	BennyJoseph	Environmental Science And Engineering	Tata Mc Graw-Hill, NewDelhi,	2016
3	GilbertM.Masters	Introduction to Environmental Engineering and Science	2ndedition, Pearson Education	2004
4	Allen, D. T. and Shonnard, D. R.	Sustainability Engineering: Concepts, Design and Case Studies	Prentice Hall	2012
5	Bradley. A.S; Adebayo, A.O., Maria, P.	Engineering applications in sustainable design and development	Cengage learning	2022

REFERENCE BOOKS:

1	Rajagopalan, R	Environmental Studies- From Crisis to Cure'	Prentice hall of India PVT. LTD, New Delhi	2015
2	Erach Bharucha	Textbook of Environmental Studies for Undergraduate Courses	Orient Blackswan Pvt. Ltd.	2013

WEB LEARNING RESOURCES:

1. https://archive.nptel.ac.in/courses/127/106/127106004/
2. https://archive.nptel.ac.in/courses/127/105/127105018/
3. https://www.youtube.com/watch?v=705DHheuG6w&t=16s
4. https://www.youtube.com/watch?v=CYJECfoP-3E&t=863s
5. https://www.youtube.com/watch?v=t7Q7y_xjR5E&t=14s

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	2	1	-	-	-	2	3	-	-	-	-	2	-	-	-
CO2	3	2	-	-	-	3	3	-	-	-	-	2	-	-	-
CO3	3	-	1	-	-	2	2	-	-	-	-	2	-	-	-
CO4	3	2	1	1	-	2	2	-	-	-	-	2	-	-	-
CO5	3	2	1	-	-	2	2	-	-	-	-	1	-	-	-
Avg	3	2	1	1	-	2	2	-	-	-	-	2	-	-	-

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: IV
24AE403	MECHANICS OF MACHINES	L	T	P	C	PC
		3	1	0	4	

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand the principles of mechanisms and their kinematics.
2. To learn the basic concepts of toothed gears and kinematics of gear trains.
3. To analyze the forces and torque acting on simple mechanical systems
4. To understand the importance of balancing and vibration
5. To study the effect of friction in different machine elements.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Understand the fundamentals of mechanisms and perform kinematic analysis of linkages and Cams
CO2: Analyze different types of gears and gear trains, and determine velocity ratios and torque transmission
CO3: Determine the forces on members of mechanisms during static and dynamic equilibrium conditions.
CO4: Determine the balancing masses on rotating machineries and the natural frequencies of vibratory systems
CO5: Evaluate the effects of friction in mechanical systems and analyze the performance of brakes and clutches.

UNIT: I	KINEMATICS OF MECHANISMS	9+3

Mechanisms – Terminology and definitions – kinematics inversions and analysis of 4 bar and slide crank chain – velocity and acceleration polygons – Degrees of freedom (Gruebler's criterion) - Cams – classifications – displacement diagrams - layout of plate cam profiles.

UNIT: II	GEARS AND GEAR TRAINS	9+3

Classification of gears: spur, helical, bevel, worm - Gear tooth terminology and gear profiles - law of toothed gearing – involute gearing – Gear tooth action - Interference and undercutting – gear trains – parallel axis gear trains – Epicyclic Gear Trains.

UNIT: III	DYNAMICS OF MACHINES	9+3

Free body diagrams – Static equilibrium conditions – Static Force analysis in simple mechanisms – Dynamic Force Analysis in simple machine members – Inertia Forces and Inertia Torque – D'Alembert's principle.

UNIT: IV	BALANCING OF ROTATING MASSES AND VIBRATION	9+3

Static and Dynamic balancing – Balancing machines – Free vibrations – natural Frequency – Damped Vibration – bending critical speed of simple shaft – Forced vibration – harmonic Forcing – Vibration isolation.

UNIT: V	FRICTION AND BRAKES	9+3

Surface contacts – Sliding and Rolling friction – Friction drives – Friction in screw threads – Friction clutches – Belt drives – Friction aspects in brakes. - Applications in aerospace systems

Total Periods: 60

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Uicker, J.J., Pennock G.R and Shigley.J.E	Theory of Machines and Mechanisms	Oxford University Press	2017
2	Rattan, S.S	Theory of Machines	McGraw-Hill Education	2014

3	Cleghorn. W. L., Nikolai Dechev	Mechanisms of Machines	Oxford University Press	2015

REFERENCE BOOKS:				
1	Rao.J.S. and Dukkipati.R.V.	Mechanism and Machine Theory	New Age International Pvt. Ltd	2006
2	Thomas Bevan	The Theory of Machines	Pearson Education	2010

WEB LEARNING RESOURCES:

1. https://archive.nptel.ac.in/courses/112/105/112105268/
2. https://archive.nptel.ac.in/courses/112/106/112106270/
3. https://www.youtube.com/watch?v=1BtBk69m8w0&list=PL8EJqJaNKpjUfKW__pRtPI9PVgjF0hEp6
4. https://www.youtube.com/watch?v=GuIYMVHvnhs&list=PLAjJ9epAHCkz-vpeUJatstaKOtBHHyZI_
5. https://www.youtube.com/watch?v=eLawX815nz0&list=PLMrpXL7ZxXYU7lqnLGgkpbG22K9dv-fUj

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	2	2	3	2	-	1	1	-	-	1	-	3	1	1
CO2	3	2	2	3	2	-	1	1	-	-	1	-	3	1	1
CO3	3	2	2	3	3	-	1	1	-	-	1	-	3	1	1
CO4	3	3	2	3	2	-	1	1	-	-	1	-	3	1	1
CO5	3	3	2	3	2	-	1	1	-	-	1	-	3	1	1
Avg	3	3	2	3	2	-	1	1	-	-	1	-	3	1	1

<table>
<tr><td>R 2024</td><td colspan="5">AERONAUTICAL ENGINEERING</td><td>SEMESTER: IV</td></tr>
<tr><td rowspan="2">24AE404</td><td rowspan="2">AIRCRAFT STRUCTURES</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">PC</td></tr>
<tr><td>3</td><td>0</td><td>2</td><td>4</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To familiarize the generalized theory of pure bending and work out problems in the calculation of bending stress involving different methods.
2. To provide an understanding on energy methods to statically determinate and indeterminate structures
3. To gain knowledge in the concept of shear flow in thin-walled sections and open sections
4. To Impart theoretical knowledge on the behaviour of thin plates and thin-walled columns.
5. To carry out basic stress analysis procedures involving aircraft structural components

COURSE OUTCOMES:

At the end of this course, students can be able to

CO1: Explain the linear static analysis of determinate and indeterminate aircraft structural components

CO2: Apply the energy methods and failure theories to determine the reactions of structure.

CO3: Analyze the normal stress variation on unsymmetrical sections, subjected to bending moments.

CO4: Investigate the behaviour of buckling of simply supported beams

CO5: Analyze the shear and bending moment variation of aircraft wing and fuselage

UNIT: I	ANALYSIS OF TRUSSES AND INDETERMINATE BEAMS	9

Overview of aircraft structural components and configurations. Types of structural loads - Plane truss analysis – method of joints – method of sections – method of shear – 3-D trusses – principle of super position, Clapeyron’s 3 moment equation and moment distribution method for indeterminate beams.

UNIT: II	ENERGY METHODS AND FAILURE THEORIES	9

Introduction to energy principles in structural analysis, Castigliano’s theorems, and unit load method. Energy methods applied to statically determinate and indeterminate beams, frames, rings & trusses.: Maximum Stress Failure theory, Maximum Strain Failure theory - Application to composite materials used in aircraft.

UNIT: III	UNSYMMETRICAL BENDING AND SHEAR FLOW	9

Unsymmetrical bending of beams – Neutral axis method, ‘k’ method, and the principal axis method – Open and Closed Sections - Shear flow in thin-walled Open sections with and without stiffening elements - Shear flow due to bending and torsion in single-cell and multi-cell structures, Effect of structural idealization - Stress analysis of thin-webbed beams using Wagner’s theory.

UNIT: IV	STABILITY AND STRUCTURAL COMPONENTS	9

Buckling of columns and beams. Euler’s theory and critical load calculations - Theory of beam columns - Local buckling in thin-walled structures - Behaviour of a rectangular plate under compression, governing equation for plate buckling, buckling analysis of sheets and stiffened panel under compression, Concept of the effective sheet width, Buckling due to shear and Combined Loading - Crippling.

UNIT: V	STRUCTURAL ANALYSIS OF AIRCRAFT COMPONENTS	9

Classification and function of primary and secondary aircraft structures - Loading and analysis of aircraft wing, fuselage, and tail unit. Use of V-n diagram for sizing the aircraft wing, fuselage, and tail unit. - Concepts of structural efficiency and weight optimization

Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

Practical Exercises:	**Periods: 30**

1. Deflection of Beams
2. Verification of superposition theorem
3. Verification of Maxwell's reciprocal theorem
4. Unsymmetrical Bending of a Cantilever Beam
5. Shear Centre of a Channel Section
6. Free Vibration of a Cantilever Beam

Total Periods: 75

LIST OF EQUIPMENTS:

1.	Beams with weight hangers	:	1 Nos
2.	Dial Gauges	:	4 Nos
3.	Unsymmetrical Bending Set Up	:	1 Nos
4.	Cantilever Beam Set Up	:	1 Nos
5.	Set up for combined bending and torsion	:	1 Nos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Megson T H G	Aircraft Structures for Engineering Students	Butterworth-Heinemann	2012
2	Peery, D.J., and Azar, J.J.	Aircraft Structures	McGraw – Hill	2007
3	Bruhn. E.H.	Analysis & Design of Flight Vehicles Structures	Tri-state off-set company	2015

REFERENCE BOOKS:

1	René Alderliesten	Introduction to Aerospace Structures and Materials	TU Delft Open	2018
2	Howard D Curtis	Fundamentals of Aircraft Structural Analysis	WCB-McGraw Hill	2010

WEB LEARNING RESOURCES:

1. http://nptel.ac.in/courses/112106141/
2. https://www.edx.org/course/introduction-to-aerospace-structures-and-materials
3. https://cosmolearning.org/courses/introduction-aerospace-structures/
4. https://www.youtube.com/watch?v=Yh7S4lSlXgQ&list=PLv4AhJns8bvCP7FGrBsNtE-YDYlD1koUM
5. https://www.youtube.com/watch?v=H8Hxz_Yyc-U

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	3	2	3	2	-	1	-	-	-	1	3	3	1	1
CO2	3	3	2	3	2	-	1	-	-	-	1	3	3	1	1
CO3	3	3	2	3	2	-	1	-	-	-	1	3	3	1	1
CO4	3	3	2	3	2	-	1	-	-	-	1	3	3	1	1
CO5	3	3	2	3	2	-	1	-	-	-	1	3	3	1	1
Avg	3	3	2	3	2	-	1	-	-	-	1	3	3	1	1

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: IV
24AE405	PROPULSION	L	T	P	C	PC
		3	0	2	4	

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To establish fundamental approach and application of jet engine components.
2. To learn about the analysis of various engine components
3. To introduce about the application of various equations in Gas Turbine Engines.
4. To learn the concepts of rocket propulsion
5. To acquire knowledge on advanced propulsion systems

COURSE OUTCOMES:

At the end of this course, students can be able to

CO1: Understand the fundamental principles of propulsion systems and evaluate the performance metrics of various air-breathing and non-air-breathing propulsion technologies.

CO2: Explain the aerodynamic behavior of engine components & assess their impact on engine performance.

CO3: Analyze the working and performance of gas turbine engines under different operating conditions.

CO4: Describe the principles of rocket propulsion calculate their performance parameters

CO5: Explore advanced propulsion technologies and assess the environmental implications

UNIT: I	FUNDAMENTALS OF PROPULSION SYSTEMS	9

Introduction to propulsion and its role in aircraft. Classification of propulsion systems: air-breathing and non-air-breathing engine - Operating principles of piston engines - Working of gas turbine engines – factors affecting thrust – methods of thrust augmentation – performance parameters of jet engines.

UNIT: II	PROPULSION ENGINE COMPONENTS	9

Aerodynamic design of subsonic and supersonic inlets. Types of nozzles: convergent, convergent-divergent, and variable area nozzles. Flow characteristics in nozzles and thrust generation - Diffuser performance – modes of operation - thrust reversal

UNIT: III	COMBUSTION CHAMBER & COMPRESSOR OF JET ENIGNE	9

Combustion equations and process - Classification of combustion chambers – combustion chamber performance - Cooling process, Materials, Aircraft fuels, HHV, LHV, Orsat apparatus - Axial and centrifugal compressors - performance parameters axial flow compressors– stage efficiency.

UNIT: IV	JET ENGINE TURBINES	9

Principle of operation of axial flow turbines– limitations of radial flow turbines- Work done and pressure rise – Velocity diagrams – degree of reaction – constant nozzle angle designs – performance parameters of axial flow turbine– turbine blade cooling methods – stage efficiency calculations – basic blade profile design considerations – matching of compressor and turbine

UNIT: V	ADVANCED PROPUSLION SYSTEMS	9

Introduction to ramjets, scramjets, and pulse detonation engines. Basic working principles and challenges.. Recent developments in green propulsion technologies and alternative fuels. Future trends in propulsion systems including electric and hybrid-electric propulsion.

Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

Practical Exercises:	Periods: 30

1. Study of aircraft piston and its components.
2. Study of an aircraft jet engine and its components
3. Flame stabilization studies using conical and hemispherical flame holders.
4. Cascade testing of compressor blades.
5. Wall pressure measurements of a subsonic diffusers
6. Performance analysis of propeller.

Total Periods: 75

LIST OF EQUIPMENTS:

1. Jet Engine : 1 Nos
2. Piston Engine : 1 Nos
3. Conical / Hemispherical Flame Holder Model : 1 Nos
4. Compressor Cascade Blade Setup : 1 Nos
5. Propeller Blade Model : 1 Nos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Hill, P.G. & Peterson, C.R.	Mechanics & Thermodynamics of Propulsion	Pearson Education	2009
2	Saravanamuttoo, H.I.H., Paul Straznicky	Gas Turbine Theory	Pearson Education	2008
3	V.Ganesan	Gas Turbines	Tata McGraw-Hill,	2017

REFERENCE BOOKS:

1	Saeed Farokhi	Aircraft Propulsion	John Wiley and Sons	2014
2	Mathur, M.L., and Sharma, R.P.	Gas Turbine, Jet and Rocket Propulsion	Standard Publishers and Distributors	2014

WEB LEARNING RESOURCES:

1. https://nptel.ac.in/courses/101101002/
2. https://www.grc.nasa.gov/www/k-12/UEET/StudentSite/engines.html
3. https://www.rolls-royce.com/products-and-services/civil-aerospace.aspx
4. https://www.youtube.com/watch?v=nZmvLNhcK_w&list=PLbMVogVj5nJTCNrek8ZsX1-YOe-O5NVI-
5. https://www.youtube.com/watch?v=O2AM8jPomDo&list=PLfn0GFFdt-vkIe2mCL87cx5csL0-LUG7i

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	1	1	2	3	-	1	-	2	-	1	1	3	1	1
CO2	3	1	1	2	3	-	1	-	2	-	1	1	3	1	1
CO3	3	3	1	2	3	-	1	-	2	-	1	1	3	1	1
CO4	3	2	1	2	3	-	1	-	1	-	1	1	3	1	1
CO5	3	1	1	2	3	-	1	-	2	-	1	1	3	1	1
Avg	3	2	1	2	3	-	1	-	2	-	1	1	3	1	1

<table>
<tr><td rowspan="2">R 2024</td><td colspan="4" rowspan="2">AERONAUTICAL ENGINEERING</td><td rowspan="2">SEMESTER: IV</td></tr>
<tr></tr>
<tr><td rowspan="2">24AE406</td><td rowspan="2">AERODYNAMICS LAB</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">PC</td></tr>
<tr><td>0</td><td>0</td><td>4</td><td>2</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To analyze the pressure distribution and aerodynamic behavior of airfoils and bluff bodies subjected to airflow.
2. To evaluate the aerodynamic forces and moments on an airfoil at varying angles of attack using a wind tunnel balance system.
3. To observe and interpret flow patterns around objects using different flow visualization techniques.

COURSE OUTCOMES:

At the end of this course, students can be able to

CO1: To calibrate and operate a subsonic wind tunnel for aerodynamic testing applications.

CO2: To analyse and interpret lift characteristics of different airfoil sections using experimental methods.

CO3: To evaluate pressure distribution over various aerodynamic and bluff body shapes under subsonic flow conditions.

CO4: To conduct flow visualization experiments to study flow patterns over airfoils, cylinders, and bluff bodies

CO5: To correlate theoretical fluid dynamics concepts to understand real-world aerodynamic behaviour.

PRACTICAL EXERCISES:

1. Calibration of a subsonic Wind tunnel.
2. Determination of lift for the given airfoil section.
3. Pressure distribution over a smooth / rough circular cylinder.
4. Pressure distribution over a symmetric aerofoil
5. Pressure distribution over a cambered aerofoil.
6. Flow over a flat plate at different angles of incidence.
7. Flow visualization studies in low-speed flows over cylinders.
8. Flow visualization studies in low-speed flows over airfoil with different angle of incidence.
9. Flow visualization on bluff bodies using water flow channel
10. Flow visualization using Hele-shaw apparatus.

Total Periods: 60

LIST OF EQUIPMENTS:

1.	Subsonic Wind tunnel	:	1 Nos
2.	Models (Rough and Smooth cylinder Symmetric Aerofoil, Cambered Aerofoil and thin Aerofoil) with pressure tapings	:	1 Each
3.	Multi Tube Manometer	:	1 Nos
4.	Pitot-Static Tubes	:	1 Nos
5.	Flat plate	:	1 Nos
6.	Wind Tunnel balances (3 or 6 components)	:	1 Nos
7.	Smoke Generator	:	1 Nos
8.	Schlieren system	:	1 Nos
9.	Water flow channel	:	1 Nos
10.	Heleshaw apparatus	:	1 Nos

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	3	2	1	-	-	1	-	2	-	-	1	3	2	3
CO2	3	3	2	1	-	-	1	-	2	-	-	1	3	2	3
CO3	3	3	2	1	-	1	1	-	2	-	-	1	3	2	3
CO4	3	3	2	1	-	1	1	-	2	-	-	1	3	2	3
CO5	3	3	2	1	-	1	1	-	2	-	-	1	3	2	3
Avg	3	3	2	1	-	1	1	-	2	-	-	1	3	2	3

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: IV
24AE407	AERO ENGINES & AERO FRAME LAB	L	T	P	C	PC
		0	0	4	2	

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To gain an understanding of maintenance and repair procedures involved in the overhaul of aero engines.
2. To develop knowledge and skills in preparing glass epoxy composite laminates and test specimens.
3. To learn the fundamentals of welding techniques and sheet metal repair processes

COURSE OUTCOMES:

At the end of this course, students can be able to

CO1: To gain ability to dismantle and reassemble the aero engines and to perform maintenance activities

CO2: To learn and execute rivetted and welded joints

CO3: To fabricate given models in sheet metal and perform patch repair

CO4: To analyze welded joint defects using Dye penetrant test

CO5: To perform tube bending and flaring

PRACTICAL EXERCISES:

1. Dismantling and Assembling of an aircraft piston engine.
2. Dimensional Inspection of Aero Engine Components using Precision Measuring Instruments
3. Valve Timing diagrams.
4. Preparation of Riveted Lap joint / Butt joint of Aluminium Plates
5. Preparation of Welded Lap joint of MS Plates
6. Sheet metal forming for the given shape.
7. Sheet metal - Riveted Patch Repair.
8. Dye penetrant test - NDT
9. Tube bending and flaring

Total Periods: 60

LIST OF EQUIPMENTS:

1. Aircraft Piston Engine	:	1 Nos
2. Set of basic tools for dismantling and assembly	:	1 Each
3. Liquid Penetrant Testing Kit	:	1 Nos
4. Micrometers, Depth gauges, Vernier Calipers	:	1 Nos
5. Valve timing Disc	:	1 Nos
6. Shear cutter pedestal type	:	1 Nos
7. Drilling Machine	:	1 Nos
8. Bench Vice	:	1 Nos
9. Radius Bend bars	:	1 Nos
10. Pipe Flaring Tools	:	1 Nos
11. Welding Machine	:	1 Nos

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	3	1	1	-	1	2	-	2	-	-	1	2	2	3
CO2	2	3	1	1	-	1	2	-	2	-	-	1	2	2	3
CO3	2	3	1	1	-	1	2	-	1	-	-	1	1	2	3
CO4	3	3	1	1	-	1	2	-	1	-	-	2	2	2	3
CO5	3	3	1	1	-	1	2	-	2	-	-	1	2	2	3
Avg	3	3	1	1	-	1	2	-	2	-	-	1	2	2	3

M.A.M SCHOOL OF ENGINEERING

(An Autonomous Institution)

(Accredited by NAAC || Approved by AICTE || Affiliated to Anna University)

Trichy – Chennai Trunk Road, Siruganur, Tiruchirappalli – 621 105

B.E. AERONAUTICAL ENGINEERING

SEMESTER V

SEMESTER V										
S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1	24AE501	Flight Dynamics	3	1	0	4	40	60	100	PC
2	24AE502	Rocket Propulsion	3	0	0	3	40	60	100	PC
3	-	Professional Elective - I	3	0	0	3	40	60	100	PE
4	-	Professional Elective - II	3	0	0	3	40	60	100	PE
5	-	Open Elective - I	3	0	0	3	40	60	100	OE
THEORY COURSE WITH LABORATORY COMPONENT										
6	24AE503	Finite Element Analysis	3	0	2	4	50	50	100	PC
LABORATORY COURSES										
7	24AE504	Aircraft Systems Lab	0	0	4	2	60	40	100	PC
8	24AE505	CAD Modelling Lab	0	0	2	1	60	40	100	PC
9	24TP501	Aptitude Skills IV & Technical Skills II	0	0	2	1	60	40	100	EEC
		TOTAL	18	1	10	24				

REGULATIONS 2024

Recommended by 3rd BOS held on 12.02.2026 and approved by 3rd Academic Council held on 17.03.2026

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AE501	FLIGHT DYNAMICS	L	T	P	C	PC
		3	1	0	4	

PRE-REQUISITES:

Basics of engineering mathematics, physics, engineering mechanics, aerodynamics and aircraft fundamentals.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand the forces, moments, and equations governing steady and maneuvering flight performance of fixed-wing aircraft.
2. To analyze aircraft performance parameters such as drag, thrust, power, range, endurance, climb, glide, take-off, and landing.
3. To evaluate static longitudinal, lateral, and directional stability characteristics and their dependence on aircraft configuration and center of gravity.
4. To examine the effectiveness of aircraft control surfaces and their influence on stability, handling qualities, and maneuvering limits.
5. To explain dynamic stability modes of aircraft and assess longitudinal and lateral-directional motion behavior during disturbances.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Analyze forces, moments, and equations governing steady and maneuvering aircraft flight conditions.

CO2: Evaluate aircraft performance including drag, thrust, power, range, endurance, climb, glide characteristics.

CO3: Determine static stability and assess effects of center of gravity, power, controls.

CO4: Explain lateral-directional stability and assess aircraft response to control inputs, asymmetry effects.

CO5: Identify and interpret dynamic stability modes and predict aircraft response to disturbances.

UNIT: I	STEADY FLIGHT PERFORMANCE	9+6

Forces and moments acting on an aircraft, Rigid-body equations of motion, Aerodynamic drag components and drag polar. Estimation of parasite drag coefficient using equivalent flat-plate area method. Variation of thrust and power with velocity and altitude for air-breathing engines. Performance in steady level flight. Power available and power required curves. Maximum level flight speed. Conditions for minimum drag and minimum power required.

UNIT: II	AIRCRAFT PERFORMANCE IN MANEUVERS	9+6

Range and endurance. Climb and glide performance: rate of climb, climb angle, rate of sink, glide angle. Take-off and landing performance. Turning flight performance: turn radius and turn rate. Bank angle and load factor. Limitations on turning performance. V–n diagram and operational load factor constraints.

UNIT: III	STATIC LONGITUDINAL STABILITY AND CONTROL	9+6

Degrees of freedom of aircraft motion. Concept of static and dynamic stability. Purpose of aircraft controls. Inherently stable, marginally stable aircraft. Static longitudinal stability criteria. Stick-fixed stability and neutral point. Effect of center of gravity location. Stick-free stability, and stick-free neutral point. Stick force gradients and maneuver stability. Aerodynamic balancing of control surfaces.

UNIT: IV	LATERAL AND DIRECTIONAL STABILITY	9+6

Dihedral effect and lateral stability. Lateral control and roll–yaw coupling. Adverse yaw and aileron effectiveness. Aileron reversal. Static directional stability and weathercock stability. Rudder control and requirements. One-engine-inoperative (OEI) condition and rudder limitations. Rudder lock considerations.

UNIT: V	DYNAMIC STABILITY OF AIRCRAFT	9+6

Introduction to dynamic stability. Longitudinal dynamic modes: short-period and phugoid. Effect of freeing the control stick on stability. Overview of lateral and directional dynamic modes. Dutch roll, spiral mode, and spiral divergence. Autorotation and spin: basic concepts and recovery considerations

Total Periods: 75

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of Publication**
1	Nelson R.C	Flight Stability and Automatic Control	McGraw-Hill Book Co	2018
2	Perkins, C.D., and Hage, R.E.	Airplane Performance stability and Control	John Wiley &Son	2020
3	Michael Cook	Flight Dynamics Principles	Elsevier	2021

REFERENCE BOOKS:				
1	Etkin, B.	Dynamics of Flight Stability and Control	John Wiley	2019
2	Mc Cornick B. W	Aerodynamics, Aeronautics and Flight Mechanics	John Wiley	2021

WEB LEARNING RESOURCES:

1. https://nptel.ac.in/courses/101106041
2. https://ocw.mit.edu/courses/16-333-aircraft-stability-and-control-fall-2004/
3. https://www.grc.nasa.gov/www/k-12/airplane/
4. https://www.faa.gov/regulations_policies/handbooks_manuals/aviation/phak
5. https://www.aiaa.org/education/online-learning

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	2	1	1	3	-	-	1	3	1	2
CO2	3	2	2	1	1	1	1	2	-	-	1	3	1	1
CO3	3	2	1	-	1	-	-	2	-	-	-	3	2	1
CO4	3	2	2	1	1	-	-	1	-	-	-	3	1	1
CO5	3	2	1	1	2	-	-	1	-	-	1	3	1	1
Avg	**3**	**2**	**1.6**	**1**	**1.4**	**1**	**1**	**1.8**	**-**	**-**	**1**	**3**	**1.2**	**1.2**

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AE502	ROCKET PROPULSION	L	T	P	C	PC
		3	0	0	3	

PRE-REQUISITES:

Basics of Thermodynamics, Fluid mechanics, air breathing propulsion and combustion.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand principles and classification of space propulsion systems and rocket fundamentals.
2. To analyze performance parameters governing solid, liquid, and hybrid rocket propulsion systems.
3. To study combustion, feed systems, and design aspects of chemical rocket engines.
4. To evaluate hybrid rocket propulsion characteristics and compare with conventional rocket systems.
5. To introduce advanced and electric propulsion systems for future space mission applications.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain fundamental principles and classify space propulsion systems including chemical rockets.

CO2: Analyze solid rocket propulsion including propellants, combustion, grain design, and performance.

CO3: Evaluate liquid rocket engines focusing on feed systems, cooling, and thrust chambers.

CO4: Assess hybrid rocket propulsion performance and compare with solid and liquid rockets.

CO5: Interpret advanced propulsion technologies and evaluate their potential space mission applications

UNIT: I	INTRODUCTION TO SPACE PROPULSION SYSTEMS	9

Classification of propulsion systems, Basic principles of rocket propulsion, Rocket momentum equation, Thrust, specific impulse, and characteristic velocity, Effective exhaust velocity, Thrust Augmentation and losses, Comparison of rockets and air-breathing engines - Rocket Safety Handling and Launch Vehicle Failure Case Studies

UNIT: II	SOLID ROCKET PROPULSION	9

Selection criteria of solid propellants, Types of solid propellants, Solid propellant regression rate, Solid propellant grain configurations, Basics of solid propellant combustion, Erosive burning, Solid rocket motor components, Ignition systems and thrust termination

UNIT: III	LIQUID ROCKET PROPULSION	9

Liquid propellant classification, Pressure-fed and pump-fed Feed systems, Liquid rocket injectors and water testing, Liquid rocket cooling methods, Basic aspects of thrust chamber design, Advantages of liquid rockets over solid rockets, Cryogenic rocket engines, Propellant slosh

UNIT: IV	HYBRID ROCKET PROPULSION	9

Standard and reverse hybrid systems, Combustion mechanism in hybrid rockets, Material combinations for hybrid rocket propellants- Estimation of hybrid rocket performance – Applications and operational challenges – Functions and types of rocket nozzles - Contour design of rocket nozzles

UNIT: V	ADVANCED PROPULSION SYSTEMS	9

Electric rocket propulsion, Types of electric propulsion techniques, Ion propulsion, Nuclear rocket, comparison of performance of these propulsion systems with chemical rocket propulsion systems, Future applications of electric propulsion systems, Case Studies of advanced propulsion projects worldwide.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Ramamurthi K,	Rocket Propulsion	Macmillian publishers India Ltd	2021
2	Mathur, M.L. and Sharma, R.P	Gas Turbine, Jet and Rocket Propulsion	Standard Publishers	2017

3	David H. Heiser and David T. Pratt.,	Hypersonic Air breathing Propulsion	AIAA Education Series	2022
REFERENCE BOOKS:				
1	Sutton, G.P	Rocket Propulsion Elements	John Wiley & Sons Inc.,	2018
2	Robert G. Jahn,	Physics of Electric Propulsion	Dover Publications	2022
WEB LEARNING RESOURCES:				

1. https://elearn.nptel.ac.in/shop/nptel/rocket-propulsion/
2. https://www.nptelprep.in/courses/101104078/materials
3. https://www.grc.nasa.gov/www/k-12/rocket/
4. https://ocw.mit.edu
5. https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/19710019929.pdf

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	2	1	-	-	-	1	1	2	1	1
CO2	3	2	1	1	2	1	-	-	-	1	1	2	1	1
CO3	3	2	2	1	2	-	-	-	-	1	-	2	1	1
CO4	3	2	1	1	2	1	-	-	-	1	1	2	1	1
CO5	2	2	1	1	2	1	-	-	-	1	1	2	1	1
Avg	**2.8**	**2**	**1.4**	**1**	**2**	**1**	**-**	**-**	**-**	**1**	**1**	**2**	**1**	**1**

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AE503	FINITE ELEMENT ANALYSIS	L	T	P	C	PC
		3	0	0	3	

PRE-REQUISITES:

Knowledge of matrix methods, mechanics of solids, numerical methods, & basic programming.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand approximate numerical methods applied to structural mechanics problems
2. To formulate finite element models using discrete elements for structures and analysis.
3. To analyze continuum problems using plane stress strain axisymmetric elements in mechanics.
4. To develop iso-parametric elements & compute matrices using numerical integration techniques efficiently.
5. To solve field problems and understand solution techniques software limitations and errors.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Apply variational & weighted residual methods to structural mechanics problems effectively

CO2: Model and analyze truss beam frame elements under mechanical thermal loads conditions.

CO3: Analyze plane stress strain and axisymmetric problems using continuum elements accurately efficiently.

CO4: Formulate iso-parametric finite elements & evaluate matrices using numerical integration methods.

CO5: Solve field problems and assess FEM solutions software features errors critically systematically.

UNIT: I	FUNDAMENTAL CONCEPTS	9

Review of various approximate methods – variational approach and weighted residual approach- application to structural mechanics problems. finite difference methods- governing equation and convergence criteria of finite element method.

UNIT: II	DISCRETE ELEMENTS	9

Bar elements, uniform section, mechanical and thermal loading, varying section, 2D and 3D truss element. Beam element - problems for various loadings and boundary conditions – 2D and 3D Frame elements - longitudinal and lateral vibration. Use of local and natural coordinates.

UNIT: III	CONTINUUM ELEMENTS	9

Plane stress, plane strain and axisymmetric problems. Derivation of element matrices for constant and linear strain triangular elements and axisymmetric element.

UNIT: IV	ISOPARAMETRIC ELEMENTS	9

Definitions, Shape function for 4, 8 and 9 nodal quadrilateral elements, stiffness matrix and consistent load vector, evaluation of element matrices using numerical integration.

UNIT: V	FIELD PROBLEM AND METHODS OF SOLUTIONS	9

Heat transfer problems, steady state fin problems, derivation of element matrices for two dimensional problems, torsion problems. bandwidth- elimination method and method of factorization for solving simultaneous algebraic equations – Features of software packages, sources of error.

Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

Practical Exercises: **Periods: 30**

1. Simulation of flow over an aero foil
2. Internal flow simulation of subsonic and supersonic flow through a CD nozzle
3. Structural analysis of bar and beam
4. Structural analysis of truss.
5. Structural analysis of fuselage structure.
6. Heat transfer analysis of structures

Total Periods: 75

TEXT BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of Publication**
1	Reddy J.N.	An Introduction to Finite Element Method	McGraw Hill	2019
2	Seshu, P,	Text Book of Finite Element Analysis	Prentice-Hall of India	2020
3	Rao, S.S	The Finite Element Method in Engineering	Butterworth Heinemann	2018
REFERENCE BOOKS:				
1	Krishnamurthy, C.S	Finite Element Analysis	Tata McGraw Hill	2021
2	Tirupathi.R, Chandrapatha and Ashok D. Belegundu	Introduction to Finite Elements in Engineering	Prentice-Hall of India	2020

WEB LEARNING RESOURCES:

1. https://nptel.ac.in/courses/105106133
2. https://ocw.mit.edu/courses/2-092-finite-element-procedures-for-solids-and-structures-fall-2009/
3. https://www.coursera.org/learn/finite-element-method
4. https://www.simscale.com/docs/tutorials/
5. https://support.3ds.com/education

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	2	1	1	2	-	1	1	3	1	2
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CO3	3	2	2	1	1	1	-	2	-	-	-	2	2	2
CO4	3	2	1	1	1	-	-	1	-	-	-	2	2	1
CO5	3	2	2	1	2	1	-	1	-	1	1	3	1	1
Avg	**3**	**2**	**1.8**	**1**	**1.6**	**1**	**1**	**1.4**	**-**	**1**	**1**	**2.6**	**1.6**	**1.4**

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AE504	AIRCRAFT SYSTEMS LABORATORY	L	T	P	C	PC
		0	0	4	2	

PRE-REQUISITES

Basic knowledge of aircraft systems, instrumentation, aerodynamics, electrical circuits, and safety procedures

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To familiarize students with standard aircraft maintenance, inspection, and ground handling procedures.
2. To understand aircraft jacking, levelling, symmetry, and weighing techniques safely accurately.
3. To perform control system rigging checks and functional testing of aircraft systems.
4. To evaluate hydraulic and fuel system health using flow and pressure testing methods.
5. To apply standard testing procedures for aircraft braking and wheel systems.

COURSE OUTCOMES:

CO1: Perform aircraft jacking, levelling, symmetry checks, and weighing procedures safely.

CO2: Conduct control system rigging checks and verify correct control surface operation.

CO3: Assess hydraulic system condition using flow, pressure, and functional testing techniques.

CO4: Evaluate fuel system components through pressure testing and leakage assessment methods.

CO5: Analyze brake torque load test results to ensure wheel brake system performance.

PRACTICAL EXERCISES:

1. Aircraft “Jacking Up” procedure
2. Aircraft “Levelling” procedure
3. Control System “Rigging check” procedure
4. Aircraft “Symmetry Check” procedure
5. “Flow test” to assess of filter element clogging
6. “Pressure Test” To assess hydraulic Leakage
7. “Functional Test” to adjust operating pressure
8. “Pressure Test” on fuel system components
9. “Brake Torque Load Test” on wheel brake units
10. Aircraft weighing procedure

Total Periods: 60

LIST OF EQUIPMENTS:

1. Serviceable aircraft	:	1 No
2. Hydraulic Jack / Screw Jack	:	5 Nos
3. Trestle adjustable	:	5 Nos
4. Spirit Level	:	2 Nos
5. Levelling Boards	:	2 Nos
6. Cable Tensiometer	:	1 No
7. Plumb Bob	:	1 No
8. Weighing machine	:	1 No

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	3	2	1	1	1	1	2	2	-	1	3	2	3
CO2	3	2	1	1	-	1	1	2	2	-	1	2	2	1
CO3	3	2	1	1	-	1	1	2	2	-	1	2	2	3
CO4	3	2	1	1	1	1	1	2	2	-	1	2	2	2
CO5	3	3	2	1	-	1	1	2	2	-	1	2	2	2
Avg	**3**	**2.4**	**1.4**	**1**	**1**	**1**	**1**	**2**	**2**	**-**	**1**	**2.2**	**2**	**2.2**

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AE505	CAD MODELLING LABORATORY	L	T	P	C	PC
		0	0	2	1	

PRE-REQUISITES:

Basic knowledge of engineering drawing, computer fundamentals, geometry, drafting principles, and CAD concepts.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To develop computer-aided models of aerospace components using appropriate CAD tools.
2. To understand geometric design considerations of aerospace vehicles and propulsion components.
3. To apply computational fluid dynamics techniques for analysing external and internal flows.
4. To perform structural and thermal analysis of aerospace structures using CAE software.
5. To interpret simulation results and validate them against fundamental engineering principles

COURSE OUTCOMES:

CO1: Create accurate CAD models of diffusers, nozzles, wings, fuselages, missiles, and satellites.

CO2: Simulate aerodynamic flow over airfoils and through subsonic and supersonic nozzles.

CO3: Analyze structural behavior of bars, beams, trusses, and fuselage structures.

CO4: Evaluate thermal performance of aerospace structures using heat transfer simulations.

CO5: Assess CFD and structural results for accuracy, convergence, and engineering relevance.

PRACTICAL EXERCISES:

Modelling:

1. Computer aided design of subsonic diffusers.
2. Computer aided design of a Converging-diverging nozzle.
3. Computer aided design of typical aircraft wing.
4. Computer aided design of typical fuselage structure.
5. Computer aided design of a Missiles.
6. Computer aided design of a Satellites

Total Periods: 30

LIST OF EQUIPMENTS:

1. Desktop Computer with accessories	:	30 Nos
2. Modelling and simulation Software – Open Source	:	30 Nos
3. Printer	:	1 No

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	2	2	2	1	1	1	1	-	2	-	1	2	2	2
CO2	2	2	2	1	1	1	1	-	2	-	1	2	2	2
CO3	2	2	2	1	1	1	1	2	2	-	1	2	2	2
CO4	2	2	1	1	1	1	1-	2	2	-	1	2	2	1
CO5	2	2	2	1	1	1	1	2	2	-	1	2	2	2
Avg	**2**	**2**	**1.8**	**1**	**1**	**1**	**1**	**2**	**2**	**-**	**1**	**2**	**2**	**1.8**

M.A.M SCHOOL OF ENGINEERING

(An Autonomous Institution)

(Accredited by NAAC || Approved by AICTE || Affiliated to Anna University)

Trichy – Chennai Trunk Road, Siruganur, Tiruchirappalli – 621 105

B.E. AERONAUTICAL ENGINEERING

SEMESTER VI

S.No	Course Code	Course	L	T	P	C	Maximum Marks			Category
THEORY COURSES										
							CA	ES	Total	
1.	24AE601	Aircraft Design	3	0	0	3	40	60	100	PC
2.	24AE602	Airport Management	3	0	0	3	40	60	100	PC
3.	-	Professional Elective - III	3	0	0	3	40	60	100	PE
4.	-	Professional Elective - IV	3	0	0	3	40	60	100	PE
5.	-	Open Elective - II	3	0	0	3	40	60	100	OE
THEORY COURSES WITH LABORATORY COMPONENT										
6.	24AE603	Basics and Applications of AI & ML in Aeronautical Engg	3	0	2	4	50	50	100	PC
LABORATORY COURSES										
7	24TP601	Aptitude Skills V & Technical Skills III	0	0	2	1	60	40	100	EEC
		TOTAL	18	0	10	20				

REGULATIONS 2024

Recommended by 3rd BOS held on 12.02.2026 and Approved by 3rd Academic Council held on 17.03.2026

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VI
24AE601	AIRCRAFT AND UAV DESIGN	L	T	P	C	PC
		3	0	0	3	

PRE-REQUISITES

Knowledge of aerodynamics, aircraft structures, propulsion systems, flight mechanics, and engineering mathematics.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand fundamental concepts, stages, and considerations involved in aircraft and UAV design.
2. To analyze preliminary aircraft design procedures including mission profile, weight estimation, and loading.
3. To examine design principles of engines, wings, fuselage, and related aerodynamic parameters.
4. To evaluate tail configuration, landing gear design, and control surface stability considerations.
5. To understand UAV system components, propulsion selection, avionics integration, and safe operational guidelines

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain aircraft design stages, configuration factors, manufacturability, maintenance, and operational cost considerations clearly.

CO2: Apply weight estimation methods, mission analysis, wing loading, and thrust loading calculations.

CO3: Analyze engine placement, wing parameters, fuselage loads, and aerodynamic design considerations.

CO4: Design tail surfaces, landing gear systems, and control surfaces considering aircraft stability.

CO5: Identify UAV components, propulsion systems, avionics integration, and apply safe flight guidelines.

UNIT: I	INTRODUCTION TO AIRCRAFT DESIGN	9

Classification of airplanes based on purpose and configuration, Purpose and scope of airplane design, Factors affecting configuration, Merits of different plane layouts. Stages in Airplane design. Designing for manufacturability, Maintenance, Operational costs, State of art in airplane design

UNIT: II	PRELIMINARY AIRCRAFT DESIGN AND WEIGHT ESTIMATION	9

Data collection and 3-view drawings, Weight estimation, Weight equation method, Development & procedures for evaluation of component weights. Mission Profile, Weight fractions for various segments of mission. Choice of wing loading and thrust Loading.

UNIT: III	DESIGN OF AIRCRAFT ENGINE, WINGS AND FUSELAGE	9

Classification of Aircraft Engines, Choices available, Location of power plants, Functions dictating the locations, Engine Installation Process, Trade-off studies and decision-making approach - Aero foil nomenclature, Selection of Wing parameters, Effect of Aspect ratio, V-n diagram, Elements of fuselage design, Loads on fuselage,

UNIT: IV	DESIGN OF TAIL, LANDING GEAR & CONTROL SURFACES	9

Determination of tail surface areas, Various Tail configurations, Structural features, Landing Gear Design, loads on landing gear, Special consideration in configuration lay-out, Control Surfaces. Stability aspects on the design of control surface. Digital Design & CAD/CAE Applications

UNIT: V	DESIGN OF UAV SYSTEMS	9

Introduction to Unmanned Aerial Vehicles (UAVs), Classification and types of UAVs, Mission requirements and system design considerations, Overview of the main drone parts- Technical characteristics of the parts -Function of the component parts - Assembling a drone- Propulsion system selection for UAVs, Avionics and onboard electronics integration - The safety risks- Guidelines to fly safely

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Raymer, D.P.	Aircraft conceptual Design	AIAA series,	2022

2	Torenbeck	Synthesis of Subsonic Airplane Design	Delft University Press	2020
3	John D Anderson	Aircraft Performance & Design	Tata McGraw Hill	2019
REFERENCE BOOKS:				
1	Daniel Tal and John Altschuld	Drone Technology	John Wiley & Sons, Inc.	2021
2	Kuechemann, D	The Aerodynamic Design of Aircraft	AIAA Series	2020
WEB LEARNING RESOURCES:				

1. https://onlinecourses.nptel.ac.in/noc24_ae24/preview
2. https://www.classcentral.com/course/swayam-introduction-to-aircraft-design-19903
3. https://www.nptelprep.in/courses/101101083/videos
4. https://alison.com/course/introduction-to-aircraft-design
5. https://www.sanfoundry.com/aircraft-design-tutorial/

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	1	3	1	2	2	1	1	-	-	1	1	3	1	-
CO2	2	3	2	2	2	1	1	2	-	-	1	3	2	3
CO3	2	3	3	1	2	1	1	2	-	-	1	2	1	3
CO4	2	3	2	1	2	1	-	1	-	-	1	2	1	2
CO5	1	3	2	2	2	1	-	1	-	1	1	3	1	2
Avg	**1.6**	**3**	**2**	**1.6**	**2**	**1**	**1**	**1.5**	**-**	**1**	**1**	**2.6**	**1.2**	**2.5**

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VI
24AE602	AIRPORT MANAGEMENT	L	T	P	C	PC
		3	0	0	3	

PRE-REQUISITES:

Basic knowledge of aviation operations, air transport systems and communication skills.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To acquire solid background of managerial skills in airport management
2. To develop personality to face business difficulties.
3. To control multicultural conditions.
4. To identify the relevant analytical and logical skills to deal with problems in the airline industry.
5. To learn the concepts of performing well in teams, professionalism, and the knowledge acquired in the field of airport planning, airport security, passengers forecasting, aerodromes work etc

COURSE OUTCOMES:

At the end of this course, students can able to

CO1:	Understand structure and evolution of Indian and global aviation industry.
CO2:	Analyze airport infrastructure planning, management models, and privatization approaches.
CO3:	Evaluate air transport services, tariffs, and regulatory frameworks in India.
CO4:	Examine institutional roles of DGCA, AAI, and ATC operations.
CO5:	Assess competition, challenges, and strategic growth opportunities in aviation.

UNIT: I	INTRODUCTION	9

History of aviation - organization, global social & ethical environment - history of aviation in India - major players in the airline industry - swot analysis of the different airline companies in India - market potential of airline industry in India - new airport development plans - current challenges in the airline industry - competition in the airline industry - domestic and international from an Indian perspective

UNIT: II	AIRPORT INFRASTRUCTURE AND MANAGEMENT	9

Airport planning - terminal planning design and operation - airport operations - airport functions - organization structure in an airline - airport authority of India - comparison of global and Indian airport management - role of AAI - airline privatization - full privatization - gradual privatization - partial privatization

UNIT: III	AIR TRANSPORT SERVICES	9

Various airport services - international air transport services - Indian scenario - an overview of airports in Delhi, Mumbai, Hyderabad and Bangalore - the role of private operators - airport development fees, rates, tariffs.

UNIT: IV	INSTITUTIONAL FRAMEWORK	9

Role of DGCA - slot allocation - methodology followed by ATC and DGCA - management of bilaterals - economic regulations

UNIT: V	CONTROLLING	9

Role of air traffic control - airspace and navigational aids - control process - case studies in airline industry - Mumbai Delhi airport privatization - Navi Mumbai airport tendering process - 6 cases in the airline industry

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Graham A	Managing Airports: An International Perspective	Butterworth-Heinemann, Oxford	2011
2	Wells A. 4th Edition, London, 2000.	Airport Planning and Management	McGraw-Hill	2012

REFERENCE BOOKS:

1	Doganis R	The Airport Business	Routledge	2011
2	Alexander T. Wells, Seth Young	Principles of Airport Management,	McGraw-Hill	2007

WEB LEARNING RESOURCES:

1. https://www.coursera.org/learn/airport-management
2. https://www.edx.org/course/aviation-management-basics
3. https://www.iata.org/en/training/career-development/airport-management/
4. https://www.aci.aero/learning-zone/
5. https://www.faa.gov/air_traffic/stakeholder_engagement/airport_safety/airport_management/

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	2	3	1	2	2	1	1	-	-	1	1	3	1	-
CO2	3	3	2	2	3	1	1	1	-	-	1	3	2	-
CO3	2	2	3	2	2	1	-	2	-	-	1	2	1	2
CO4	2	2	1	1	1	1	-	-	-	-	-	3	1	2
CO5	2	3	1	2	2	1	-	1	-	1	1	3	1	2
Avg	**2.2**	**2.6**	**1.6**	**1.8**	**2**	**1**	**1**	**1.3**	**-**	**1**	**1**	**2.8**	**1.2**	**2**

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VI
24AE603	BASICS & APPLICATIONS OF AI & ML IN AERONAUTICAL ENGINEERING	L	T	P	C	PC
		3	0	2	4	

PRE-REQUISITES:

Knowledge of mathematics, statistics, programming basics and aeronautical engineering fundamentals

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To introduce fundamental concepts of artificial intelligence and problem-solving search techniques.
2. To develop understanding of informed search, game theory, and constraint satisfaction problems.
3. To explain core machine learning concepts including learning types, evaluation, and regression models.
4. To provide knowledge of supervised learning algorithms for classification and prediction tasks.
5. To familiarize students with unsupervised learning, clustering techniques, and aerospace applications.

COURSE OUTCOMES:

At the end of this course, students can be able to

CO1: Explain artificial intelligence concepts and apply uninformed and informed search algorithms effectively.

CO2: Formulate problems and solve them using heuristic, adversarial, and CSP approaches.

CO3: Apply fundamental machine learning concepts to classification and regression problems.

CO4: Implement supervised learning algorithms for prediction, classification, and aerospace-related case studies.

CO5: Analyze datasets using unsupervised learning, clustering methods, and aerospace application scenarios.

UNIT: I	FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE	9

Introduction to AI: Definitions, history, applications, Foundations of AI: Intelligent agents, problem-solving agents, Nature of environments, Problem formulation and state space representation, Uninformed Search Algorithms, Breadth-First Search, Depth-First Search, Depth-Limited Search, Uniform-Cost Search / Dijkstra’s Algorithm, Applications of AI in Aeronautical Engineering

UNIT: II	PROBLEM SOLVING WITH SEARCH TECHNIQUES	9

Informed search strategies: Heuristic search, Greedy Best-First Search, A* algorithm, Adversarial search: Minimax algorithm, Alpha-Beta pruning, Game theory basics: Optimal decisions, strategy selection, Constraint Satisfaction Problems (CSP): Examples and backtracking

UNIT: III	FUNDAMENTALS OF MACHINE LEARNING	9

Introduction to Machine Learning: Definitions, classification, regression, Supervised, unsupervised, reinforcement learning, Probability and Linear Algebra essentials, Hypothesis space, inductive bias, Evaluation metrics: Training/test sets, cross-validation, overfitting, underfitting, bias-variance, Regression models: Linear Regression, Logistic Regression

UNIT: IV	SUPERVISED LEARNING	9

Neural Networks: Perceptron, Adaline, Backpropagation network, Decision Trees: Entropy, information gain, Gini impurity for aircraft diagnostics, Rule-based classification: Maintenance scheduling and fault detection, Naïve Bayes classification, Support Vector Machines (SVM), Case studies: UAV flight prediction, sensor data classification

UNIT: V	UNSUPERVISED LEARNING & CLUSTERING	9

Unsupervised Learning – Principle Component Analysis - Neural Network: Fixed Weight Competitive Nets - Kohonen Self-Organizing Feature Maps – Clustering: Definition - Types of Clustering – Hierarchical clustering algorithms – k-means algorithm, Machine Learning applications in aerospace

Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

Practical Exercises: **Periods: 30**

Programs for Problem solving with Search

1. Implement breadth first search
2. Implement depth first search
3. Analysis of breadth first and depth first search in terms of time and space
4. Implement and compare Greedy and A* algorithms.

Supervised Learning

5. Implement the non-parametric locally weighted regression algorithm in order to fit data points. Select appropriate data set for your experiment and draw graphs
6. Write a program to demonstrate the working of the decision tree-based algorithm.
7. Build an artificial neural network by implementing the back propagation algorithm and test the same using appropriate data sets.
8. Write a program to implement the naïve Bayesian classifier.

Unsupervised learning

9. Implementing neural network using self-organizing maps
10. Implementing k-Means algorithm to cluster a set of data.

Total Periods: 75

LIST OF EQUIPMENTS:

1. Desktop Computers with Accessories : 30 Nos
2. C++/JAVA/ Python or appropriate tools : 30 Nos
3. Printer : 1 Nos
4. UPS : 1 Nos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	S. Russell and P. Norvig,	Artificial Intelligence: A Modern Approach	Prentice Hall	2021
2	S.N.Sivanandam and S.N.Deepa	Principles of soft computing	Wiley India.	2020
3	Lyla B. Das, Sudhish N. George	Artificial Intelligence and Machine Learning: Theory and Practice	IK International	2022

REFERENCE BOOKS:

1	C. Muller & Sarah Alpaydin	Introduction to machine learning	MIT press	2020
2	Christopher M. Bishop	Pattern Recognition and Machine Learning	Springer	2021

WEB LEARNING RESOURCES:
1. https://ocw.mit.edu/courses/6-034-artificial-intelligence-fall-2010/ 2. https://ocw.mit.edu/courses/6-036-introduction-to-machine-learning-fall-2020/ 3. https://online.stanford.edu/courses/cs229-machine-learning 4. https://onlinecourses.nptel.ac.in/noc22_cs56/preview

CO – PO – PSO MAPPING

CO	**PO**												**PSO**		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	3	2	3	2	-	1	-	-	-	1	3	3	1	1
CO2	3	3	2	3	2	-	1	-	-	-	1	3	3	1	1
CO3	3	3	2	3	2	-	1	-	-	-	1	3	3	1	1
CO4	3	3	2	3	2	-	1	-	-	-	1	3	3	1	1
CO5	3	3	2	3	2	-	1	-	-	-	1	3	3	1	1
Avg	**3**	**3**	**2**	**3**	**2**	**-**	**1**	**-**	**-**	**-**	**1**	**3**	**3**	**1**	**1**

M.A.M SCHOOL OF ENGINEERING

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B.E. AERONAUTICAL ENGINEERING

SEMESTER VII & VIII

SEMESTER VII										
S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1	24HS701	Universal Human Values	3	0	0	3	40	60	100	PC
2	24AE701	Wind Tunnel Techniques	3	0	0	3	40	60	100	PC
3	-	Professional Elective - V	3	0	0	3	40	60	100	PE
4	-	Open Elective - III	3	0	0	3	40	60	100	OE
LABORATORY COURSES										
5	24TPS05	Internship	0	0	0	2	100	-	100	EEC
6	24AE702	Aircraft & UAV Design Project	0	0	6	3	60	40	100	EEC
		TOTAL	12	0	0	17				

SEMESTER VIII										
S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
LABORATORY COURSES										
1	24AE801	Project Work	0	0	20	10	60	40	100	EEC
		TOTAL	0	0	20	10				

REGULATIONS 2024

Recommended by 3rd BOS held on 12.02.2026 and Approved by 3rd Academic Council held on 17.03.2026

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VII
24HS701	UNIVERSAL HUMAN VALUES	L	T	P	C	HS
		3	0	0	3	

PRE-REQUISITES

NIL

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand the concept, importance and process of value education for holistic human development.
2. To analyze harmony within human being through understanding self, body and their needs.
3. To develop awareness about harmonious relationships in family and society based on trust.
4. To understand interconnectedness and mutual fulfilment among different orders existing in nature.
5. To apply holistic understanding and human values in professional ethics and life practices.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain fundamental concepts of value education and basic human aspirations for happiness.

CO2: Distinguish needs of self and body ensuring harmony and self-regulation.

CO3: Demonstrate understanding of trust, respect, justice in family and societal relationships.

CO4: Interpret harmony, interconnectedness, and mutual fulfilment among four orders of nature.

CO5: Apply human values and ethical principles in professional decisions and societal responsibilities.

UNIT: I	INTRODUCTION TO VALUE EDUCATION	9

Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education) Understanding Value Education, Self-exploration as the Process for Value Education, Continuous Happiness and Prosperity – the Basic Human Aspirations, Happiness and Prosperity – Current Scenario, Method to Fulfil the Basic Human Aspirations

UNIT: II	HARMONY IN THE HUMAN BEING	9

Understanding Human being as the Co-existence of the Self and the Body, Distinguishing between the Needs of the Self and the Body, The Body as an Instrument of the Self, Understanding Harmony in the Self, Harmony of the Self with the Body, Programme to ensure self-regulation and Health

UNIT: III	HARMONY IN THE FAMILY AND SOCIETY	9

Harmony in the Family – the Basic Unit of Human Interaction, 'Trust' – the Foundational Value in Relationship, 'Respect' – as the Right Evaluation, Other Feelings, Justice in Human-toHuman Relationship, Understanding Harmony in the Society, Vision for the Universal Human Order

UNIT: IV	HARMONY IN THE NATURE/EXISTENCE	9

Understanding Harmony in the Nature, Interconnectedness, self-regulation and Mutual Fulfilment among the Four Orders of Nature, Realizing Existence as Co-existence at All Levels, The Holistic Perception of Harmony in Existence

UNIT: V	IMPLICATIONS OF THE HOLISTIC UNDERSTANDING	9

Natural Acceptance of Human Values, Definitiveness of (Ethical) Human Conduct, A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order, Competence in Professional Ethics Holistic Technologies, Production Systems and Management Models-Typical Case Studies, Strategies for Transition towards Value-based Life and Profession

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	R.R Gaur, R Asthana, G P Bagaria	The Textbook A Foundation Course in Human Values and Professional Ethics,	2nd Revised Edition, Excel Books, New Delhi,	2019

2	Premvir Kapoor	Professional Ethics and Human Values	Khanna Book Publishing-	2022
REFERENCE BOOKS:				
1	C. S. G. Krishnamacharyulu, Lalitha Ramakrishnan	Universal Human Values	Himalaya Publishing House	2025
2	N. Raghunathan	Value Education	Sri Ramakrishna Math	2021
WEB LEARNING RESOURCES:				

1. https://fdp-si.aicte-india.org/downloads/uhv_book.pdf
2. https://uhv.org.in
3. https://nptel.ac.in/courses/109104068
4. https://aicte-india.org/education/uhv
5. https://epgp.inflibnet.ac.in/Home/ViewSubject?catid=36

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	2		-	-	1	-	-	2	1	3
CO2	3	2	2	1	2		-	-	1	-	-	2	1	3
CO3	3	2	2	1	2		-	-	1	-	-	2	1	3
CO4	3	2	2	1	2		-	-	1	-	-	2	1	3
CO5	3	2	2	1	2		-	-	1	-	-	2	1	3
Avg	3	2	2	1	2		-	-	1	-	-	2	1	3

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VII
24AE701	WIND TUNNEL TECHNIQUES	L	T	P	C	PC
		3	0	0	3	

PRE-REQUISITES

Knowledge of aerodynamics, fluid mechanics, dimensional analysis, flow measurement techniques, and instrumentation basics.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To learn the Types of low-speed Wind tunnels and non-dimensional numbers with its applications.
2. To learn the Types of high-speed Wind tunnels and with its calibration methods.
3. To Understand the Special Wind tunnels and with its calibration methods with its design methods.
4. To describe flow visualization techniques and data acquisition methods
5. To understand the functions of various instruments associated with wind tunnel

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain the uses of various types of tunnels and its losses

CO2: Experiment with calibration of different types of high speed tunnels

CO3: Make use of various special tunnels and its applications.

CO4: Make use of various measurement techniques of instruments of wind tunnel

CO5: Can use various techniques for aerodynamic data generation

UNIT: I	LOW SPEED WIND TUNNELS	9

Classification –non-dimensional numbers-types of similarities - Layout of open circuit and closed circuit subsonic wind tunnels – design parameters-energy ratio - HP calculations - Calibration methods.

UNIT: II	HIGH SPEED WIND TUNNELS	9

Blow down, in draft and induction tunnel layouts and their design features -Transonic, and supersonic tunnels-peculiar features of these tunnels and operational difficulties - sample design calculations and calibration methods.

UNIT: III	SPECIAL WIND TUNNEL TECHNIQUES	9

Types of Special Wind Tunnels – Hypersonic, Gun and Shock Tunnels – Design features and calibration methods- Intake tests – store carriage and separation tests - wind tunnel model design for these tests

UNIT: IV	WIND TUNNEL INSTRUMENTATION	9

Instrumentation and sensors required for both steady and unsteady measurements – Force measurements using three component and six component balances – calibration of measuring instruments – error estimation and uncertainty analysis.

UNIT: V	FLOW VISUALIZATION and NON-INTRUSIVE FLOW DIAGNOSTICS	9

Smoke and Tuft grid techniques – Dye injection special techniques – Oil flow visualization and PSP techniques - Optical methods of flow visualization – PIV and Laser Doppler techniques – Image processing and data deduction

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	NAL-UNI Lecture Series 12	Experimental Aerodynamics	NAL SP 98	2021
2	Rae, W.H. and Pope, A	Low Speed Wind Tunnel Testing	John Wiley Publication.	2019

REFERENCE BOOKS:				
1	Pope, A., and Goin, L	High Speed Wind Tunnel Testing	John Wiley	2021
2	Rathakrishnan, E.	Instrumentation, Measurements, and Experiments in Fluids	CRC Press – Taylor & Francis	2022

WEB LEARNING RESOURCES:

1. https://innovationspace.ansys.com/product/wind-tunnels-and-aerodynamic-studies/
2. https://innovationspace.ansys.com/product/electronic-transducers-in-wind-tunnels/
3. https://www.grc.nasa.gov/WWW/k-12/airplane/bgt.html
4. https://www.grc.nasa.gov/www/k-12/TunnelSim/index.htm
5. https://courseware.cutm.ac.in/courses/experimental-aerodynamics/

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	2	3	1	2	2	1	1	-	-	1	1	3	1	-
CO2	2	3	2	2	1	1	1	2	-	-	1	2	2	2
CO3	2	3	3	2	2	1	-	2	-	-	1	2	2	3
CO4	2	3	2	1	2	1	1	-	-	1	1	2	1	2
CO5	1	3	1	2	2	1	-	1	-	1	1	3	1	2
Avg	1.8	3	1.8	1.8	1.8	1	1	1.6	-	1	1	2.4	1.4	2.2

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VII
24AE702	AIRCRAFT & UAV DESIGN PROJECT	L	T	P	C	PC
		0	0	6	3	

PRE-REQUISITES:

Knowledge of aerodynamics, structures, propulsion, flight mechanics, aircraft design principles, and CAD.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To make the student work in groups and effectively improve their team work.
2. To understand the Concepts involved in Aerodynamic design, Performance analysis and stability aspects of different types of airplanes
3. To carry out the structural design part of the airplane / UAV

COURSE OUTCOMES:

At the end of this course, students can be able to

CO1: Analyze aircraft configurations, specifications, and performance to support preliminary aerodynamic design decisions.

CO2: Estimate aircraft weights, select design parameters, powerplant, aerofoil, geometry, and landing gear.

CO3: Perform aerodynamic analyses including drag estimation, calculations, stability assessment, and V-n diagrams.

CO4: Design aircraft wing and fuselage structures using load distributions, stresses, shear, buckling.

CO5: Prepare detailed CAD-based design reports integrating aerodynamic, structural, control surface analyses results.

DESCRIPTION:

1. **Comparative study of aircraft & UAV types** considering specifications, performance characteristics, and relevance to the selected design case.
2. **Preliminary aircraft design activities** including weight estimation, selection of design parameters, powerplant, aerofoil, wing, tail, control surfaces, and landing gear.
3. **Aircraft/ UAV layout development** involving preparation of layout drawings, balance diagrams, and three-view drawings of the selected airplane.
4. **Aerodynamic analysis** covering drag estimation, performance calculations, stability analysis, and construction of V–n diagram.
5. **Preliminary wing structural design** including Schrenk's curve, load distribution, and shear force, bending moment, and torque diagrams.
6. **Detailed wing structural design** involving spar and stringer design, bending stress, shear flow, and buckling analysis of wing panels.
7. **Fuselage and control surface structural design** including fuselage load distribution, design of bulkheads and longerons, stress, shear flow, buckling analysis, control surface loads, wing-root attachment, and landing gear design.
8. **Preparation of comprehensive design documentation** including detailed calculations, structural drawings, and CAD-based design report.

Total Periods: 60

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	1	1	1	2	2	-	1	3	2	3
CO2	3	2	1	1	-	2	1	2	2	-	1	2	1	1
CO3	3	2	2	1	-	1	1	2	2	-	1	2	1	3
CO4	3	2	2	1	1	1	1	2	2	-	1	2	2	1
CO5	3	2	2	1	-	1	1	2	2	-	1	2	2	2
Avg	**3**	**2**	**1.8**	**1**	**1**	**1.2**	**1**	**2**	**2**	**-**	**1**	**2.2**	**1.6**	**2**

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VIII
24AE801	**PROJECT WORK**	**L**	**T**	**P**	**C**	**PC**
		0	**0**	**6**	**3**	

PRE-REQUISITES:

Knowledge of aerodynamics, structures, propulsion, flight mechanics, aircraft design principles, and CAD.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To develop problem-solving skills through aeronautical engineering projects.
2. To apply theoretical knowledge to practical aerospace design challenges.
3. To enhance teamwork, communication, and project management abilities.

COURSE OUTCOMES:

At the end of this course, students can be able to

CO1: Identify and formulate real-world aeronautical engineering design problems effectively.

CO2: Apply modern engineering tools for analysis and project development.

CO3: Design, analyze, and validate aerospace systems meeting performance requirements.

CO4: Demonstrate teamwork, documentation skills, and effective technical presentation abilities.

CO5: Develop innovative solutions considering ethical and professional responsibilities.

DESCRIPTION:

1. Students shall undertake the project in groups of **3–4 members** under the guidance of a faculty supervisor with approval from the HoD.
2. The project topic must be relevant to **core aeronautical domains** such as Aerodynamics, Structures, Propulsion, Flight Mechanics, Avionics, UAVs, or related emerging technologies.
3. The work may be **design-based, experimental, analytical, computational, or industry-oriented**, focusing on practical problem-solving.
4. A **detailed literature review, clear objectives, and defined methodology** must be finalized before project execution.
5. The project progress shall be monitored through **a minimum of three internal reviews** conducted by a committee constituted by the HoD.
6. Students must maintain proper **documentation and a project logbook** throughout the project period.
7. Use of **modern engineering tools and software** (e.g., CFD, FEA, CAD, MATLAB, testing facilities) is encouraged.
8. A **comprehensive project report** shall be submitted at the end of the semester as per prescribed guidelines.
9. Final evaluation shall be based on **project quality, innovation, report submission, and oral presentation/viva-voce**.
10. The viva-voce examination shall be conducted jointly by **internal and external examiners**, and originality of work must be ensured.

Total Periods: 300

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	3	1	1	1	1	2	2	-	1	2	2	3
CO2	3	2	2	1	1	-	1	2	2	-	1	2	2	2
CO3	3	2	2	1	-	1	1	2	2	-	1	2	1	3
CO4	3	2	2	1	1	1	1	2	2	-	1	2	2	1
CO5	3	3	2	1	-	1	1	2	2	-	1	2	2	2
Avg	**3**	**2.2**	**2.2**	**1**	**1**	**1**	**1**	**2**	**2**	**-**	**1**	**2**	**1.8**	**2.2**

M.A.M SCHOOL OF ENGINEERING

(An Autonomous Institution)

(Accredited by NAAC || Approved by AICTE || Affiliated to Anna University)

Trichy – Chennai Trunk Road, Siruganur, Tiruchirappalli – 621 105

B.E. AERONAUTICAL ENGINEERING

ELECTIVES LIST

VERTICAL I - PROFESSIONAL ELECTIVE I
AERODYNAMICS & FLIGHT MECHANICS

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1	24AEX01	Highspeed Aerodynamics	3	0	0	3	40	60	100	PE
2	24AEX02	Aerodynamic Testing and Instrumentation	3	0	0	3	40	60	100	PE
3	24AEX03	Computational Fluid Dynamics	3	0	0	3	40	60	100	PE
4	24AEX04	Helicopter Theory	3	0	0	3	40	60	100	PE
5	24AEX05	Rockets and Missiles	3	0	0	3	40	60	100	PE
6	24AEX06	Aircraft Stability and Control	3	0	0	3	40	60	100	PE

REGULATIONS 2024

Recommended by 3rd BOS held on 12.02.2026 and Approved by 3rd Academic Council held on 17.03.2026

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AEX01	HIGH SPEED AERODYNAMICS	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Basic aerodynamics, compressible flow concepts, thermodynamics fundamentals, fluid mechanics, mathematics, shock waves understanding.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand fundamental principles governing compressible flow and high-speed aerodynamic phenomena.
2. To analyze shock waves, expansion waves, and related flow relations in supersonic regimes.
3. To develop theoretical understanding of two-dimensional compressible flows using linearized aerodynamic models.
4. To study high-speed flow behavior over airfoils, wings, and aircraft configurations.
5. To introduce experimental facilities and flow visualization techniques for high-speed aerodynamics.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1:	Apply compressible flow equations to analyze one-dimensional and two-dimensional high-speed flows.
CO2:	Evaluate shock and expansion wave characteristics using analytical and graphical methods.
CO3:	Analyze aerodynamic performance of airfoils and wings in transonic supersonic regimes.
CO4:	Assess shock boundary layer interaction and wave drag effects on aircraft performance.
CO5:	Identify appropriate wind tunnel facilities for transonic, supersonic, and hypersonic flow studies.

UNIT: I	PRINCIPLES OF COMPRESSIBLE FLOW	9

Compressibility Effects in Gas Flow, Governing Equations of Compressible Flow, Energy Relations and Bernoulli Equation, Mach Number and Speed of Sound, Area–Velocity–Mach Number Relationships, Mach Cone and Mach Angle, Isentropic Flow Through Variable Area Ducts, Static, Stagnation, and Critical conditions properties

UNIT: II	SHOCK AND EXPANSION WAVES	9

Normal shock relations, Prandtl's relation-Hugoniot equation, Raleigh Supersonic Pitot tube equation, Oblique shocks, θ-β-M relation, Reflection of oblique shocks, Interaction of oblique shock waves, slip line, Rayleigh flow, Fanno flow, Expansion waves, Prandtl-Meyer expansion, Maximum turning angle, Simple and non-simple regions

UNIT: III	TWO-DIMENSIONAL COMPRESSIBLE FLOW	9

Potential equation for 2-dimensional compressible flow, Linearization of potential equation, perturbation potential, Linearized Pressure Coefficient, linearized subsonic flow, Prandtl- Glauert rule, Linearized supersonic flow, Method of characteristics.

UNIT: IV	HIGH-SPEED FLOW OVER AIRCRAFT CONFIGURATIONS	9

Critical Mach number, drag divergence Mach number, Shock Formation and Shock Stall, Supercritical airfoil sections, Transonic Area Rule and Swept Wings, Airfoils for Supersonic Flow, Aerodynamic Characteristics of Supersonic Profiles, Shock-Expansion Theory and Wave Drag, Supersonic Wings and Aircraft Design Considerations

UNIT: V	ANALYSIS OF HIGH-SPEED FLOW REGIMES	9

Shock-Boundary layer interaction, Wind tunnels for transonic, Supersonic and hypersonic flows, shock tube, Gun tunnels, Supersonic flow visualization, Introduction to Hypersonic Flows.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of Publication**
1	Anderson, J.	Modern Compressible Flow: With Historical Perspective	McGraw-Hill	2019
2	SM Yahya	Fundamentals of Compressible Flow	New age Publishers	2020
3	Rathakrishnan. E,	Gas Dynamics	Prentice-Hall of India	2018

REFERENCE BOOKS:				
1	Zucrow, M. J. and Anderson, J. D	Elements of Gas Dynamics	McGraw- Hill &Co	2021
2	Shapiro, A. H	Dynamics and Thermodynamics of Compressible Fluid Flow	Ronald Press,	2002

WEB LEARNING RESOURCES:

1. https://mitocw.ups.edu.ec/courses/aeronautics-and-astronautics/16-120-compressible-flow-spring-2003/
2. https://www.getyoureducation.net/course/fundamentals-of-compressible-flow
3. https://innovationspace.ansys.com/courses/courses/understanding-compressible-flow-dynamics/
4. https://freevideolectures.com/course/3265/high-speed-aero-dynamics
5. https://www.classcentral.com/course/swayam-fundamentals-of-supersonic-and-hypersonic-flow-269708

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	3	1	1	2	1	1	3	-	-	1	3	2	1
CO2	3	2	2	1	2	1	1	3	-	-	1	3	2	1
CO3	3	2	1	-	1	-	1	2	-	-	1	2	1	-
CO4	3	3	2	1	2	1	-	1	-	-	1	2	1	1
CO5	3	1	1	1	3	1	1	1	-	-	1	2	2	1
Avg	3	2.2	1.4	1	2	1	1	2	-	-	1	2.4	1.6	1

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AEX02	AERODYNAMIC TESTING AND INSTRUMENTATION	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Basic aerodynamics, fluid mechanics knowledge, measurement techniques, sensors basics

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand fundamental principles and objectives of aerodynamic testing in various applications.
2. To analyze wind tunnel design, flow quality, and testing facility characteristics.
3. To study pressure, force, and moment measurement techniques in aerodynamic experiments.
4. To explore flow measurement and visualization methods including advanced optical techniques.
5. To learn data acquisition systems, instrumentation, and experimental validation of CFD results.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Apply aerodynamic testing principles to design and conduct accurate flow experiments.

CO2: Evaluate wind tunnel performance and select appropriate facilities for testing purposes.

CO3: Measure aerodynamic forces, moments, and pressures using calibrated instruments effectively.

CO4: Utilize flow visualization and measurement techniques to interpret complex aerodynamic phenomena.

CO5: Implement modern data acquisition systems and instrumentation for reliable experimental analysis.

UNIT: I	BASICS OF AERODYNAMIC TESTING	9

Objectives and scope of aerodynamic testing, Similarity requirements: geometric, kinematic, and dynamic similarity, Dimensional analysis and Buckingham π theorem, Reynolds number and Mach number effects, Model scaling and blockage corrections, Types of aerodynamic tests, Measurement accuracy and uncertainty analysis, Data reduction and presentation techniques

UNIT: II	WIND TUNNELS AND TEST FACILITIES	9

Classification of wind tunnels, Low-speed wind tunnels and components, High-speed wind tunnels: transonic and supersonic, Hypersonic wind tunnels, Open circuit and closed-circuit tunnels, Tunnel flow quality and calibration, Test section design and wall interference, Special facilities: shock tubes and gun tunnels

UNIT: III	PRESSURE, FORCE, AND MOMENT MEASUREMENT	9

Pressure measurement principles, Pressure taps and pressure transducers, Manometers and electronic pressure sensors, Force balance systems and types, Strain gauge balances, Lift, drag, and pitching moment measurements, Calibration of force and pressure instruments, Errors in force and pressure measurements

UNIT: IV	FLOW MEASUREMENT AND VISUALIZATION TECHNIQUES	9

Velocity measurement methods, Pitot-static probe and hot-wire anemometry, Laser Doppler velocimetry, Particle image velocimetry, Smoke, tuft, and oil-flow visualization, Schlieren and shadowgraph techniques, Interferometry methods, Visualization of shock and boundary layer flows

UNIT: V	ADVANCED INSTRUMENTATION	9

Data acquisition systems and signal conditioning, Transducers and sensors in aerodynamic testing, Dynamic pressure and unsteady measurements, High-frequency response instruments, Computerized test control systems, Model support systems and mounting techniques, Experimental validation of CFD results, Recent trends in aerodynamic testing instrumentation

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication

1	William H. Rae Jr., Alan Pope	Low-Speed Wind Tunnel Testing	John Wiley & Sons	2019
2	Colin Britcher & Drew Landman	Wind Tunnel Test Techniques	Prentice Hall	2021
3	E.Radhakrishnan	Instrumentation, Measurements, and Experiments in Fluids	Taylor & Francis	2021
REFERENCE BOOKS:				
1	Bruno Chanetz	Experimental Aerodynamics: An Introductory Guide	Tata McGraw Hill	2020
2	Mrinal Kaushik	Instrumentation and Measurements in Compressible Flows	CRC Press	2023

WEB LEARNING RESOURCES:

1. https://innovationspace.ansys.com/product/wind-tunnels-and-aerodynamic-studies/
2. https://innovationspace.ansys.com/courses/courses/particle-image-velocimetry-techniques/lessons/how-wind-tunnel-and-associated-instrumentation-are-used-for-performing-aerodynamic-studies-lesson-4/
3. https://innovationspace.ansys.com/product/electronic-transducers-in-wind-tunnels/
4. https://innovationspace.ansys.com/product/fluid-dynamics-and-measurement-techniques/
5. https://wizape.com/English/Experimental-Methods-in-Fluid-Dynamics

CO – PO – PSO MAPPING

CO	**PO**											**PSO**		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	2	1	1	2	-	2	1	3	1	2
CO2	3	2	2	-	2	1	1	2	-	-	1	2	2	1
CO3	3	2	2	1	1	1	-	1	-	-	-	2	2	3
CO4	3	2	1	1	1	-	-	1	-	-	-	2	1	1
CO5	3	2	2	1	2	-	-	1	-	1	1	3	2	2
Avg	3	2	1.8	1	1.6	1	1	1.4	-	1.5	1	2.4	1.6	1.8

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AEX03	COMPUTATIONAL FLUID DYNAMICS	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Fluid mechanics fundamentals, numerical methods basics, differential equations knowledge, programming basics, aerodynamics concepts understanding.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand fundamental concepts and governing equations of fluid dynamics numerically.
2. To learn grid generation techniques for accurate computational fluid dynamics simulations.
3. To apply panel methods for solving incompressible and compressible flow problems.
4. To study time-dependent numerical methods for unsteady transonic and heat conduction flows.
5. To implement finite volume techniques for accurate discretization and solution of fluid equations.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Analyze fluid dynamics problems using numerical methods and partial differential equation discretization.

CO2: Generate structured and unstructured grids for complex geometries like aerofoils and nozzles.

CO3: Apply two-dimensional and three-dimensional panel methods to subsonic and supersonic flows.

CO4: Solve unsteady flow and heat conduction problems using explicit and time-split methods.

CO5: Implement finite volume techniques for accurate solution of compressible and viscous flows.

UNIT: I	FUNDAMENTAL CONCEPTS	9

Introduction – Basic Equations of Fluid Dynamics – Mathematical properties of Fluid Dynamics, Equations – Elliptic, Parabolic and Hyperbolic equations – Well posed problems – discretization of partial Differential Equations – Transformations and grids – Explicit finite difference methods of subsonic, supersonic and viscous flows.

UNIT: II	GRID GENERATION	9

Need for grid generation – Various grid generation techniques – Algebraic, conformal and numerical grid generation – importance of grid control functions – boundary point control – orthogonality of grid lines at boundaries – Elliptic grid generation using Laplace's equations for geometries like aerofoil and CD nozzle.

UNIT: III	PANEL METHODS	9

Elements of two and three-dimensional panels, panel singularities – Application of panel methods to incompressible, compressible, subsonic and supersonic flows – Numerical solution of flow over a cylinder using 2D panel methods using both vertex and source panel methods for lifting and non-lifting cases respectively.

UNIT: IV	TIME DEPENDENT METHODS	9

Stability of solution – Explicit methods – Time split methods – Approximate factorization scheme – Unsteady transonic flow around aerofoils – Sometime dependent solutions of gas dynamic problems – Numerical solution of unsteady 2D heat conduction problems using SLOR methods.

UNIT: V	FINITE VOLUME TECHNIQUES	9

Finite Volume Techniques – Cell Centred Formulation – Lax-Vendoroff Time Stepping – Runge- Kutta Time Stepping – Multi-stage Time Stepping – Accuracy – Cell Vertex Formulation – Multistage Time Stepping – FDM-like Finite Volume Techniques – Central and Up-wind Type Discretization – Treatment of Derivatives.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Fletcher, C.A.J	Computational Techniques for Fluid Dynamics	Springer - Verlag	2009

2	Anderson J. D	Fundamentals of Aerodynamics	McGraw-Hill	2020
3	Blazek, J.	Computational Fluid Dynamics: Principles and Applications	Elsevier	2019
REFERENCE BOOKS:				
1	Charles Hirsch	Numerical Computation of Internal and External Flows	Butterworth-Heinemann	2017
2	John F. Wendt	Computational Fluid Dynamics - An Introduction	Springer - Verlag	2014

WEB LEARNING RESOURCES:

1. https://archive.nptel.ac.in/courses/112/107/112107080/
2. https://www.classcentral.com/course/swayam-computational-fluid-dynamics-using-finite-volume-method-19826
3. https://www.classcentral.com/course/youtube-cfd-basics-514402
4. https://www.classcentral.com/course/youtube-introduction-to-cfd-92348
5. https://www.cfd-online.com/Links/modeling_and_numerics.html

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	2	1	-	-	-	1	1	2	1	1
CO2	3	2	1	1	2	1	-	-	-	1	1	2	1	1
CO3	3	2	1	1	2	-	-	-	-	1	1	2	1	1
CO4	2	2	1	1	2	1	-	-	-	1	-	2	1	1
CO5	2	2	1	1	2	1	-	-	-	1	1	2	1	-
Avg	2.6	2	1.2	1	2	1	-	-	-	1	1	2	1	1

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AEX04	HELICOPTER THEORY	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Basic aerodynamics, flight mechanics fundamentals, rotorcraft principles awareness, mathematics basics, aircraft performance knowledge.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand helicopter configuration, rotor systems, and fundamental design parameters.
2. To analyze rotor aerodynamics in hover, forward flight, and critical regimes.
3. To evaluate helicopter power plants and performance characteristics under various operating conditions.
4. To explain helicopter stability, control response, and disturbance effects.
5. To study rotor dynamics, vibration phenomena, and vibration control techniques.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1:	Apply rotor system concepts to analyze helicopter configuration and performance.
CO2:	Analyze aerodynamic behavior of rotor blades in hover and forward flight.
CO3:	Estimate helicopter performance parameters including climb, range, and endurance.
CO4:	Assess stability and control characteristics of helicopters compared to airplanes.
CO5:	Analyze rotor vibrations and propose suitable vibration control and blade design methods.

UNIT: I	HELICOPTER CONFIGURATION AND ROTOR SYSTEMS	9

Introduction to Helicopters: Basic Configuration, Rotor Fundamentals, Rotor Types and Controls, Blade Design Parameters, Performance Losses, Fenestron and NOTAR systems, composite rotor blades

UNIT: II	ROTOR AERODYNAMICS AND FLIGHT REGIMES	9

Aerofoil characteristics in forward flight, Hovering and Vortex ring state, Blade stall, maximum lift of the helicopter calculation of Induced Power, High speed limitations; parasite drag, power loading, ground effect.

UNIT: III	POWER PLANTS AND HELICOPTER PERFORMANCE	9

Helicopter Power Plants, Transmission and Power Matching, Performance Parameters, Rate of climb, best climb speed, service and absolute ceiling. Range and Endurance, Autorotation, Emerging Trends: Hybrid-electric and electric propulsion concepts

UNIT: IV	STABILITY AND CONTROL OF HELICOPTERS	9

Physical description of effects of disturbances, Stick fixed Longitudinal and lateral dynamic stability, lateral stability characteristics, control response. Differences between stability and control of airplane and helicopter.

UNIT: V	ROTOR DYNAMICS AND VIBRATION	9

Rotor Dynamic Modeling, Rigid and articulated blade models., Flapping, lagging (lead–lag), and feathering dynamics. Properties of vibrating systems, sources of rotor vibration. Transmission of vibrations to fuselage and crew. Vibration Control: Dampers, absorbers, and active vibration control methods. In-flight vibration measurement and monitoring. Design considerations, airfoil selection, structural & aerodynamic trade-offs.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Lalit Gupta,	Helicopter Engineering	Himalayan Books	2008

2	John Fay	The Helicopter and How It Flies	Himalayan Books	2018
3	E. Rathakrishnan	Helicopter Aerodynamics	Prentice Hall	2019
REFERENCE BOOKS:				
1	C. Venkatesan	Fundamentals of Helicopter Dynamics	CRC Press	2021
2	Joseph Schafer	Basic Helicopter Maintenance	Jeppesen	2020

WEB LEARNING RESOURCES:

1. https://www.nptelprep.in/courses/101104017
2. https://freevideolectures.com/course/3128/introduction-to-helicopter-aerodynamics-and-dynamics
3. https://kumar-sumeet.github.io/HeliAeroNotes/Lectures/0_CourseOverview.html
4. https://aiaa.org/courses/rotorcraft-and-propeller-aerodynamics-and-aeroacoustics/
5. https://www.brainscape.com/subjects/rotorcraft

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	2	1	-	-	-	1	1	2	1	1
CO2	3	2	1	1	2	1	-	-	-	1	1	2	1	1
CO3	3	2	1	1	2	-	-	-	-	1	-	2	1	1
CO4	2	2	1	1	2	1	-	-	-	1	1	2	1	1
CO5	2	1	1	1	2	1	-	-	-	1	1	2	1	-
Avg	2.6	1.8	1.2	1	2	1	-	-	-	1	1	2	1	1

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AEX05	ROCKETS AND MISSILES	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Basic aerodynamics, propulsion fundamentals, thermodynamics concepts, orbital mechanics basics, mathematics and physics understanding.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand the history, classification, and current status of rockets and missiles.
2. To analyze rocket flight mechanics in free space and gravitational fields.
3. To study aerodynamics of rockets and missiles including forces and moments.
4. To examine rocket control systems, multi-staging, and thrust vectoring methods.
5. To explore rocket propulsion systems, combustion chambers, propellant systems, and material requirements.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Classify rockets and missiles and explain their history and Indian programs.

CO2: Determine trajectories, burnout velocity, range, and altitude for various rocket motions.

CO3: Calculate aerodynamic forces, moments, drag, lift, and damping for missile analysis.

CO4: Design and analyze rocket control systems, stage separation, and thrust vectoring.

CO5: Understand propulsion system components, ignition, propellant feed, and material performance requirements.

UNIT: I	OVERVIEW OF ROCKET AND MISSILE SYSTEMS	9

History of rockets and missiles, Classification of rockets and missiles, Basic aerodynamic characteristics of surface to surface, surface to air, air to surface and air to air missiles – Examples of various Indian space launch vehicles and missiles – Current status of Indian rocket and missile Programmes

UNIT: II	ROCKET FLIGHT MECHANICS	9

One Dimensional and Two Dimensional rocket Motions in Free Space and Homogeneous Gravitational Fields – description of Vertical, Inclined and Gravity Turn Trajectories – Determination of range and Altitude, Simple Approximations to Burnout Velocity and altitude-estimation of culmination time and altitude.

UNIT: III	AERODYNAMICS OF ROCKETS	9

Airframe Components of Rockets and Missiles – Forces Acting on a Missile While Passing Through Atmosphere – Classification of Missiles – methods of Describing Aerodynamic Forces and Moments – Lateral Aerodynamic Moment – Lateral Damping Moment and Longitudinal Moment of a Rocket – lift and Drag Forces – Drag Estimation.

UNIT: IV	ROCKET CONTROL SYSTEMS	9

Multi-staging of rockets and ballistic missiles – Multistage Vehicle Optimization – Stage Separation Dynamics – Stage Separation Techniques in atmosphere and in space, Various types of aerodynamic control methods for tactical and short range missiles, Various types of rocket thrust vector control methods.

UNIT: V	ROCKET PROPULSION SYSTEMS	9

Ignition System in rockets, Design Consideration of liquid Rocket Combustion Chamber, Injector Propellant Feed Lines, Valves, Propellant Tanks Outlet and propellant feed Systems – Propellant Slash and Propellant Hammer – Elimination of Geysering Effect in Missiles – Selection of Materials – Special Requirements of Materials to Perform under Adverse Conditions.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:				
Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	George P. Sutton & Oscar Biblarz	Rocket Propulsion Elements	John Wiley & Sons	2022
2	Mathur, M., and Sharma, R.P	Gas Turbines and Jet and Rocket Propulsion	Standard Publishers	2018
3	Don Edberg & Willie Costa	Design of Rockets and Space Launch Vehicles	AIAA	2020
REFERENCE BOOKS:				
1	Cornelisse, J.W	Rocket Propulsion and Space Dynamics	Freeman & Co.	2021
2	Parker, E.R.,	Materials for Missiles and Spacecraft	McGraw-Hill	2019

WEB LEARNING RESOURCES:

1. https://nptel.ac.in/courses/101104016
2. https://www.coursera.org/learn/space-missions
3. https://www.freevideolectures.com/course/3376/rockets-and-missiles
4. https://link.springer.com/book/10.1007/978-981-97-5644-5
5. https://www.aiaa.org/courses/rocket-propulsion-and-missile-system

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	2	1	-	-	-	1	1	2	1	1
CO2	3	2	1	1	2	1	-	-	-	1	1	2	1	1
CO3	3	2	2	1	2	-	-	-	-	1	1	2	1	1
CO4	2	2	1	1	2	1	-	-	-	1	1	2	1	1
CO5	2	2	2	1	2	1	-	-	-	1	1	2	1	1
Avg	2.6	2	1.6	1	2	1	-	-	-	1	1	2	1	1

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AEX06	AIRCRAFT STABILITY AND CONTROL	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Flight mechanics fundamentals, aerodynamics basics, dynamics principles, aircraft performance knowledge, mathematics understanding.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand fundamental concepts of aircraft stability, equilibrium conditions, and aerodynamic moments during flight.
2. To analyze longitudinal stability characteristics and control effectiveness of elevator and horizontal tail.
3. To study lateral and directional stability including roll, yaw coupling and control surfaces.
4. To examine dynamic stability modes and aircraft response to small disturbances.
5. To understand stability augmentation systems and modern automatic flight control technologies.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain fundamental concepts of aircraft stability, equilibrium conditions, and aerodynamic moment characteristics clearly.

CO2: Analyze longitudinal stability parameters and determine trim conditions for steady flight operations.

CO3: Evaluate lateral and directional stability characteristics including roll, yaw coupling effects accurately.

CO4: Interpret aircraft dynamic stability modes such as phugoid, dutch roll and spiral motion.

CO5: Describe stability augmentation systems and basic principles of automatic aircraft flight control.

UNIT: I	INTRODUCTION TO AIRCRAFT STABILITY	9

Types of stability, static and dynamic stability, longitudinal, lateral and directional stability, equilibrium conditions of flight, forces and moments acting on aircraft, stability derivatives, aerodynamic center and center of gravity, influence of aircraft configuration on stability.

UNIT: II	LONGITUDINAL STABILITY AND CONTROL	9

Longitudinal static stability criteria, stick fixed and stick free stability, tail volume ratio and its significance, elevator control and effectiveness, trim conditions in steady flight, pitching moment characteristics, effect of power on longitudinal stability, neutral point determination, longitudinal stability during maneuvers.

UNIT: III	LATERAL AND DIRECTIONAL STABILITY	9

Lateral stability concepts, dihedral effect and its importance, directional stability due to vertical tail, rolling moment derivatives, yawing moment derivatives, adverse yaw phenomenon, aileron control effectiveness, rudder control characteristics, coupling between roll and yaw motions.

UNIT: IV	DYNAMIC STABILITY OF AIRCRAFT	9

Equations of motion for aircraft, small disturbance theory, longitudinal dynamic modes, short period oscillation characteristics, phugoid motion behavior, lateral dynamic modes, dutch roll oscillation characteristics, spiral divergence phenomenon, roll subsidence motion.

UNIT: V	STABILITY AUGMENTATION AND CONTROL SYSTEMS	9

Need for stability augmentation systems, automatic flight control concepts, stability augmentation system components, autopilot principles and operation, fly-by-wire control systems, artificial stability in modern aircraft, control surface actuators and sensors, feedback control concepts, applications in modern aircraft control systems.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of Publication**
1	Nelson, R. C.	Flight Stability and Automatic Control	McGraw-Hill Education	2018
2	Etkin, B. and Reid, L. D.	Dynamics of Flight: Stability and Control	John Wiley & Sons	2021
REFERENCE BOOKS:				
1	Perkins, C. D. and Hage, R. E.	Airplane Performance, Stability and Control	John Wiley & Sons	2020
2	Cook, M. V.	Flight Dynamics Principles	Butterworth-Heinemann	2022
WEB LEARNING RESOURCES:				

1. https://ocw.mit.edu/courses/16-333-aircraft-stability-and-control-fall-2004/
2. https://nptel.ac.in/courses/101106046
3. https://www.aircraftflightmechanics.com
4. https://www.faa.gov/regulations_policies/handbooks_manuals/aviation
5. https://www.grc.nasa.gov/www/k-12/airplane/airplane.html

CO – PO – PSO MAPPING

CO	**PO**											**PSO**		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	1	1	2	1	-	-	-	1	1	3	1	1
CO2	3	2	1	-	2	1	-	-	-	1	1	2	1	1
CO3	3	2	-	1	2	-	-	-	-	1	1	3	1	1
CO4	3	2	1	1	2	1	-	-	-	1	1	2	1	1
CO5	3	2	1	-	2	1	-	-	-	1	1	3	1	1
Avg	3	2	1	1	2	1	-	-	-	1	1	2.4	1	1

M.A.M SCHOOL OF ENGINEERING

(An Autonomous Institution)

(Accredited by NAAC || Approved by AICTE || Affiliated to Anna University)

Trichy – Chennai Trunk Road, Siruganur, Tiruchirappalli – 621 105

B.E. AERONAUTICAL ENGINEERING

ELECTIVES LIST

VERTICAL II - PROFESSIONAL ELECTIVE II

PROPULSION

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1	24AEX07	Advanced Propulsion Systems	3	0	0	3	40	60	100	PE
2	24AEX08	Design of Gas Turbine Engine Components	3	0	0	3	40	60	100	PE
3	24AEX09	Turbo Machines	3	0	0	3	40	60	100	PE
4	24AEX10	Cryogenic Engineering	3	0	0	3	40	60	100	PE
5	24AEX11	Combustion Techniques	3	0	0	3	40	60	100	PE
6	24AEX12	Aircraft Fuels and Propellants	3	0	0	3	40	60	100	PE

REGULATIONS 2024

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AEX07	ADVANCED PROPULSION SYSTEMS	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Basic propulsion concepts, thermodynamics, fluid mechanics, gas dynamics, combustion fundamentals, mathematics understanding.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand the hypersonic air-breathing propulsion
2. To make students understand the chemical rocket propulsion
3. To impart knowledge to students on the basic aspects of nuclear rocket propulsion
4. To impart knowledge to students on the basic aspects of advanced rocket propulsion
5. Expose the students on obtaining knowledge on launch vehicles

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain hypersonic air-breathing propulsion systems and combined-cycle engine operations.

CO2: Analyze chemical rocket propulsion types, propellants, performance, and thrust vector control.

CO3: Describe nuclear rocket propulsion principles, reactor operation, fuel elements, and applications.

CO4: Evaluate advanced, electric, micro, and propellant-less propulsion systems for space missions.

CO5: Understand launch vehicle design, staging, propulsion challenges, guidance, control, and case studies.

UNIT: I	HYPERSONIC AIR-BREATHING PROPULSION	9

Ramjets at high speeds, Need for supersonic combustion, Implications criticality of efficient diffusion and acceleration, problems of combustion in high-speed flow, scramjet engine- construction, flow process-description, Combined cycle engines- turbo-ramjet, Air turbo-rocket (ATR), ejector ramjet, Liquid-air collection engine (LACE) - need, principle, construction, operation, performance.

UNIT: II	CHEMICAL ROCKET PROPULSION	9

Classification of rocket engine, chemical rocket engine types, working principle, schematic diagram, applications, types, advantages and disadvantages- solid, liquid and hybrid propellant rocket engine, TVC. Solid propellant rocket motors, principle, applications, Solid propellant types, composition, properties, Propellant grain, properties, structural design

UNIT: III	NUCLEAR ROCKET PROPULSION	9

Nuclear propulsion history, Power, thrust, energy. Nuclear fission- basics, sustainable chain reaction, neutron leakage, control, reflection, prompt and delayed neutrons, thermal stability. Principles and fuel elements. The nuclear thermal rocket engine, start-up and shutdown. Development status of nuclear engines, alternative reactor types, safety issues in nuclear propelled missions.

UNIT: IV	ADVANCED ROCKET PROPULSION	9

Electric propulsion systems - structure, types, generation of thrust. Electrostatic thrusters, electromagnetic thrusters, applications to space missions, pulsed plasma thrusters (PPT) for micro-spacecraft, solar electric propulsion. Micro-propulsion, application of MEMS, chemical, electric micro- thrusters, principle, description, Propellant-less propulsion, Photon rocket, beamed energy propulsion, solar, magnetic sails.

UNIT: V	LAUNCH VEHICLES	9

Role and military functions of space launch vehicle, Types, missions, mission profile, staging employed in the vehicle, guidance and control requirements. Successful launch vehicles - Case Studies, Description of space shuttle engine, Propellant slosh - Propellant hammer, Geysering effect in cryogenic rocket engines, SSTO.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of Publication**
1	Cornelisse, J. W., Schoyer H.F.R	Rocket Propulsion and Space Flight Dynamics	Pitman	2018
2	Turner, M.J.L	Rocket and Spacecraft Propulsion	Springer	2010
3	T. W. Lee	Aerospace Propulsion	Wiley	2022

REFERENCE BOOKS:				
1	Goteti Satyanarayana	Propulsion Systems for Space Applications	B.S.Publications	2020
2	Joachim Kurzke	Propulsion and Power	Springer	2008

WEB LEARNING RESOURCES:

1. https://aiaa.org/courses/fundamentals-of-high-speed-air-breathing-and-space-propulsion/
2. https://classcentral.com/course/youtube-aerospace-propulsion-92334
3. https://classcentral.com/course/swayam-rocket-propulsion-17764
4. https://wizape.com/English/Advanced-Aerospace-Propulsion-Systems
5. https://elevify.com/en/courses/engineering-construction-and-technology/rocket-propulsion-course-2dbab

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	3	2	2	2	-	-	-	-	-	-	2	1	-
CO2	3	3	2	2	-	-	-	-	-	-	2	3	1	-
CO3	3	2	2	2	2	-	-	-	-	-	-	3	1	-
CO4	2	1	1	2	3	-	-	-	-	-	2	-	-	1
CO5	3	2	2	2	3	-	-	-	-	-	-	-	2	1
Avg	2.8	2.2	1.8	2.0	2.5	-	-	-	-	-	2.0	2.8	1.4	1.0

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AEX08	DESIGN OF GAS TURBINE ENGINE COMPONENTS	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Thermodynamics fundamentals, fluid mechanics, turbomachinery basics, strength of materials, gas turbine principles understanding.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. Introduce gas turbine engine cycles, configurations, performance parameters, and preliminary design methodology.
2. Develop understanding of intake and compressor aerodynamic, mechanical, and performance design aspects.
3. Explain combustion chamber design principles, combustion processes, emissions, cooling, and thermal management.
4. Familiarize students with turbine aerodynamics, blade design, cooling techniques, and material considerations.
5. Provide knowledge of nozzle design, accessories, structural integrity, reliability, and maintenance practices.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Analyze gas turbine cycles, configurations, mission requirements, and off-design operational behavior.

CO2: Design and evaluate intake systems and compressors considering aerodynamics, losses, stall, surge.

CO3: Assess combustion chamber performance, fuel injection, flame stabilization, and thermal stresses.

CO4: Design turbine stages considering efficiency, cooling, stresses, creep, materials, and manufacturing.

CO5: Evaluate nozzle performance, accessories, structural design, reliability, life estimation, and maintenance.

UNIT: I	ENGINE CYCLES AND CONFIGURATIONS	9

Gas turbine engine classification and applications, Ideal and actual Brayton cycle analysis, Performance parameters: thrust, power, efficiency, Design considerations for aircraft gas turbines, Selection of design point and off-design operation, Introduction to preliminary gas turbine design methodology, Mission analysis - Aircraft weight and fuel consumption data

UNIT: II	INTAKE AND COMPRESSOR DESIGN	9

Air intake types and performance requirements, Intake pressure recovery and distortion effects, Axial compressor working principle, Compressor aerodynamics and velocity triangles, Blade and cascade design parameters, Compressor losses, stall, and surge, Multistage axial compressor design, Centrifugal compressor design principles, Mechanical design considerations of compressor blades

UNIT: III	COMBUSTION CHAMBER DESIGN	9

Functions and types of combustion chambers, Combustion process in gas turbines, Fuel injection and atomization techniques, Flame stabilization and ignition systems, Combustion efficiency and pressure loss, Temperature distribution and pattern factor, Emission characteristics and control methods, Liner cooling and thermal stresses, Design considerations for aircraft combustors

UNIT: IV	TURBINE DESIGN	9

Impulse and reaction turbine principles, Turbine velocity diagrams and stage loading, Axial turbine blade design, Degree of reaction and efficiency, Turbine losses and cooling requirements, High-pressure and low-pressure turbine design, Blade cooling techniques, Mechanical stresses and creep considerations, Turbine materials and manufacturing aspects

UNIT: V	NOZZLE, ACCESSORIES, AND STRUCTURAL DESIGN	9

Exhaust nozzle types and performance, Convergent and convergent–divergent nozzle design, Thrust calculation and expansion efficiency, Afterburner concept and design aspects, Bearings, seals, and lubrication systems, Shaft dynamics and vibration issues, Thermal expansion and casing design, Materials selection for gas turbine components, Reliability, life estimation, and maintenance considerations

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Meinhard T. Schobeiri	Gas Turbine Design, Components and System Design Integration	Springer Nature	2018
2	Je-Chin Han, Sandip Dutta, Srinath Ekkad	Gas Turbine Heat Transfer and Cooling Technology	CRC Press	2013

REFERENCE BOOKS:

1	Saravanamuttoo H.I.H and Rogers, G.F.C.	Gas Turbine Technology	Pearson Education	2019
2	Mattingly J.D., Heiser, W.H. and Pratt D.T	Aircraft Engine Design	AIAA Education Series	2022

WEB LEARNING RESOURCES:

1. https://www.classcentral.com/course/swayam-gas-turbines-jet-engines-503001
2. https://www.classcentral.com/course/swayam-aeroengine-gas-turbine-cycle-502935
3. https://alison.com/course/principles-and-constructional-features-of-gas-turbines
4. https://wizape.com/English/Jet-Engine-Design-and-Analysis
5. https://www.classcentral.com/course/youtube-lecture-08-introduction-to-gas-turbines-437329

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	3	2	2	3	-	-	-	-	-	-	2	1	-
CO2	3	3	2	1	-	-	-	-	-	-	2	3	1	-
CO3	3	3	2	1	3	-	-	-	-	-	-	3	1	-
CO4	3	3	1	2	3	-	-	-	-	-	2	-	-	1
CO5	3	3	2	2	3	-	-	-	-	-	-	-	2	1
Avg	3	3	1.8	1.6	3	-	-	-	-	-	2	2.8	1.4	1

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AEX09	TURBO MACHINES	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Fluid mechanics fundamentals, thermodynamics basics, energy transfer principles, turbomachinery concepts, mathematics understanding.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand turbomachinery principles applied to aircraft propulsion systems design.
2. To analyze axial and centrifugal compressor performance, and design aspects.
3. To study turbine operation, efficiency, cooling, stresses, and material selection.
4. To evaluate turbomachine losses, efficiencies, scaling laws, and off-design performance.
5. To introduce special turbomachines, aeroelasticity, noise, reliability, and aerospace trends.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1:	Explain turbomachinery fundamentals, energy transfer, velocity triangles, efficiencies, and aircraft propulsion relevance.
CO2:	Analyze axial and centrifugal compressors considering performance, losses, stall, surge, and design.
CO3:	Apply turbine principles to evaluate stage loading, efficiency, cooling, stresses, and materials.
CO4:	Assess turbomachine matching, off-design behavior, aeroelastic effects, noise, reliability, and maintenance requirements.
CO5:	Interpret emerging aerospace turbomachinery technologies, performance trends, sustainability challenges, and future applications.

UNIT: I	FUNDAMENTALS OF TURBOMACHINES	9

Role of turbomachines in aircraft propulsion systems, Classification of turbomachines for aerospace applications Energy transfer between fluid and rotor, Euler turbine equation and its physical significance, Velocity triangles for compressors and turbines, Degree of reaction and stage work, Specific speed and scaling laws, Performance parameters and efficiencies, Losses in turbomachines and their impact on aircraft performance

UNIT: II	AXIAL FLOW COMPRESSORS	9

Axial compressor construction and working, Velocity triangles and blade loading, Compressor stage and multistage design, Degree of reaction and efficiency, Compressor losses and performance characteristics, Stall and surge phenomena, Methods of stall and surge control, Materials and structural aspects of compressor blades

UNIT: III	CENTRIFUGAL COMPRESSORS AND FANS	9

Construction and working of centrifugal compressors, Velocity triangles and energy transfer, Slip factor and diffuser design, Performance characteristics and efficiency, Aircraft engine fans and bypass ratio, Multistage centrifugal compressors, Comparison of axial and centrifugal compressors, Design considerations for aerospace applications

UNIT: IV	AXIAL AND RADIAL FLOW TURBINES	9

Role of turbines in aircraft gas turbine engines, Impulse and reaction turbine principles, Velocity diagrams and stage loading, Degree of reaction and turbine efficiency, Multistage axial turbine design, Blade cooling requirements in aircraft turbines, Mechanical stresses and creep in turbine blades, Radial flow turbines - Types – Elements - Stage velocity diagrams, Turbine materials and manufacturing techniques

UNIT: V	SPECIAL TURBOMACHINES AND APPLICATIONS	9

Aircraft gas turbine power plants overview, Matching of compressor and turbine in jet engines, Performance of turbomachines at off-design conditions, Ram air turbines and emergency power units, Turbochargers and turbofans, Noise generation and reduction in turbomachinery, Aeroelastic effects and vibration in rotating blades Reliability, life estimation, and maintenance, Future trends in aerospace turbomachinery design

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	S.M. Yahya	Turbines, Compressors and	Tata McGraw-Hill	2019
2	Venkanah, B.K	Fundamentals of Turbomachinery	PHI Learning	2022
3	Dixon, S.L.,	Fluid Mechanics and Thermodynamics of Turbomachinery	Elsevier	2021

REFERENCE BOOKS:

1	Ganesan, V	Gas Turbines	Tata McGraw HilL	2021
2	Appu Kuttan	Turbomachines	Wiley	2020

WEB LEARNING RESOURCES:

1. https://ocw.mit.edu/courses/16-50-introduction-to-propulsion-systems-fall-2012/
2. https://nptel.ac.in/courses/112106200
3. https://www.grc.nasa.gov/www/k-12/airplane/turbine.html
4. https://www.engineeringtoolbox.com/turbomachinery-d_1230.html
5. https://www.aiaa.org/education/online-courses

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	3	2	2	2	-	-	-	-	-	2	2	1	-
CO2	3	1	2	2	-	-	1	-	-	-	-	3	1	-
CO3	3	2	2	2	2	-	-	-	-	-	-	3	1	-
CO4	2	1	2	2	3	-	-	-	-	-	2	-	-	1
CO5	3	2	2	2	3	-	1	-	-	-	-	-	2	1
Avg	2.8	2	2	2.0	2.5	-	1	-	-	-	2.0	2.8	1.4	1.0

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AEX10	CRYOGENIC ENGINEERING	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Thermodynamics fundamentals, heat transfer basics, fluid mechanics knowledge, low temperature physics concepts.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand cryogenic properties materials fluids and low temperature behavior
2. To analyze gas liquefaction principles minimum work and refrigeration cycles
3. To design safe storage transfer systems for cryogenic liquids space
4. To evaluate measurement techniques and heat exchangers in cryogenic environments
5. To explore superconducting devices applications materials and system level optimization

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain low temperature material behavior and thermophysical properties accurately clearly

CO2: Calculate liquefaction work and select appropriate cryogenic refrigeration cycles systems

CO3: Design storage vessels transfer lines and valves for cryogenic service

CO4: Apply cryogenic measurements heat exchangers compressors pumps and expanders effectively

CO5: Assess superconducting technologies and cryogenic applications across aerospace industries

UNIT: I	INTRODUCTION TO CRYOGENIC ENGINEERING	9

Mechanical and thermal Properties at low temperatures. Properties of Cryogenic Fluids. Gas Liquefaction: Minimum work for liquefaction. Methods to protect low temperature. Super conducting materials, thermo electric materials, composite materials, Cryo metallurgy

UNIT: II	STORAGE OF CRYOGENIC LIQUIDS	9

Design considerations of storage vessel; Dewar vessels; Industrial storage vessels; Storage of cryogenic fluids in space; Transfer systems and Lines for cryogenic liquids; Cryogenic valves in transfer lines; Two phase flow in Transfer system; Cooldown of storage and transfer systems.

UNIT: III	MEASUREMENT IN CRYOGENICS	9

Measurement of strain, pressure, flow, liquid level and Temperature in cryogenic environment; Cryostats. Cryogenic heat exchangers – recuperative and regenerative; Variables affecting heat exchanger and system performance; Cryogenic compressors, Pumps, expanders; Turbo alternators; Effect of component inefficiencies; System Optimization.

UNIT: IV	SUPER CONDUCTIVE DEVICES	9

Super conductive devices such as bearings, motors, cryotrons, magnets, D.C. transformers, tunnel diodes, space technology, space simulation, cryogenics in biology and medicine, food preservation and industrial applications, nuclear propulsions, chemical propulsions

UNIT: V	CRYOGENIC REFRIGERATION SYSTEM	9

Types of cryogenic refrigerants, classification of cryogenic fluids, thermophysical properties of cryogenic refrigerants, boiling point and critical properties, storage and handling characteristics, environmental impact of refrigerants, safety considerations in cryogenic fluids, application-based selection criteria for refrigerants, industrial and aerospace applications of cryogenic refrigerants.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication

1	Mamata Mukhopadhyay	Fundamentals of Cryogenic Engineering	CBS publisher	2020
2	Thomas M Flynn	Cryogenic Engineering	Tata McGraw Hill	2021
3	Zuyu Zhao	Cryogenic Engineering and Principles	CRC press	2019

REFERENCE BOOKS:				
1	S.S.Thipse	Cryogenic Engineering	Alpha	2013
2	Alyson	Cryogenic Engineering	Nova Publishers	2020

WEB LEARNING RESOURCES:

1. https://nptel.ac.in/courses/112106266
2. https://ocw.mit.edu/courses/16-520-cryogenic-engineering-spring-2005/
3. https://www.csa.gov.in/knowledge/cryogenics
4. https://www.cryogenicsociety.org/resources/educational-resources
5. https://www.energy.gov/science/ascr/cryogenic-engineering-and-superconductivity

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	3	2	2	2	-	-	-	1	-	-	3	2	1
CO2	3	3	2	2	-	-	-	-	-	-	2	3	1	-
CO3	3	2	2	2	2	-	-	-	1	-	-	3	2	-
CO4	3	2	2	2	3	-	-	-	-	-	2	-	-	1
CO5	3	2	2	2	3	-	-	-	1	-	-	3	2	1
Avg	3	2.3	2	2.0	2.5	-	-	-	**1**	-	2.0	3	1.6	1.0

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AEX11	COMBUSTION TECHNIQUES	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Thermodynamics basics, chemical reactions fundamentals, combustion principles, fluid mechanics knowledge, heat transfer understanding.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand premixed combustion principles, reaction rates, flame propagation
2. To analyze SI engine combustion phenomena, knock control chambers
3. To evaluate CI engine spray combustion delay heat release
4. To investigate engine emissions formation mechanisms, control strategies effects
5. To explore aeronautical combustor design stabilization cooling future trends

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain premixed flame behavior equilibrium dissociation laminar turbulent combustion processes

CO2: Differentiate SI engine combustion stages abnormal phenomena and mitigation techniques

CO3: Analyze CI engine spray formation delay heat release injection effects

CO4: Assess emission formation control technologies environmental and human health impacts

CO5: Apply aeronautical combustion concepts to combustor configuration stabilization cooling systems

UNIT: I	PREMIXED COMBUSTION	9

Combustion Principles- Combustion - Combustion equations, heat of combustion - Theoretical flame temperature, Chemical equilibrium and dissociation - Theories of Combustion - Pre-flame reactions, Reaction rates - Laminar and Turbulent, Flame Propagation in Engines.

UNIT: II	COMBUSTION IN SI ENGINE	9

Initiation of combustion, stages of combustion, normal and abnormal combustion, knocking combustion, pre-ignition, knock and engine variables, features and design consideration of combustion chambers.- Flame structure and speed, Cycle by cycle variations, Lean burn combustion, stratified charge combustion ystems. Heat release correlations. After treatment devices for SI engines.

UNIT: III	COMBUSTION IN CI ENGINE	9

Stages of combustion, vaporization of fuel droplets and spray formation, air motion, swirl measurement, knock and engine variables, features and design considerations of combustion chambers- delay period correlations, heat release correlations, and influence of the injection system on combustion. Direct and indirect injection systems. After treatment devices for diesel engines.

UNIT: IV	EMISSIONS	9

Main pollutants in engines, Kinetics of NO formation, Nox formation in SI and CI engines. Unburned-hydrocarbons, sources, formation in SI and CI engines, Soot formation and oxidation, Particulates in diesel engines, Emission control measures for SI and CI engines, Effect of emissions on Environment and human beings.

UNIT: V	AERONAUTICAL COMBUSTION	9

Components and Mechanisms-Atomization and Spray-Flame Stabilization: - Air-Fuel Zonation-Cooling Systems: Common Combustor Configurations-Annular, Can-Annular-Can Future Trends in Aeronautical combustion

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication

1	Kenneth W. Ragland	Combustion Engineering	CRC press	2022
2	Allan T. Kirkpatrick	Principles of Combustion	Wiley	2018
3	Stephen R. Turns	An Introduction to Combustion: Concepts and Applications	Tata McGraw Hill	2021

REFERENCE BOOKS:

1	Maximilian Lackner	Combustion: From Basics to Applications	Wiley	2019
2	O.P. Gupta	Elements of Fuel & Combustion Technology	Kamal Publishing House	2020

WEB LEARNING RESOURCES:

1. https://www.classcentral.com/course/youtube-fundamentals-of-combustion-i-92342
2. https://www.classcentral.com/course/swayam-fundamentals-of-combustion-22986
3. https://nptel.ac.in/courses/112106419
4. https://alison.com/course/mechanical-engineering-internal-combustion-engine-basics
5. https://www.getyoureducation.net/course/fundamentals-of-combustion

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	2	3	2	2	2	-	-	-	-	-	-	3	1	-
CO2	2	3	2	-	-	-	-	-	-	-	2	3	1	1
CO3	2	3	2	2	2	-	-	-	-	-	-	3	1	-
CO4	2	3	-1	2	2	-	-	-	-	-	2	3	1	1
CO5	2	3	2	2	2	-	-	-	-	-	-	3	1	1
Avg	2	3	2	2	2	-	-	-	-	-	2	3	1	1.0

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AEX06	AIRCRAFT FUELS AND PROPELLANTS	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Basic chemistry concepts, propulsion fundamentals, thermodynamics basics, fuel properties knowledge, combustion principles understanding

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand different types of aircraft fuels and their properties used in aviation.
2. To study production, refining, storage and quality control methods of aviation fuels.
3. To analyze various types of rocket propellants and their performance characteristics.
4. To examine combustion processes and factors influencing fuel performance in propulsion systems.
5. To understand alternative aviation fuels and their environmental and technological implications.

COURSE OUTCOMES:

At the end of this course, students can able to

- CO1: Explain properties, classification and performance requirements of fuels used in aircraft engines.
- CO2: Describe refining processes, additives, storage methods and quality control of aviation fuels.
- CO3: Classify rocket propellants and explain characteristics of solid, liquid and hybrid propellants.
- CO4: Analyze combustion characteristics affecting efficiency, stability and emissions in propulsion systems.
- CO5: Discuss alternative aviation fuels, environmental impacts and future sustainable aviation fuel technologies.

UNIT: I — AIRCRAFT FUELS — 9

Classification of aircraft fuels, properties of aviation gasoline, properties of aviation turbine fuels, fuel performance requirements in aircraft engines, fuel specifications and standards, volatility and viscosity characteristics, calorific value and combustion characteristics, safety and handling of aviation fuels.

UNIT: II — FUEL PRODUCTION AND REFINING — 9

Petroleum sources for aviation fuels, crude oil refining processes, distillation and cracking processes, blending of aviation fuels, additives used in aircraft fuels, anti-knock and anti-icing additives, fuel contamination and prevention methods, storage and transportation of aviation fuels, quality control and testing procedures.

UNIT: III — PROPELLANTS FOR ROCKETS AND MISSILES — 9

Classification of propellants, solid propellants and their characteristics, liquid propellants and their types, hybrid propellants and applications, oxidizers used in rocket propulsion, propellant performance parameters, propellant grain configurations, safety and handling of propellants.

UNIT: IV — COMBUSTION CHARACTERISTICS OF FUELS — 9

Fundamentals of combustion processes, ignition characteristics of aviation fuels, flame propagation in fuels, combustion efficiency in aircraft engines, detonation and knocking phenomena, combustion instability in propulsion systems, factors affecting combustion performance, emissions from aircraft fuels, methods to improve combustion efficiency.

UNIT: V — ADVANCED AND ALTERNATIVE AVIATION FUELS — 9

Need for alternative aviation fuels, biofuels for aviation applications, synthetic aviation fuels, hydrogen as aircraft fuel, environmental impact of aviation fuels, fuel economy and energy considerations, fuel system compatibility issues, future trends in aviation fuels, sustainability in aviation fuel technology.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Sutton, G. P. and Biblarz, O.	Rocket Propulsion Elements	John Wiley & Sons	2021

2	Huzel, D. K. and Huang, D. H.	Modern Engineering for Design of Liquid-Propellant Rocket Engines	AIAA	2020
REFERENCE BOOKS:				
1	Lefebvre, A. H. and Ballal, D. R.	Gas Turbine Combustion	CRC Press	2020
2	Speight, J. G.	Handbook of Petroleum Product Analysis	John Wiley & Sons	2019

WEB LEARNING RESOURCES:

1. https://www.grc.nasa.gov/www/k-12/airplane/airplane.html
2. https://www.grc.nasa.gov/WWW/K-12/BGP/PAT/Fuel_and_Air_int.htm
3. https://www.nasa.gov/testing-and-analysis/propellants-and-aerospace-fluids/
4. https://www.naa.edu/aviation-fuel/
5. https://www.teachengineering.org/lessons/view/cub_rockets_lesson04

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	1	-	-	1	-	-	-	1	1	3	1	1
CO2	3	2	1	1	2	1	-	-	-	1	1	2	1	1
CO3	3	2	1	1	2	-	-	-	-	1	1	2	1	1
CO4	3	2	1	-	2	1	-	-	-	1	1	2	1	1
CO5	3	2	1	1	-	1	-	-	-	1	1	3	1	1
Avg	3	2	1	1	2	1	-	-	-	1	1	2.2	1	1

M.A.M SCHOOL OF ENGINEERING

(An Autonomous Institution)

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Trichy – Chennai Trunk Road, Siruganur, Tiruchirappalli – 621 105

B.E. AERONAUTICAL ENGINEERING

ELECTIVES LIST

VERTICAL III - PROFESSIONAL ELECTIVE III
AIRCRAFT STRUCTURES & MATERIALS

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1	24AEX13	Fatigue and Fracture Mechanics	3	0	0	3	40	60	100	PE
2	24AEX14	Composite Materials and Structures	3	0	0	3	40	60	100	PE
3	24AEX15	Additive Manufacturing	3	0	0	3	40	60	100	PE
4	24AEX16	Aerospace Materials	3	0	0	3	40	60	100	PE
5	24AEX17	Experimental Stress Analysis	3	0	0	3	40	60	100	PE
6	24AEX18	Smart Materials and Structures	3	0	0	3	40	60	100	PE

REGULATIONS 2024

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VI
24AEX13	FATIGUE AND FRACTURE MECHANICS	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Strength of materials concepts, stress analysis, material properties, failure theories, and basic mechanics.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand fatigue mechanisms and failure in aerospace structures
2. To analyze cyclic loading effects using S–N and mean stress
3. To evaluate crack initiation propagation using fracture mechanics concepts
4. To apply elasto-plastic fracture parameters in damage tolerance
5. To design fatigue resistant aerospace structures through testing methodologies

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain fatigue failure mechanisms and stress life relationships in structures

CO2: Analyze crack initiation propagation and variable amplitude loading effects

CO3: Evaluate fracture toughness and cracked body behavior accurately

CO4: Apply damage tolerance and safe life design philosophies effectively

CO5: Assess fatigue performance of metallic and composite aerospace structures

UNIT: I	FATIGUE OF STRUCTURES	9

Nature and significance of fatigue failure, Fatigue failure in aircraft structures, Cyclic stresses and strain parameters, S–N curves and endurance limit, Goodman, Gerber and Soderberg relations and diagrams - Notches and stress concentrations - Neuber's stress concentration factors - Plastic stress concentration factors - Notched S.N. curves – Fatigue of composite materials.

UNIT: II	FATIGUE CRACK INITIATION AND PROPAGATION	9

Microscopic mechanisms of fatigue damage, Crack initiation in metals and alloys, Slip band formation and surface effects, Fatigue crack growth stages, Paris law and crack growth models, Threshold stress intensity factor, Load interaction and retardation effects, Environmental effects on fatigue, Aircraft fatigue case studies

UNIT: III	ELASTO-PLASTIC AND FATIGUE FRACTURE	9

Limitations of LEFM, Plastic zone size and correction models, Crack tip opening displacement (CTOD), J-integral concept, Elastic–plastic fracture criteria, Fatigue fracture interaction, Crack closure phenomenon, Variable amplitude loading, Damage tolerance approach in aircraft

UNIT: IV	FRACTURE MECHANICS	9

Strength of cracked bodies - Potential energy and surface energy - Griffith's theory - Irwin - Orwin extension of Griffith's theory to ductile materials - stress analysis of "cracked bodies - Effect of thickness on fracture toughness" - stress intensity factors for typical 'geometries.

UNIT: V	FATIGUE DESIGN AND TESTING	9

Safe life and Fail-safe design philosophies - Importance of Fracture Mechanics in aerospace structures - Application to composite materials and structures

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Barrois W, Ripely, E.L.,	Fatigue of Aircraft Structures	Pergamon Press	2021
2	Vladimir V. Bolotin	Mechanics of Fatigue	CRC Press	2019

3	Prasanth Kumar	Elements of Fracture Mechanics	Wheater publication	2020

REFERENCE BOOKS:				
1	Kare Hellan	Introduction to Fracture Mechanics	McGraw-Hill	2023
2	Knott, J.F	Fundamentals of Fracture Mechanics	Buterworth & Co., Ltd.	2019

WEB LEARNING RESOURCES:

1. https://ocw.mit.edu/courses/materials-science-and-engineering/3-35-fracture-and-fatigue-fall-2003/
2. https://www.am.chalmers.se/~anek/teaching/fatfract/
3. https://wizape.com/English/Fatigue-and-Fracture-Mechanics-in-Aerospace
4. https://ocw.mit.edu/courses/3-11-mechanics-of-materials-fall-1999/resources/mit3_11f99_fatigue/
5. https://www.nafems.org/publications/resource_center/el_ffm/

CO – PO – PSO MAPPING

CO	**PO**											**PSO**		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	2	2	-	-	-	-	-	-	2	1	-
CO2	3	2	2	-	-	-	-	-	-	-	2	3	1	-
CO3	2	2	2	2	2	-	-	1	-	-	-	3	1	-
CO4	2	2	1	2	3	-	-	-	-	-	2	-	-	1
CO5	3	2	2	-	3	-	-	1	-	-	-	-	2	1
Avg	2.6	2	1.8	2.0	2.5	-	-	1	-	-	2.0	2.8	1.4	1.0

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VI
24AEX14	COMPOSITE MATERIALS AND STRUCTURES	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Mechanics of materials, basic materials science, stress analysis, structural mechanics, and engineering mathematics fundamentals.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand micro-mechanics and macro-mechanics of composite materials
2. To analyze laminated plate theory for aerospace structural applications
3. To evaluate sandwich constructions behaviour under aerospace loading conditions
4. To study fabrication processes and repair methods of composites
5. To apply composite design concepts to aeronautical structures

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain micro and macro level behavior of composite materials clearly
CO2: Analyze lamina laminate and sandwich structural responses effectively
CO3: Evaluate failure mechanisms including hygrothermal and impact effects
CO4: Apply laminated plate theory to aerospace structural design problems
CO5: Assess fabrication processes defects and repair standards in aerospace

UNIT: I	MICROMECHANICS OF COMPOSITES	9

Classification and role of composites in aerospace structures, Fiber, matrix and interface constituents, Micromechanics of continuous fiber composites, Rule of mixtures and effective property estimation, Load transfer and fiber–matrix interaction, Elastic properties of unidirectional lamina, Thermal and moisture effects at micro level, Failure mechanisms in fiber and matrix, . Effect of voids in composites.

UNIT: II	MACROMECHANICS AND LAMINA BEHAVIOR	9

Behavior of composite lamina, Stress–strain relations for anisotropic materials, Transformation of stresses and strains, Engineering constants of lamina, Strength and stiffness characterization, Hygrothermal effects on lamina properties, Failure criteria for composite lamina, Lamina testing methods, Design considerations at macroscopic level

UNIT: III	LAMINATED PLATE THEORY	9

Governing differential equation for a laminate. Stress – strain relations for a laminate. Different types of laminates. in plane and flexural constants of a laminate. hygrothermal stresses and strains in a laminate. Failure analysis of a laminate. Impact resistance and interlaminar stresses. Netting analysis.

UNIT: IV	SANDWICH CONSTRUCTIONS AND STRUCTURAL ELEMENTS	9

Concept of sandwich structures, Face sheet and core materials, Honeycomb and foam core configurations, Bending, shear and buckling of sandwich panels, Failure modes in sandwich constructions, Vibration and impact behavior, Design considerations for aerospace sandwich structures, Applications in aircraft and spacecraft structures, Case studies of sandwich panels in aviation

UNIT: V	FABRICATION PROCESS AND REPAIR METHODS	9

Hand lay-up and spray lay-up processes, Filament winding and pultrusion, Resin transfer molding and vacuum infusion, Automated fiber placement and tape laying, Curing processes and autoclave techniques, Types of Structural Failures and prevention methods, Certification and repair standards in aerospace.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of Publication**
1	Autar K Kaw	Mechanics of Composite Materials’	CRC Press	2023
2	Isaac M. Daniel & Ori Ishai	Mechanics of Composite Materials	OUP USA Publishers	2019
3	Agarwal, B.D., and Broutman, L.J	Analysis and Performance of Fibre Composites	John Wiley & Sons	2021

REFERENCE BOOKS:				
1	Allen Baker	Composite Materials for Aircraft Structures	AIAA Series, 2ndEdition	2024
2	Krishan K. Chawla	Composite Materials: Science and Engineering	Springer	2019

WEB LEARNING RESOURCES:

1. https://archive.nptel.ac.in/courses/101/104/101104010/
2. https://www.nptelprep.in/courses/101104010
3. https://www.classcentral.com/course/swayam-mechanics-of-composite-materials-and-structures-452130
4. https://wizape.com/English/Structural-Applications-of-Composite-Materials
5. https://www.getyoureducation.net/course/composite-materials-overview-for-engineers

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	3	2	2	3	-	1	-	-	-	-	2	1	-
CO2	3	3	2	1	-	-	1	-	-	-	2	3	1	1
CO3	3	3	2	1	3	-	-	-	-	-	-	3	1	-
CO4	3	3	2	2	3	-	1	-	-	-	2	-	-	1
CO5	3	3	2	2	3	-	-	-	-	-	-	3	2	1
Avg	3	3	2	1.6	3	-	1	-	-	-	2	2.9	1.4	1

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VI
24AEX15	ADDITIVE MANUFACTURING	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Basic manufacturing processes, engineering materials, CAD modeling, engineering graphics, and mechanical design fundamentals.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand fundamentals, evolution, classification and applications of additive manufacturing
2. To apply design principles and tools for additive manufacturing products
3. To analyze vat polymerization, powder bed fusion and material extrusion
4. To evaluate additive manufacturing processes, materials, benefits, limitations and applications
5. To understand opportunities, studies and future directions in additive manufacturing

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain additive manufacturing processes, classifications, materials and applications

CO2: Design optimized parts using DfAM principles, topology optimization, lattices and generative approaches

CO3: Select appropriate additive manufacturing processes based on materials, quality requirements

CO4: Analyze process capabilities, limitations & defects in major additive manufacturing technologies systems

CO5: Evaluate industrial case studies and identify business opportunities in additive manufacturing sectors

UNIT: I	BASICS OF ADDITIVE MANUFACTURING	9

Overview - Need - Development of Additive Manufacturing (AM) Technology: Rapid Prototyping- Rapid Tooling - Rapid Manufacturing - Additive Manufacturing. AM Process Chain- ASTM/ISO 52900 Classification - Benefits. Applications: Building Printing - Bio Printing - Food Printing- Electronics Printing. Business Opportunities and Future Directions – Case studies: Automobile, Aerospace, Healthcare.

UNIT: II	DESIGN FOR ADDITIVE MANUFACTURING	9

Concepts and Objectives - AM Unique Capabilities - Part Consolidation – Topology Optimization- Generative design - Lattice Structures - Multi-Material Parts and Graded Materials - Data Processing: CAD Model Preparation - AM File formats: STL-Problems with STL- AMF Design for Part Quality Improvement: Part Orientation - Support Structure - Slicing - Tool Path Generation – Design rules for Extrusion based AM.

UNIT: III	VAT POLYMERIZATION & DIRECTED ENERGY DEPOSITION	9

Photo polymerization: Stereolithography Apparatus (SLA)- Materials -Process – top down and bottom up approach - Advantages - Limitations - Applications. Digital Light Processing (DLP) - Process - Advantages - Applications. Continuous Liquid Interface Production (CLIP) Technology. Directed Energy Deposition: Laser Engineered Net Shaping (LENS)- Process - Material Delivery - Materials -Benefits -Applications.

UNIT: IV	POWDER BED FUSION AND MATERIAL EXTRUSION	9

Powder Bed Fusion: Selective Laser Sintering (SLS): Process - Powder Fusion Mechanism - Materials and Application. Selective Laser Melting (SLM), Electron Beam Melting (EBM): Materials - Process - Advantages and Applications. Material Extrusion: Fused Deposition Modeling (FDM)- Process-Materials -Applications and Limitations.

UNIT: V	ADVANCED ADDITIVE MANUFACTURING PROCESSES	9

Binder Jetting: Three-Dimensional Printing - Materials - Process - Benefits- Limitations - Applications. Material Jetting: Multijet Modeling- Materials - Process - Benefits - Applications. Sheet Lamination: Laminated Object Manufacturing (LOM)- Basic Principle- Mechanism: Gluing or Adhesive Bonding - Thermal Bonding- Materials- Application and Limitation.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of Publication**
1	Ian Gibson, David Rosen, Brent Stucker, Mahyar Khorasani	Additive Manufacturing Technologies	Springer	2021
2	Andreas Gebhardt and Jan-Steffen Hötter	Additive Manufacturing: 3D Printing for Prototyping and Manufacturing	Hanser Publications	2015
3	Andreas Gebhardt	Understanding Additive Manufacturing: Rapid Prototyping, Rapid Manufacturing	Hanser Gardner Publication	2011

REFERENCE BOOKS:				
1	Manu Srivastava, Sandeep Rathee, Sachin Maheshwari	Additive Manufacturing Fundamentals and Advancements	CRC Press	2020
2	Panagiotis Stavropoulos	Additive Manufacturing: Design, Processes and Applications	Springer	2023

WEB LEARNING RESOURCES:

1. https://www.coursera.org/specializations/additive-manufacturing
2. https://www.coursera.org/learn/additive-manufacturing-3d-printing
3. https://alison.com/course/additive-manufacturing-architecture
4. https://alison.com/course/the-future-of-additive-manufacturing-technologies
5. https://www.open.edu/openlearn/science-maths-technology/additive-manufacturing

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	3	2	2	3	-	-	1	-	-	-	3	1	-
CO2	3	3	2	1	-	-	-	-	-	-	2	3	1	-
CO3	2	3	2	1	3	-	-	-	-	-	-	3	1	-
CO4	2	3	1	2	3	-	-	1	-	-	2	-	-	1
CO5	3	3	2	2	3	-	-	-	-	-	-	3	1	1
Avg	2.8	3	1.8	1.6	3	-	-	1	-	-	2	3	1	1

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AEX16	AEROSPACE MATERIALS	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Materials science fundamentals, crystal structures, phase diagrams, mechanical properties, and basic metallurgy concepts.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand classification, properties, and selection criteria of materials used in aerospace structures.
2. To study metallic materials and their processing methods for aerospace engineering applications.
3. To analyze characteristics and applications of polymeric and ceramic materials in aerospace systems.
4. To understand fundamentals, manufacturing methods and applications of composite materials in aerospace structures.
5. To explore advanced aerospace materials and their role in modern aerospace technologies.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain classification, properties, and selection criteria of materials used in aerospace structures.

CO2: Analyze characteristics and applications of metallic materials used in aerospace structural components.

CO3: Interpret properties and uses of polymeric and ceramic materials in aerospace engineering.

CO4: Evaluate composite materials, manufacturing techniques, and their advantages in aerospace structural design.

CO5: Assess advanced aerospace materials and their applications in modern aerospace technologies.

UNIT: I	INTRODUCTION TO AEROSPACE MATERIALS	9

Overview of aerospace materials, classification into metals, polymers, ceramics and composites, required properties, crystal structure basics, mechanical properties, environmental effects, processing techniques and aerospace structural applications.

UNIT: II	METALLIC MATERIALS FOR AEROSPACE APPLICATIONS	9

Study of aerospace metallic materials including aluminum, titanium, magnesium and superalloys, heat treatment processes, strengthening mechanisms, corrosion behavior, and structural applications in aircraft and engine components.

UNIT: III	POLYMERIC AND CERAMIC MATERIALS	9

Introduction to aerospace polymers and ceramics, classification of thermoplastics and thermosets, high temperature ceramics, processing methods, properties, advantages, limitations and applications in insulation and protection.

UNIT: IV	COMPOSITE MATERIALS IN AEROSPACE	9

Fundamentals of composite materials, matrix and reinforcement types, fiber reinforced composites, manufacturing methods, laminate structures, mechanical properties, failure mechanisms and aerospace structural applications.

UNIT: V	ADVANCED AEROSPACE MATERIALS AND APPLICATIONS	9

Overview of advanced aerospace materials including smart materials, nano-materials and functionally graded materials, high temperature alloys, environmental effects, testing techniques, design considerations and future developments.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Michael F. Ashby, David R. H. Jones	*Engineering Materials 1: An Introduction to Properties, Applications and Design*	Butterworth-Heinemann	2022

2	B. C. Hosford	Materials for Engineers	Cambridge University Press	2022
REFERENCE BOOKS:				
1	T. H. G. Megson	*Aircraft Structures for Engineering Students*	Elsevier	2017
2	Ian Baker, S. R. Nutt	*Composite Materials for Aircraft Structures*	American Institute of Aeronautics and Astronautics (AIAA)	2016
WEB LEARNING RESOURCES:				

1. https://nptel.ac.in/courses/101106044
2. https://ocw.mit.edu/courses/materials-science-and-engineering
3. https://www.grc.nasa.gov/www/k-12/airplane/matls.html
4. https://www.sciencedirect.com/topics/engineering/aerospace-materials
5. https://www.azom.com/article.aspx?ArticleID=12021

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	2	1	-	1	-	1	1	2	1	1
CO2	3	2	1	1	2	1	-	-	-	1	1	2	1	1
CO3	3	2	2	1	2	-	-	-	-	1	1	2	1	1
CO4	3	2	1	1	2	1	-	1	-	-	1	2	1	1
CO5	3	2	2	1	2	1	-	-	-	1	1	2	1	1
Avg	3	2	1.6	1	2	1	-	1	-	1	1	2	1	1

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AEX17	**EXPERIMENTAL STRESS ANALYSIS**	**L**	**T**	**P**	**C**	**PE**
		3	**0**	**0**	**3**	

PRE-REQUISITES:

Nil

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand principles, characteristics, and requirements of mechanical measurement systems in engineering.
2. To study construction, working principles and applications of electrical resistance strain gauges.
3. To analyze strain gauge circuits and instrumentation techniques for accurate strain measurement.
4. To understand photo-elastic methods used for experimental stress analysis in engineering structures.
5. To explore various non-destructive testing techniques for material evaluation and defect detection.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain principles, characteristics, and errors associated with mechanical measurement systems and instrumentation.

CO2: Describe construction, working principles, and applications of electrical resistance strain gauges.

CO3: Analyze Wheatstone bridge circuits and instrumentation used for accurate strain measurement applications.

CO4: Interpret photoelastic techniques for determining stress distribution in engineering components experimentally.

CO5: Evaluate non-destructive testing methods for detecting defects and assessing structural material integrity.

UNIT: I	**BASICS OF MECHANICAL MEASUREMENTS**	**9**

. Basic Characteristics and Requirements of a Measuring System – Principles of Measurements – Precision, Accuracy, Sensitivity and Range of Measurements – Sources of Error – Statistical Analysis of Experimental Data – Contact Type Mechanical Extensometers – Advantages and Disadvantages – Examples of Non -Contact Measurement Techniques.

UNIT: II	**ELECTRICAL-RESISTANCE STRAIN GAUGES**	**9**

Strain Sensitivity in Metallic Alloys – Gage Construction – Gage Sensitivities and Gage Factor – Corrections for Transverse Strain Effects – Performance Characteristics of Foil Strain Gages – Materials Used for Strain Gauges – Environmental Effects – The Three-Element Rectangular - Rosette for Strain Measurement – Other Types of Strain Gages – Semiconductor Strain Gages - Grid & Brittle Coating Methods of Strain Analysis

UNIT: III	**STRAIN-GAUGE CIRCUITS & INSTRUMENTATION**	**9**

The Potentiometer Circuit and Its Application to Strain Measurement – Variations from Basic Circuit –Circuit Output – The Wheatstone Bridge Circuit – Current and Constant Voltage Circuits – Analog to Digital Conversion – Calibrating Strain-Gage Circuits – Effects of Lead Wires and Switches – Electrical Noise -– Strain Measurement in Bars, Beams and Shafts – Circuit Sensitivity & Circuit Efficiency**.**

UNIT: IV	**PHOTOELASTIC METHODS OF STRESS ANALYSIS**	**9**

Introduction to Photo elastic Methods – Stress-Optic Law – Effects of a Stressed Model in a Plane Polariscope – Effects of a Stressed Model in a Circular Polariscope - Tardy Compensation - Two-Dimensional Photo elastic Stress Analysis – Fringe Multiplication and Fringe Sharpening - Materials for Two-Dimensional Photo elasticity - Properties and Calibration of Commonly Employed Photo elastic Materials – Introduction to Three-Dimensional Photo elasticity.

UNIT: V	**NON-DESTRUCTIVE TESTING**	**9**

Different types of NDT Techniques - Acoustic Emission Technique – Ultrasonic – Pulse-Echo – Through Transmission – Eddy Current Testing – Magnetic Particle Inspection – X-Ray Radiography – Challenges in Non-Destructive Evaluation – Non-Destructive Evaluation in Composites – Image Processing Basics.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of Publication**
1	Dally, J.W., and Riley, W.F.,	Experimental Stress Analysis	McGraw Hill Inc	2021
2	Sadhu Singh	Experimental Stress Analysis	Khanna Publishers	2019
REFERENCE BOOKS:				
1	James F. Doyle and James W. Phillips	Manual on Experimental Stress Analysis	McGraw Hill	2021
2	Albert S. Kobayashi	Handbook on Experimental Mechanics	Prentice Hall Publishers	2022

WEB LEARNING RESOURCES:

1. https://nptel.ac.in/courses/112106179
2. https://www.nde-ed.org
3. https://ocw.mit.edu/courses/mechanical-engineering/
4. https://www.efunda.com/formulae/solid_mechanics/strain_gages/strain_gage_intro.cfm
5. https://www.asnt.org/what-is-ndt

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	2	1	-	-	-	1	1	3	1	1
CO2	3	2	1	-	2	1	-	-	-	-	1	3	1	1
CO3	3	-	2	1	2	-	-	-	-	1	1	3	1	1
CO4	2	-	1	-	2	1	-	-	-	1	1	2	1	1
CO5	2	2	2	1	2	1	-	-	-	1	1	2	1	1
Avg	2.6	2	1.6	1	2	1	-	-	-	1	1	2.6	1	1

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: V
24AEX18	SMART MATERIALS AND STRUCTURES	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Basic mechanics, materials science fundamentals, stress strain concepts, vibrations, and electrical principles basics.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand fundamentals and applications of smart materials used in engineering systems.
2. To study piezoelectric material behavior, constitutive relations and actuator sensor characteristics.
3. To analyze beam modeling techniques using piezoelectric actuators for smart structural systems.
4. To understand shape memory alloy behavior, phase transformation and actuation mechanisms.
5. To explore constitutive modeling and thermomechanical behavior of shape memory alloy systems.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain fundamentals, classifications and engineering applications of smart materials and adaptive structures.

CO2: Analyze piezoelectric material behavior, coupling coefficients, actuator performance and nonlinear effects.

CO3: Apply beam modeling concepts to analyze piezoelectric actuator based smart structures.

CO4: Interpret shape memory alloy behavior including pseudo-elasticity and martensitic phase transformation mechanisms.

CO5: Evaluate constitutive models and thermomechanical behavior of shape memory alloy actuators.

UNIT: I	OVERVIEW AND INTRODUCTION	9

Piezoelectric materials fundamentals, crystal structure, shape memory alloys basics, phase transformation, electro-strictive materials, ER and MR fluids, and major applications in aerospace, automotive, medical systems, robotics, electronics and energy harvesting.

UNIT: II	PIEZOELECTRIC THEORY	9

Electromechanical constitutive equations, piezoceramic actuator and sensor equations, coupling coefficients, load line analysis, hysteresis and nonlinear behavior, static and dynamic excitation, depoling effects, Curie temperature, equivalent circuits.

UNIT: III	BEAM MODELLING WITH PIEZOELECTRIC MATERIAL	9

Stress, strain and displacement in beams, Euler–Bernoulli beam theory, transverse deflection, piezoelectric actuator beam models, symmetric and asymmetric actuation, embedded actuators, and experimental testing of surface mounted piezoelectric beams.

UNIT: IV	UNDERSTANDING SHAPE MEMORY ALLOYS	9

Stress–strain characteristics, one-way and two-way shape memory effects, pseudo-elasticity, martensitic transformation, R-phase transformation, constrained and free recovery behavior, SMA actuator load lines, transformation temperature characteristics.

UNIT: V	CONSTITUTIVE MODELLING AND SMA BEHAVIOUR	9

Constitutive models including Tanaka, Liang–Rogers and Brinson models, stress–strain–temperature relations, SMA wire testing, thermomechanical energy equilibrium, power requirements, resistance behavior, heat dissipation and damping characteristics.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Inderjit Chopra and Jayant Sirohi	Smart Structures Theory	Cambridge University Press,	2022

2	Martin, J.W.	Engineering Materials Their properties and Applications	Wykedham Publications	2021
REFERENCE BOOKS:				
1	M. V. Gandhi, B. S. Thompson	Smart Materials and Structures	Springer (Chapman & Hall)	2022
2	Mel Schwartz	Smart Materials	CRC Press	2019
WEB LEARNING RESOURCES:				

1. https://nptel.ac.in/courses/112104031
2. https://ocw.mit.edu/courses/mechanical-engineering/
3. https://www.sciencedirect.com/topics/engineering/smart-materials
4. https://www.azom.com/article.aspx?ArticleID=1565
5. https://www.comsol.com/multiphysics/piezoelectric-effect

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	2	2	2	-	2	1	-	-	-	1	1	2	1	1
CO2	3	2	1	-	2	1	-	-	-	1	1	2	1	1
CO3	3	2	2	-	2	-	1	-	-	1	1	2	1	1
CO4	2	2	2	-	2	1	-	-	-	1	1	2	1	1
CO5	2	2	2	-	2	1	1	-	-	1	1	2	1	1
Avg	2.4	2	1.8	**-**	2	1	1	-	-	1	1	2	1	1

M.A.M SCHOOL OF ENGINEERING

(An Autonomous Institution)
(Accredited by NAAC || Approved by AICTE || Affiliated to Anna University)
Trichy – Chennai Trunk Road, Siruganur, Tiruchirappalli – 621 105

B.E. AERONAUTICAL ENGINEERING

ELECTIVES LIST

VERTICAL IV - PROFESSIONAL ELECTIVE IV
DIVERSIFIED GROUP

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1	24AEX19	Aircraft General Engineering & Maintenance Practices	3	0	0	3	40	60	100	PE
2	24AEX20	Design of UAV systems	3	0	0	3	40	60	100	PE
3	24AEX21	Non-Destructive Testing and Evaluation	3	0	0	3	40	60	100	PE
4	24AEX22	Aircraft Hydraulics Systems	3	0	0	3	40	60	100	PE
5	24AEX23	Air Traffic Control	3	0	0	3	40	60	100	PE
6	24AEX24	Aircraft Rules and Regulations	3	0	0	3	40	60	100	PE
7	24AEX25	MEMS	3	0	0	3	40	60	100	PE

REGULATIONS 2024

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VI
24AEX19	AIRCRAFT GENERAL ENGINEERING & MAINTENANCE PRACTICES	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Basic knowledge of aircraft systems, engineering mechanics, materials, workshop practices, tools, and fundamental aircraft maintenance procedures and safety regulations.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand fundamental aircraft ground handling procedures and associated support equipment.
2. To learn servicing practices of aircraft subsystems including air conditioning, oxygen and oil systems.
3. To develop knowledge of aircraft maintenance shop safety, tools and standard maintenance practices.
4. To understand aircraft inspection procedures, documentation, manuals and airworthiness regulations.
5. To study aircraft hardware, materials, fittings and cable maintenance including swaging techniques.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain aircraft ground handling procedures, support equipment usage and safety precautions effectively.

CO2: Describe servicing procedures of aircraft subsystems including oxygen, oil, pressurization and air conditioning.

CO3: Apply aircraft maintenance shop safety practices using appropriate tools and precision instruments.

CO4: Interpret aircraft inspection procedures, manuals, airworthiness directives and standard maintenance documentation.

CO5: Identify aircraft hardware, materials, fittings and perform cable swaging procedures correctly.

UNIT: I	AIRCRAFT GROUND OPERATIONS	9

Aircraft ground handling procedures including mooring, jacking, leveling and towing operations. Preparation steps, ground support equipment usage, safety precautions during handling, engine starting procedures for piston, turboprop and turbojet engines.

UNIT: II	AIRCRAFT SUBSYSTEM SERVICING	9

Servicing practices of aircraft subsystems including air conditioning, cabin pressurization, oxygen supply and oil systems. Ground servicing units used for subsystem maintenance, operational procedures, handling precautions and routine maintenance requirements.

UNIT: III	MAINTENANCE OF AIRCRAFT SAFETY SYSTEMS	9

Aircraft maintenance shop safety procedures, environmental cleanliness and operational precautions. Usage of hand tools, precision measuring instruments and special maintenance equipment. Identification terminology and standard practices followed in aircraft maintenance workshops.

UNIT: IV	INSPECTION AND AIRWORTHINESS DOCUMENTATION	9

Aircraft inspection processes, objectives and classification of inspection types. Inspection intervals, techniques and checklist procedures. Maintenance documentation including manuals, service bulletins, airworthiness directives, type certificate data sheets and ATA specifications.

UNIT: V	AIRCRAFT HARDWARE MAINTENANCE PRACTICES	9

Aircraft hardware specifications including nuts, bolts, rivets and screws. American and British specification systems, threads, gears and bearings. Use of drills, taps and reamers, fluid line fittings, materials, plumbing connectors, cables and swaging procedures.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Kroes Watkins	Aircraft Maintenance and Repair	McGraw Hill	2023
2	Dale Crane	Aviation Maintenance Technician Handbook	Aviation Supplies & Academics (ASA)	2018

REFERENCE BOOKS:				
1	Jeppesen	Aviation Maintenance Technician:	Jeppesen Sanderson Inc.	2019
2	Larry Reithmaier	Standard Aircraft Handbook for Mechanics and Technicians	McGraw-Hill Education	2024

WEB LEARNING RESOURCES:

1. https://www.faa.gov/regulations_policies/handbooks_manuals/aviation
2. https://www.easa.europa.eu/en/document-library
3. https://www.aviationmaintenance.edu
4. https://skybrary.aero
5. https://www.aircraftsystemstech.com

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	2	-	-	-	-	-	-	-	-	-	-	2	-	-
CO2	2	-	1	-	-	-	1	-	1	-	-	2	-	-
CO3	3	1	-	-	-	-	1	-	1	1	-	3	2	1
CO4	3	1	-	-	-	-	-	-	1	-	-	2	1	1
CO5	2	-	-	-	-	-	-	-	-	-	-	2	2	1
Avg	2.4	1	1	-	-	-	1	-	1	1	-	2.2	1.6	1

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VI
24AEX20	DESIGN OF UAV SYSTEMS	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Basic knowledge of aerodynamics, aircraft structures, flight mechanics, electronics fundamentals and control systems

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand the history, classification and applications of unmanned aerial vehicles.
2. To study aerodynamic design principles and airframe configurations used in UAV systems.
3. To learn avionics hardware components including sensors, actuators and autopilot systems.
4. To understand communication systems, payload integration and control mechanisms in UAV operations.
5. To analyze UAV system development, navigation methods, testing procedures and operational challenges.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain UAV history, classification, components and applications in civil and military domains.

CO2: Analyze aerodynamic design principles and airframe configurations used in various UAV systems.

CO3: Describe avionics hardware including sensors, actuators, autopilot systems and integration procedures.

CO4: Evaluate UAV communication systems, telemetry links, payload integration and control algorithms effectively.

CO5: Apply waypoint navigation, ground control software and testing procedures for UAV development.

UNIT: I	FUNDAMENTALS OF UNMANNED AERIAL VEHICLES	9

Evolution and history of unmanned aerial vehicles, classification of UAV types, introduction to unmanned aircraft systems architecture, system components, prototype development, operational concepts, mission roles, and civilian, military, surveillance applications.

UNIT: II	UAV AERODYNAMIC DESIGN AND CONFIGURATION	9

Principles of UAV aerodynamic design, airframe configuration selection, performance characteristics of different aircraft types, design standards, regulatory considerations in USA, UK and Europe, stealth concepts, control surfaces and specification requirements.

UNIT: III	UAV AVIONICS AND SENSOR HARDWARE SYSTEMS	9

Avionics architecture for UAVs including autopilot systems, AGL and pressure sensors, servos, accelerometers, gyroscopes, actuators, onboard power supply units, processors, hardware integration, installation procedures, configuration and system testing.

UNIT: IV	COMMUNICATION, PAYLOAD AND CONTROL SYSTEMS	9

Payload integration including cameras and sensors, telemetry systems, communication links, tracking mechanisms, aerial photography systems, control algorithms, PID feedback mechanisms, radio control frequencies, modems, onboard memory and data management.

UNIT: V	UAV DEVELOPMENT, TESTING AND APPLICATION	9

Waypoint navigation techniques, ground control software operation, system ground testing procedures, in-flight testing methods, troubleshooting approaches, future prospects of UAV technology, operational challenges, and case studies of mini and micro UAV systems.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Reg Austin	Unmanned Aircraft Systems UAV design, development and deployment	Wiley	2020
2	Paul G Fahlstrom, Thomas J Gleason	Introduction to UAV Systems	UAV Systems, Inc	2019

REFERENCE BOOKS:				
1	Dr. Armand J. Chaput	Design of Unmanned Air Vehicle Systems	Lockheed Martin	2021
2	Robert C. Nelson	Flight Stability and Automatic Control	McGraw-Hill	2018

WEB LEARNING RESOURCES:

1. https://in.mathworks.com/academia/student-competitions/resources/uavs.html
2. https://aiaa.org/courses/uav-aircraft-design-by-dan-raymer/
3. https://www.udemy.com/course/fixed-wing-uav-design/
4. https://www.udbharat.com/drone-engineering
5. https://www.academyinnovaworld.org/uav

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	1	1	2	-	-	-	-	-	-	1	1	-	-
CO2	2	-	-	-	-	1	-	-	-	-	-	2	-	-
CO3	2	3	1	-	-	1	-	-	-	-	1	1	1	-
CO4	3	2	-	-	-	1	1	-	-	-	-	1	1	-
CO5	2	-	1	1	3	-	-	-	1	-	1	-	-	-
Avg	**2.4**	**2**	**1**	**1.5**	**3**	**1**	**1**	-	**1**	-	**1**	**1.2**	**1**	-

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VI
24AEX21	NON-DESTRUCTIVE TESTING AND EVALUATION	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To acquaint the students with the overview of NDT
2. To elaborate the concept and procedure for liquid and magnetic penetrant testing and evaluate through practical study
3. To introduce the concept and procedure for radiograph testing methods and evaluate through practical study
4. To brief the concepts and procedures for Ultrasonic testing methods and their applications
5. To impart knowledge in other methods of NDT and electrical method with case study

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain principles of non destructive testing methods used for defect detection and evaluation.

CO2: Apply liquid penetrant and magnetic particle testing methods for surface defect identification.

CO3: Analyze thermography and eddy current testing techniques for material inspection applications.

CO4: Demonstrate ultrasonic testing and acoustic emission techniques for internal flaw detection.

CO5: Evaluate radiographic testing methods for identifying defects in engineering materials and components.

UNIT: I	FUNDAMENTALS OF NON-DESTRUCTIVE TESTING	9

Introduction to non-destructive testing and comparison with mechanical testing methods. Overview of common NDT techniques for detecting manufacturing defects and material characterization. Material properties, advantages, limitations, and visual inspection methods.

UNIT: II	LIQUID PENETRANT & MAGNETIC PARTICLE TESTING	9

Liquid penetrant testing principles, types of penetrants, developers, procedures, advantages, limitations and interpretation of results. Magnetic particle testing fundamentals including magnetization methods, inspection materials, defect detection, evaluation of indications and demagnetization.

UNIT: III	THERMOGRAPHY AND EDDY CURRENT TESTING	9

Thermography principles including contact and non-contact inspection techniques, infrared radiation detectors, instrumentation and applications. Eddy current testing fundamentals including generation of currents, probes, sensing elements, instrumentation, advantages, limitations and evaluation.

UNIT: IV	ULTRASONIC TESTING AND ACOUSTIC EMISSION	9

Ultrasonic testing principles including transducers, pulse echo and transmission techniques, straight and angle beam inspection methods, instrumentation and scan types. Acoustic emission testing fundamentals, parameters, monitoring techniques and engineering applications.

UNIT: V	RADIOGRAPHIC TESTING METHODS	9

Radiographic testing principles using X-rays, interaction with materials, film and digital imaging techniques, filters and screens. Radiographic parameters including film characteristics, exposure methods, penetrameters, and advanced imaging techniques including computed tomography.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Ravi Prakash	Non-Destructive Testing Techniques	New Age International Publishers,	2020
2	Baldev Raj, T.Jayakumar	Practical Non-Destructive Testing	Narosa Publishing House	2022

REFERENCE BOOKS:				
1	J. Prasad	Non-Destructive Test and Evaluation of Materials	McGraw-Hill Education	2023
2	Paul E. Mix	Introduction to Non-Destructive Testing: A Training Guide	Wiley	2022

WEB LEARNING RESOURCES:

1. https://www.nde-ed.org
2. https://www.asnt.org
3. https://www.ndeaa.com
4. https://www.twi-global.com/technical-knowledge
5. https://www.faa.gov/aircraft/air_cert/design_approvals/ndt

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	2	1	1	2	2	2	2	3	2	2
CO2	3	2	2	1	2	1	1	2	2	2	2	3	2	2
CO3	3	2	2	1	2	1	1	2	3	2	2	3	2	2
CO4	2	2	2	1	2	1	1	2	3	2	2	3	1	2
CO5	3	2	2	1	2	1	1	2	3	1	1	3	2	2
Avg	**2.8**	**2**	**2**	**1**	**2**	**1**	**1**	**2**	**2.6**	**1.8**	**1.8**	**3**	**1.8**	**2**

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VI
24AEX22	AIRCRAFT HYDRAULICS SYSTEMS	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Basic knowledge of aircraft systems, fluid mechanics, engineering mechanics and aircraft maintenance fundamentals.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand fundamentals and working principles of aircraft hydraulic systems.
2. To study hydraulic fluids, pumps, valves and system components used in aircraft.
3. To analyze aircraft hydraulic circuits used for landing gear, brakes and flight controls.
4. To understand operation, maintenance and troubleshooting of aircraft hydraulic systems.
5. To study safety practices and modern developments in aircraft hydraulic technology.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain principles, components and operation of aircraft hydraulic systems clearly.

CO2: Describe hydraulic fluids, pumps, actuators and control valves used in aircraft.

CO3: Analyze hydraulic circuits used for landing gear, braking and flight control systems.

CO4: Apply maintenance procedures and troubleshooting techniques for aircraft hydraulic system components.

CO5: Evaluate safety practices and modern developments in aircraft hydraulic technology.

UNIT: I	INTRODUCTION TO AIRCRAFT HYDRAULIC SYSTEMS	9

Basic principles of hydraulics, Pascal’s law, advantages of hydraulic systems in aircraft. Layout of aircraft hydraulic systems, system components, hydraulic reservoirs, pumps, actuators, filters and accumulators.

UNIT: II	HYDRAULIC FLUIDS AND SYSTEM COMPONENTS	9

Types of aircraft hydraulic fluids, properties, selection and handling procedures. Hydraulic pumps, valves, actuators, pressure regulators, accumulators, filters and system accessories used in aircraft hydraulic systems.

UNIT: III	AIRCRAFT HYDRAULIC CIRCUITS	9

Hydraulic circuit design and operation. Application of hydraulic systems in landing gear operation, braking systems, nose wheel steering, flaps, spoilers and flight control systems.

UNIT: IV	OPERATION AND MAINTENANCE	9

Hydraulic system servicing procedures, inspection methods, maintenance practices, troubleshooting techniques, contamination control and safety precautions followed in aircraft hydraulic system maintenance.

UNIT: V	ADVANCED AIRCRAFT HYDRAULIC SYSTEMS	9

Modern hydraulic systems, power-by-wire concepts, redundancy in aircraft hydraulics, system monitoring and diagnostics. Case studies of hydraulic systems in modern commercial and military aircraft.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Thomas W. Wild	Aircraft Power Systems	John Wiley & Sons	2024
2	Jeppesen	Aviation Maintenance Technician Handbook – Airframe	FAA / Jeppesen	2021
3	Dale Crane	Aircraft Systems for Pilots	Aviation Supplies & Academics	2017

REFERENCE BOOKS:				
1	Ian Moir, Allan Seabridge	Aircraft Systems: Mechanical, Electrical and Avionics Subsystems Integration	Wiley	2019
2	Michael J. Kroes	Aircraft Maintenance and Repair	McGraw-Hill Education	2021

WEB LEARNING RESOURCES:

1. https://www.faa.gov/regulations_policies/handbooks_manuals/aviation
2. https://www.aircraftsystemstech.com
3. https://nptel.ac.in/courses/101
4. https://skybrary.aero
5. https://www.easa.europa.eu/en/document-library

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	1	-	-	-	-	-	-	-	-	2	1	-
CO2	3	2	2	1	-	-	-	-	-	-	-	2	1	-
CO3	3	3	2	1	-	-	-	-	-	-	-	3	2	-
CO4	2	2	2	2	1	-	-	-	-	-	-	2	2	-
CO5	2	2	1	1	1	-	-	-	-	1	1	2	2	1
Avg	**2.6**	**2.2**	**1.6**	**1**	**1**	-	-	-	-	**1**	**1**	**2.2**	**1.6**	**1**

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VI
24AEX23	AIR TRAFFIC CONTROL	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Basics of aircraft operations, navigation principles and communication systems.

COURSE OBJECTIVES:

The objectives of learning this course are:
1. To introduce the fundamentals and evolution of air traffic control systems.
2. To understand communication, navigation and surveillance systems used in ATC.
3. To study airspace organization and air traffic management procedures.
4. To learn airport control operations and separation standards.
5. To familiarize students with modern air traffic management technologies and safety practices.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain the basic concepts, history and structure of air traffic control systems.

CO2: Describe communication, navigation and surveillance technologies used in ATC operations.

CO3: Analyze airspace classification, flight rules and air traffic management procedures.

CO4: Interpret airport traffic control operations, separation standards and control tower functions.

CO5: Evaluate modern air traffic management systems and safety measures in aviation operations.

UNIT: I	INTRODUCTION TO AIR TRAFFIC CONTROL	9

History and evolution of air traffic control, need for ATC, objectives of air traffic services, ICAO regulations, organization of air traffic services, classification of airspace, flight rules – VFR and IFR, structure of air traffic control system.

UNIT: II	COMMUNICATION AND NAVIGATION SYSTEMS	9

Aeronautical communication systems, VHF communication, data link communication, navigation aids – VOR, DME, ILS, GNSS, radar systems – primary and secondary radar, Automatic Dependent Surveillance – Broadcast (ADS-B), surveillance technologies used in ATC.

UNIT: III	AIRSPACE AND AIR TRAFFIC MANAGEMENT	9

Airspace classification, airway system, control areas and terminal areas, flight planning procedures, air traffic flow management, flight progress strips, coordination between ATC units, responsibilities of air traffic controllers.

UNIT: IV	AIRPORT TRAFFIC CONTROL	9

Airport control tower operations, ground control, approach and departure control, runway operations, taxiway management, aircraft separation standards, wake turbulence separation, emergency handling and contingency procedures.

UNIT: V	MODERN AIR TRAFFIC MANAGEMENT SYSTEMS	9

Satellite based navigation systems, Performance Based Navigation (PBN), NextGen and SESAR concepts, digital ATC systems, automation in ATC, safety management systems, future trends in global air traffic management.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	M. Nolan	Fundamentals of Air Traffic Control	Cengage Learning	2019
2	A. K. Bhatia	Air Traffic Control	Sterling Publishers	2021
3	J. G. Ward	Air Traffic Control	McGraw Hill	2020

REFERENCE BOOKS:				
1	I. Moir & A. Seabridge	Aircraft Systems: Mechanical, Electrical and Avionics Subsystems	Wiley	2018
2	R. W. Hurst	Air Traffic Control Systems	Prentice Hall	2023

WEB LEARNING RESOURCES:

1. https://www.faa.gov/air_traffic
2. https://www.icao.int
3. https://www.eurocontrol.int
4. https://www.skybrary.aero
5. https://www.aai.aero

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	1	-	-	-	-	1	-	1	-	2	1	-
CO2	3	3	1	2	2	-	-	1	-	1	-	3	2	1
CO3	2	3	2	2	1	-	-	1	-	1	-	2	2	1
CO4	2	2	2	2	1	-	-	1	1	1	-	2	2	1
CO5	2	2	2	2	2	-	1	1	1	1	1	2	2	2
Avg	**2.4**	**2.4**	**1.6**	**1.6**	**1.2**	**-**	**1**	**1**	**1**	**1**	**1**	**2.2**	**1.8**	**1.0**

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VI
24AEX24	AIRCRAFT RULES AND REGULATIONS	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

Basic knowledge of aviation operations, aircraft systems, air navigation and aviation safety principles.

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To introduce national and international aviation regulatory frameworks.
2. To understand aircraft certification procedures and regulatory requirements.
3. To study aviation safety regulations and airworthiness standards.
4. To familiarize students with airport, airline and air navigation regulations.
5. To understand the role of aviation regulatory authorities and legal requirements.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain the structure and functions of international and national aviation regulatory bodies.

CO2: Describe aircraft certification procedures and airworthiness requirements.

CO3: Interpret aviation operational rules related to aircraft operations and licensing.

CO4: Analyze airport and air navigation service regulations governing aviation operations.

CO5: Evaluate aviation safety regulations and legal frameworks for safe aircraft operations.

UNIT: I INTRODUCTION TO AVIATION REGULATIONS 9

History of aviation regulations, need for regulatory framework, international aviation organizations, role of the International Civil Aviation Organization, structure of global aviation regulatory systems, national aviation authorities, overview of aviation law and regulatory policies.

UNIT: II AIRCRAFT CERTIFICATION AND AIRWORTHINESS 9

Aircraft certification procedures, type certification, production certification, airworthiness certification, continued airworthiness, maintenance regulations, certification standards and regulatory approval processes.

UNIT: III AIRCRAFT OPERATIONS AND FLIGHT RULES 9

Rules governing aircraft operations, visual flight rules (VFR) and instrument flight rules (IFR), flight crew licensing requirements, pilot responsibilities, operational limitations, aviation operational documentation and compliance procedures.

UNIT: IV AIRPORT AND AIR NAVIGATION REGULATIONS 9

Airport certification requirements, aerodrome standards, runway safety regulations, air navigation service regulations, communication and navigation standards, regulatory oversight for air traffic services.

UNIT: V AVIATION SAFETY AND LEGAL FRAMEWORK 9

Aviation safety regulations, accident investigation procedures, safety management systems (SMS), aviation liability laws, regulatory compliance and enforcement, future trends in aviation regulation. Regulatory role of the DGCA, FAA, CAA

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	K. K. Chitkara	Aircraft Rules and Regulations	Tata McGraw Hill	2020
2	Michael H. Gerardi	Aviation Law and Regulations	McGraw Hill	2021
3	R. D. Campbell	Introduction to Aviation Law	Routledge	2019

REFERENCE BOOKS:				
1	I. Moir & A. Seabridge	Aircraft Systems: Mechanical, Electrical and Avionics Subsystems	Wiley	2018
2	Paul Stephen Dempsey	Public International Air Law	McGill University Press	2022

WEB LEARNING RESOURCES:

1. https://www.icao.int
2. https://www.dgca.gov.in
3. https://www.faa.gov
4. https://www.easa.europa.eu
5. https://nptel.ac.in

CO – PO – PSO MAPPING

CO	**PO**											**PSO**		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	1	–	–	1	–	2	–	1	–	2	1	–
CO2	3	3	1	1	–	1	–	2	–	1	–	2	2	1
CO3	2	3	2	1	–	1	–	2	–	1	–	2	2	1
CO4	2	2	2	1	–	1	–	2	1	1	–	2	2	1
CO5	2	2	1	1	–	2	1	2	1	1	1	2	2	1
Avg	**2.4**	**2.4**	**1.4**	**0.8**	**–**	**1.2**	**0.2**	**2.0**	**0.4**	**1.0**	**0.2**	**2.0**	**1.8**	**0.8**

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VI
24AEX25	MEMS	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISITES:

NIL

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To introduce the fundamentals and working principles of MEMS devices.
2. To understand microfabrication techniques used in MEMS technology.
3. To study various MEMS sensors and actuators.
4. To analyze design considerations and modelling of MEMS devices.
5. To explore applications of MEMS in aerospace, biomedical and industrial systems.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain the basic concepts, principles and components of MEMS technology.

CO2: Describe microfabrication techniques used in MEMS device manufacturing.

CO3: Analyze different MEMS sensors and actuators and their working principles.

CO4: Interpret modelling and design aspects of MEMS devices.

CO5: Evaluate applications of MEMS technology in aerospace and engineering systems.

UNIT: I	INTRODUCTION TO MEMS	9

Introduction to MEMS, history and development of MEMS technology, micro systems and micro sensors, advantages and limitations of MEMS devices, scaling laws in MEMS, materials used in MEMS fabrication.

UNIT: II	MICROFABRICATION TECHNIQUES	9

Photolithography process, thin film deposition techniques, oxidation, diffusion and ion implantation, etching techniques – wet etching and dry etching, surface micromachining and bulk micromachining processes.

UNIT: III	MEMS SENSORS	9

Principles of MEMS sensing, pressure sensors, accelerometers, gyroscopes, temperature sensors, chemical and biological sensors, signal conditioning and interfacing circuits.

UNIT: IV	MEMS ACTUATORS	9

Electrostatic actuators, thermal actuators, piezoelectric actuators, magnetic actuators, micro motors and micro pumps, design considerations and performance characteristics.

UNIT: V	MEMS APPLICATIONS	9

Applications of MEMS in aerospace systems, inertial navigation sensors, micro propulsion devices, biomedical MEMS devices, automotive MEMS sensors, RF MEMS devices, future trends in MEMS technology.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Tai-Ran Hsu	MEMS and Microsystems: Design and Manufacture	McGraw Hill	2018
2	Mohamed Gad-el-Hak	MEMS: Design and Fabrication	CRC Press	2022
3	Nadim Maluf & Kirt Williams	An Introduction to Microelectromechanical Systems Engineering	Artech House	2021

REFERENCE BOOKS:				
1	Stephen D. Senturia	Microsystem Design	Springer	2021
2	Julian W. Gardner & Vijay Varadan	Microsensors, MEMS and Smart Devices	Wiley	2021

WEB LEARNING RESOURCES:

1. https://nptel.ac.in/courses/117105082
2. https://ocw.mit.edu
3. https://www.memsexchange.org
4. http://www.people.cornell.edu/pages/akt1/memsmain.html
5. http://www.memsnet.org

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	2	1	-	-	-	1	1	2	1	1
CO2	3	2	1	1	2	1	-	-	-	1	1	2	1	1
CO3	3	2	2	1	2	-	-	-	-	1	1	2	1	1
CO4	2	2	1	1	2	1	-	-	-	1	1	2	1	1
CO5	2	2	2	1	2	1	-	-	-	1	1	2	1	1
Avg	2.6	2	1.6	1	2	1	-	-	-	1	1	2	1	1

M.A.M SCHOOL OF ENGINEERING

(An Autonomous Institution)

(Accredited by NAAC || Approved by AICTE || Affiliated to Anna University)

Trichy – Chennai Trunk Road, Siruganur, Tiruchirappalli – 621 105

B.E. AERONAUTICAL ENGINEERING

ELECTIVES LIST

VERTICAL V - PROFESSIONAL ELECTIVE V

MANAGEMENT ELECTIVES

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1	24MGT01	Total Quality Management	3	0	0	3	40	60	100	PE
2	24MGT02	Principles of Management	3	0	0	3	40	60	100	PE
3	24MGT03	Industrial Management	3	0	0	3	40	60	100	PE
4	24MGT04	Safety Management	3	0	0	3	40	60	100	PE
5	24MGT05	Operational Management	3	0	0	3	40	60	100	PE

REGULATIONS 2024

Recommended by 3rd BOS held on 12.02.2026 and Approved by 3rd Academic Council held on 17.03.2026

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VII
24MGT01	TOTAL QUALITY MANAGEMENT	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISTES

Nil

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand concepts, principles and importance of Total Quality Management in organizations.
2. To study quality tools and techniques used for process improvement.
3. To analyze statistical methods for monitoring and controlling industrial processes.
4. To learn quality management systems, standards and auditing practices.
5. To apply continuous improvement methods such as Kaizen, Lean and Six Sigma.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Explain principles, philosophy and benefits of Total Quality Management in modern organizations effectively.

CO2: Apply basic quality tools for identifying, analyzing and solving organizational quality problems.

CO3: Analyze process variation using statistical techniques and interpret control charts for improvement.

CO4: Evaluate quality management systems, ISO standards and auditing practices for organizational implementation.

CO5: Implement continuous improvement techniques including Kaizen, Lean practices and Six Sigma concepts.

UNIT: I	INTRODUCTION TO TOTAL QUALITY MANAGEMENT	9

Concept of quality and quality management. Evolution of quality practices. Principles of Total Quality Management. Contributions of Deming, Juran and Crosby. Customer focus and satisfaction. Leadership and management commitment. Employee involvement and teamwork. Strategic quality planning. Benefits and barriers in TQM implementation.

UNIT: II	TQM TOOLS AND TECHNIQUES	9

Basic quality tools. Check sheets for data collection. Cause and effect diagram. Pareto analysis for problem priority. Histogram for data distribution. Scatter diagram for relationship study. Control charts for process monitoring. Flow charts for process analysis. Applications of quality tools.

UNIT: III	STATISTICAL PROCESS CONTROL	9

Process control and process variation. Types of variation. Statistical Process Control concepts. Control charts for variables. Control charts for attributes. Process capability and capability indices. Process performance analysis. Acceptance sampling methods. Industrial applications of SPC.

UNIT: IV	QUALITY MANAGEMENT SYSTEMS	9

Quality management system concepts. Quality documentation practices. ISO 9000 standards overview. ISO 9001 requirements. Quality auditing principles. Internal and external audits. Total productive maintenance basics. Benchmarking methods. Role of quality certification.

UNIT: V	CONTINUOUS IMPROVEMENT TECHNIQUES	9

Continuous improvement concepts. Kaizen philosophy. Quality circles. Lean manufacturing basics. Six Sigma methodology. DMAIC process. Failure Mode and Effects Analysis. Quality Function Deployment. Cost of quality and applications.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication

1	Dale H. Besterfield, Carol Besterfield-Michna, Glen H. Besterfield	Total Quality Management	Pearson Education	2022
2	K. C. Arora	Total Quality Management	S. K. Kataria & Sons	2022
3	Poornima M. Charantimath	Total Quality Management	Pearson Education	2021

REFERENCE BOOKS:				
1	P. N. Mukherjee	Total Quality Management	PHI Learning	2022
2	John S. Oakland	Total Quality Management and Operational Excellence	Routledge	2021

WEB LEARNING RESOURCES:

1. https://uniathena.com/short-courses/basics-of-total-quality-management
2. https://asq.org/quality-resources/total-quality-management
3. https://www.coursera.org/learn/six-sigma-principles
4. https://www.managementstudyguide.com/total-quality-management.htm

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	2	-	-	-	-	-	-	3	1	-
CO2	3	2	2	2	-	-	-	-	-	-	2	3	1	-
CO3	3	2	2	2	2	-	-	-	-	-	-	3	1	-
CO4	3	2	1	1	3	-	-	-	-	-	2	-	-	1
CO5	3	2	2	2	3	-	-	-	-	-	-	-	2	1
Avg	**3**	**2**	**1.8**	**1.6**	**2.5**	**-**	**-**	**-**	**-**	**-**	**2.0**	**3**	**1.4**	**1.0**

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VII
24MGT02	PRINCIPLES OF MANAGEMENT	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISTES

Nil

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand basic concepts, principles and functions of management in organizations.
2. To learn planning and decision-making techniques for effective organizational management.
3. To study organizing structures, authority relationships and coordination within modern organizations.
4. To understand leadership, motivation and communication for improving employee performance and productivity.
5. To learn controlling techniques for monitoring organizational performance and achieving planned objectives.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1:	Explain basic management principles, functions and their role in effective organizational administration.
CO2:	Apply planning and decision making techniques for solving managerial problems in organizations.
CO3:	Analyze organizational structures, authority relationships and coordination mechanisms for effective management practices.
CO4:	Demonstrate leadership, motivation and communication skills for improving employee performance and teamwork.
CO5:	Evaluate control techniques for monitoring organizational performance and achieving managerial objectives effectively.

UNIT: I	INTRODUCTION TO MANAGEMENT	9

Concept of management and organization. Nature and importance of management. Functions of management. Levels of management. Management roles and skills. Evolution of management thought. Contributions of Taylor and Fayol. Modern management approaches. Importance of management in organizations.

UNIT: II	PLANNING AND DECISION MAKING	9

Concept and importance of planning. Types of plans and planning process. Objectives and goal setting. Strategic planning concepts. Forecasting techniques. Decision making process. Management by objectives. Policies and procedures. Planning tools and techniques.

UNIT: III	ORGANIZATION STRUCTURE AND DESIGN	9

Concept and importance of organizing. Organization structure and design. Departmentation principles. Authority and responsibility. Delegation of authority. Span of control. Centralization and decentralization. Line and staff organization. Coordination in organizations.

UNIT: IV	LEADERSHIP AND MOTIVATION	9

Concept and importance of directing. Leadership and leadership styles. Motivation concepts and theories. Communication process in organizations. Barriers to communication. Supervision and guidance. Teamwork and group dynamics. Employee morale and discipline. Managerial decision making.

UNIT: V	PERFORMANCE AND CONTROL SYSTEMS	9

Concept and need for control. Steps in controlling process. Types of control techniques. Budgetary control methods. Performance standards and measurement. Variance analysis. Quality control basics. Corrective actions in control. Importance of effective control systems.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication

1	Harold Koontz, Heinz Weihrich	Essentials of Management	McGraw Hill Education	2019
2	P. C. Tripathi, P. N. Reddy	Principles of Management	Tata McGraw Hill	2023

REFERENCE BOOKS:				
1	Stephen P. Robbins, Mary Coulter	Management	Pearson Education	2020
2	L. M. Prasad	Principles and Practice of Management	Sultan Chand & Sons	2022

WEB LEARNING RESOURCES:

1. https://open.lib.umn.edu/principlesmanagement/
2. https://www.managementstudyguide.com/principles-of-management.htm
3. https://www.tutorialspoint.com/management_principles/index.htm
4. https://www.lumenlearning.com/courses/principles-of-management/
5. https://saylordotorg.github.io/text_principles-of-management/

CO – PO – PSO MAPPING

CO	**PO**											**PSO**		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	3	2	2	2	-	-	-	-	-	-	2	1	-
CO2	3	3	2	2	-	-	-	-	-	-	2	3	1	-
CO3	3	2	2	2	2	-	-	-	-	-	-	3	1	-
CO4	2	1	1	2	3	-	-	-	-	-	2	-	-	1
CO5	3	2	2	2	3	-	-	-	-	-	-	-	2	1
Avg	**2.8**	**2.2**	**1.8**	**2.0**	**2.5**	**-**	**-**	**-**	**-**	**-**	**2.0**	**2.8**	**1.4**	**1.0**

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VII
24MGT03	INDUSTRIAL MANAGEMENT	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISTES

Nil

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand concepts, scope and importance of industrial management in manufacturing organizations.
2. To learn plant location and layout principles for efficient industrial operations.
3. To study production planning and control techniques for effective manufacturing management.
4. To understand work study methods for improving productivity and operational efficiency.
5. To learn materials management and quality control practices in industrial production systems.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1:	Explain concepts, functions and importance of industrial management in modern manufacturing industries effectively.
CO2:	Analyze plant location and layout decisions for improving industrial productivity and operational efficiency.
CO3:	Apply production planning and control techniques for managing manufacturing processes efficiently.
CO4:	Evaluate work study methods to improve productivity, efficiency and utilization of resources.
CO5:	Apply materials management and quality control practices for improving industrial production performance.

UNIT: I	INTRODUCTION TO INDUSTRIAL MANAGEMENT	9

Concept of industrial management. Role of management in industries. Objectives and scope of industrial management. Functions of industrial management. Productivity and efficiency concepts. Industrial organization structure. Responsibilities of industrial managers. Industrial growth and development. Importance of industrial management.

UNIT: II	PLANT LOCATION AND PLANT LAYOUT	9

Concept of plant location. Factors influencing plant location. Selection of suitable plant site. Concept of plant layout. Types of plant layout. Product layout and process layout. Fixed position layout. Combination layout. Advantages of good plant layout.

UNIT: III	PRODUCTION PLANNING AND CONTROL	9

Concept of production planning. Objectives of production planning and control. Forecasting in production planning. Routing and scheduling. Loading and sequencing. Dispatching procedures. Expediting and follow-up. Production control techniques. Importance of production planning.

UNIT: IV	WORK STUDY AND INDUSTRIAL ENGINEERING	9

Concept of work study. Method study techniques. Work measurement methods. Time study and motion study. Job design and work simplification. Ergonomics in industry. Productivity improvement methods. Industrial safety practices. Applications of work study.

UNIT: V	MATERIALS AND QUALITY MANAGEMENT	9

Concept of materials management. Purchasing and inventory control. Stores management practices. Economic order quantity concept. Quality control in industries. Inspection and testing methods. Maintenance management basics. Cost control in production. Role of quality improvement in industry.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication

1	O. P. Khanna	Industrial Engineering and Management	Dhanpat Rai Publications	2021
2	K. K. Chitale, R. C. Gupta	Industrial Engineering and Management	PHI Learning	2023

REFERENCE BOOKS:				
1	T. R. Banga, S. C. Sharma	Industrial Engineering and Management	Khanna Publishers	2019
2	S. K. Sharma, Savita Sharma	Industrial Engineering and Management	S. K. Kataria & Sons	2022

WEB LEARNING RESOURCES:

1. https://www.managementstudyguide.com/industrial-management.htm
2. https://www.tutorialspoint.com/industrial_engineering/index.htm
3. https://www.scribd.com/doc/Industrial-Engineering-and-Management-Lecture-Notes
4. https://www.sharecourseware.org/
5. https://www.nptel.ac.in/courses/110/107/110107146/

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	3	1	2	3	-	-	-	-	-	-	-	1	-
CO2	2	-	1	2	-	-	-	-	-	-	2	3	1	-
CO3	3	2	1	2	-	-	-	-	-	-	-	3	1	-
CO4	2	1	1	2	3	-	-	-	-	-	2	-	1	1
CO5	2	2	1	2	3	-	-	-	-	-	-	3	1	1
Avg	**2.2**	**1.6**	**1**	**2.0**	**3**	**-**	**-**	**-**	**-**	**-**	**2.0**	**3**	**1**	**1**

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VII
24MGT04	SAFETY MANAGEMENT	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISTES

Nil

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand principles, concepts and importance of safety management in industrial workplaces.
2. To identify different types of hazards and risks present in industries.
3. To study accident prevention methods and fire safety practices in industries.
4. To learn occupational health, industrial hygiene and workplace safety improvement measures.
5. To understand safety standards, regulations and management systems used in industries.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1:	Explain basic concepts, principles and importance of safety management in industrial environments.
CO2:	Identify workplace hazards and apply risk assessment techniques for accident prevention effectively.
CO3:	Analyze accident causes and recommend suitable safety measures including fire prevention practices.
CO4:	Evaluate occupational health and industrial hygiene practices for maintaining safe working conditions.
CO5:	Apply safety standards, regulations and management systems for improving workplace safety performance.

UNIT: I	INTRODUCTION TO SAFETY MANAGEMENT	9

Concept of safety and safety management. Importance of safety in industries. Objectives of safety management. Types of industrial hazards. Causes of workplace accidents. Safety policies and procedures. Safety organization in industries. Roles of safety officers and employees. Benefits of effective safety management.

UNIT: II	INDUSTRIAL HAZARDS AND RISK ASSESSMENT	9

Types of industrial hazards. Mechanical hazards in industries. Electrical hazards and safety precautions. Chemical hazards and exposure risks. Hazard identification methods. Risk assessment techniques. Safety signs and symbols. Hazard control measures. Workplace hazard prevention practices.

UNIT: III	ACCIDENT PREVENTION AND FIRE SAFETY	9

Concept of accident prevention. Causes and analysis of accidents. Accident investigation methods. Safety inspection procedures. Personal protective equipment. Fire hazards in industries. Fire prevention and control methods. Fire extinguishers and fire safety systems. Emergency response and evacuation planning.

UNIT: IV	INDUSTRIAL HEALTH AND HYGIENE	9

Occupational health hazards. Industrial hygiene practices. Workplace environmental factors. Noise and vibration control. Ventilation and lighting systems. Ergonomics in workplace. Health monitoring of workers. Measures for healthy work environment.

UNIT: V	SAFETY STANDARDS AND MANAGEMENT SYSTEMS	9

Safety legislation and regulations. Factory safety provisions. Safety management systems in industries. OHSAS standards for occupational safety. ISO safety and health management standards. Safety audits and inspections. Documentation and safety records. Safety training programs. Continuous safety improvement.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	David L. Goetsch	Occupational Safety and Health for Technologists, Engineers and Managers	Pearson Education	2020
2	N. V. Krishnan	Safety Management in Industry	Jaico Publishing House	2018

REFERENCE BOOKS:				
1	K. U. Mistry	Fundamentals of Industrial Safety and Health	Siddharth Prakashan	2020
2	Charles D. Reese	Occupational Health and Safety Management	CRC Press	2019

WEB LEARNING RESOURCES:

1. OHSAS 18001 Occupational Health and Safety Management Course
2. https://uniathena.com/short-courses/essentials-of-safety-and-hazard-management
3. https://www.safetyhub.com/safety-training-courses/safety-basics/
4. https://www.osha.gov/training
5. https://www.ilo.org/global/topics/safety-and-health-at-work/lang--en/index.htm

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	2	3	2	2	2	-	-	-	-	-	-	3	1	-
CO2	3	1	-	2	-	-	-	-	-	-	2	3	1	-
CO3	2	2	2	2	3	-	-	-	-	-	-	3	1	-
CO4	2	1	-	2	2	-	-	-	-	-	2	3	-	1
CO5	3	2	2	2	3	-	-	-	-	-	-	3	2	1
Avg	**2.4**	**1.8**	**2.0**	**2.0**	**2.5**	**-**	**-**	**-**	**-**	**-**	**2.0**	**3**	**1.4**	**1.0**

R 2024	AERONAUTICAL ENGINEERING					SEMESTER: VII
24MGT05	OPERATIONAL MANAGEMENT	L	T	P	C	PE
		3	0	0	3	

PRE-REQUISTES

Nil

COURSE OBJECTIVES:

The objectives of learning this course are:

1. To understand concepts, scope and importance of operations management in organizations.
2. To learn product and process design for efficient production systems.
3. To study capacity planning and facility layout for effective operations management.
4. To understand production planning, scheduling and control techniques in manufacturing industries.
5. To learn inventory management and quality control methods in operational activities.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1:	Explain concepts, objectives and functions of operations management in organizational production systems.
CO2:	Analyze product and process design decisions for improving efficiency of operations systems.
CO3:	Apply capacity planning and facility layout methods for effective operations management.
CO4:	Evaluate production planning, scheduling and control techniques in manufacturing environments effectively.
CO5:	Apply inventory management and quality control methods to improve operational performance.

UNIT: I	INTRODUCTION TO OPERATIONS MANAGEMENT	9

Role of operations in organizations. Objectives of operations management. Functions of operations manager. Types of production systems. Operations strategy basics. Relationship with other departments. Productivity concepts and measurement. Importance of operations management.

UNIT: II	PRODUCT AND PROCESS DESIGN	9

Product development process. Product life cycle stages. Process selection methods. Types of production processes. Process planning techniques. Technology in production systems. Process analysis and improvement. Role of design in operations.

UNIT: III	CAPACITY AND FACILITY PLANNING	9

Capacity measurement methods. Capacity planning strategies. Facility location decisions. Factors affecting facility location. Facility layout planning. Types of layout systems. Layout design techniques. Importance of efficient facility planning.

UNIT: IV	PRODUCTION PLANNING AND SCHEDULING	9

Aggregate production planning. Master production scheduling. Material requirements planning basics. Routing and loading. Scheduling techniques in production. Dispatching procedures. Follow up and expediting. Production control methods.

UNIT: V	INVENTORY AND QUALITY MANAGEMENT	9

Types of inventory in industries. Inventory control techniques. Economic order quantity concept. Reorder level and safety stock. Quality management in operations. Inspection and testing methods. Continuous improvement practices. Role of quality in operations.

Total Periods: 45

Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, YouTube Videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	William J. Stevenson	Operations Management	McGraw Hill Education	2018
2	K. Aswathappa, K. Shridhara Bhat	Production and Operations Management	Himalaya Publishing House	2021

REFERENCE BOOKS:				
1	Nigel Slack, Stuart Chambers, Robert Johnston	Operations Management	Pearson Education	2020
2	Panneerselvam R.	Production and Operations Management	PHI Learning	2019

WEB LEARNING RESOURCES:

1. MIT Open Course Ware – Introduction to Operations Management Lecture Notes
2. https://ocw.mit.edu/courses/15-760a-operations-management-spring-2002/
3. https://www.managementstudyguide.com/operations-management.htm
4. https://www.tutorialspoint.com/production_and_operations_management/index.htm
5. https://www.edx.org/learn/operations-management

CO – PO – PSO MAPPING

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
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CO2	2	3	2	2	-	-	-	-	-	-	2	3	1	-
CO3	2	3	2	2	2	-	-	-	-	-	-	3	1	-
CO4	2	3	1	2	3	-	-	-	-	-	2	-	-	1
CO5	2	3	2	2	3	-	-	-	-	-	-	-	2	1
Avg	**2**	**3**	**1.8**	**2.0**	**2.5**	**-**	**-**	**-**	**-**	**-**	**2.0**	**3**	**2**	**1**